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Title 10 —Energy

Chapter I —Nuclear Regulatory Commission

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PART 53—RISK-INFORMED, TECHNOLOGY-INCLUSIVE REGULATORY FRAMEWORK FOR COMMERCIAL NUCLEAR PLANTS

Authority: Atomic Energy Act of 1954, secs. 11, 101, 103, 108, 122, 147, 161, 181, 182, 183, 184, 185, 186, 187, 189, 223, 234 (42 U.S.C. 2014, 2131, 2132, 2133, 2134, 2135, 2138, 2152, 2167, 2169, 2201, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2239, 2273, 2282); Energy Reorganization Act of 1974, secs. 201, 202, 206, 211 (42 U.S.C. 5841, 5842, 5846, 5851); Nuclear Waste Policy Act of 1982, sec. 306 (42 U.S.C. 10226); National Environmental Policy Act of 1969 (42 U.S.C. 4332); 44 U.S.C. 3504 note; Pub. L. 115-439, 132 Stat. 5571.

Source: 91 FR 15794, Mar. 30, 2026, unless otherwise noted.

§ 53.000 Purpose.

This part provides an optional, technology-inclusive, performance-based framework for the issuance, amendment, renewal, and termination of licenses, permits, certifications, and approvals for commercial nuclear plants licensed under section 103 of the Atomic Energy Act of 1954, as amended (the Act) (68 Stat. 919), and Title II of the Energy Reorganization Act of 1974, as amended (88 Stat. 1242). Also, this part gives notice to all persons who knowingly provide to any holder of or applicant for an approval, certification, permit, or license, or to a contractor, subcontractor, or consultant of any of them, components, equipment, materials, or other goods or services that relate to the activities of a holder of or applicant for an approval, certification, permit, or license, subject to this part, that they may be individually subject to U.S. Nuclear Regulatory Commission enforcement action for violation of the provisions in § 53.050.

Subpart A—General Provisions

§ 53.015 Scope.

Subpart A provides general provisions applicable to all applicants and licensees subject to the rules of this part.

§ 53.020 Definitions.

As used in this part:

Anticipated event sequence means event sequences expected to occur one or more times during the life of a commercial nuclear plant. Anticipated event sequences take into account the expected response of all structures, systems, and components (SSCs) within the plant, regardless of safety classification.

Applicant means a person applying for a license, permit, or other form of Commission permission or approval under this part.

Certified fuel handler means, for a commercial nuclear plant, either—

- (1) A non-licensed operator who has qualified in accordance with a fuel handler training program approved by the Commission; or
- (2) A non-licensed operator who demonstrates compliance with the following criteria:
 - (i) Has qualified in accordance with a fuel handler training program that demonstrates compliance with the same requirements as training programs for non-licensed operators required by § 53.830, and
 - (ii) Is responsible for decisions on—
 - (A) Safe conduct of decommissioning activities,
 - (B) Safe handling and storage of spent fuel; and
 - (C) Appropriate response to plant emergencies.

Combined license (COL) means a combined construction permit (CP) and operating license (OL) with conditions for a commercial nuclear plant issued under this part.

Commercial nuclear plant means a facility consisting of one or more commercial nuclear reactors and associated co-located support facilities, including the collection of buildings, radionuclide sources, and SSCs for which a license, certification, or approval is being sought under this part, that is or will be used for producing power for commercial electric power or other commercial purposes. For the purposes of requirements in this part that reference requirements in part 50 of this chapter, a commercial nuclear plant is equivalent to a nuclear power plant.

Commercial nuclear reactor means an apparatus, other than an atomic weapon, designed or used to sustain nuclear fission. For the purposes of requirements in this part that reference requirements in 10 CFR part 50, a commercial nuclear reactor is equivalent to a nuclear reactor as defined in § 50.2 of this chapter.

Commission means the U.S. Nuclear Regulatory Commission (NRC) or its duly authorized representatives.

Construction means the activities in paragraph (1) of this definition and does not mean the activities in paragraph (2) of this definition.

- (1) Activities constituting construction are those activities that are conducted on-site to build the commercial nuclear plant, including the driving of piles; subsurface preparation; placement of backfill, concrete, or permanent retaining walls within an excavation; installation of foundations; or in-place assembly, erection, fabrication, or testing, which are for—
 - (i) Safety-related (SR) SSCs and those non-safety-related but safety-significant (NSRSS) SSCs of a facility for which special treatment includes requirements on design or installation, including associated quality assurance measures;
 - (ii) SSCs necessary to comply with 10 CFR part 73; or
 - (iii) Onsite emergency facilities necessary to comply with § 53.855.
- (2) Construction does not include—
 - (i) Changes for temporary use of the land for public recreational purposes;
 - (ii) Site exploration, including necessary borings to determine foundation conditions or other preconstruction monitoring to establish background information related to the suitability of the site, the environmental impacts of construction or operation, or the protection of environmental values;
 - (iii) Preparation of a site for construction of a facility, including clearing of the site, grading, installation of drainage, erosion, and other environmental mitigation measures, and construction of temporary roads and borrow areas;
 - (iv) Erection of fences and other access control measures;
 - (v) Excavation;
 - (vi) Erection of support buildings (such as construction equipment storage sheds, warehouse and shop facilities, utilities, concrete mixing plants, docking and unloading facilities, and office buildings) for use in connection with the construction of the facility;
 - (vii) Building of service facilities (such as paved roads, parking lots, railroad spurs, exterior utility and lighting systems, potable water systems, sanitary sewage treatment facilities, and transmission lines);
 - (viii) Procurement or fabrication of components or portions of the proposed facility occurring at locations other than the final, in-place location at the facility; or
 - (ix) Manufacture of a nuclear power reactor under a manufacturing license (ML) under subpart H of this part to be installed at the proposed site and to be part of the proposed facility.

Custom combined license (custom COL) means a COL that does not reference a standard design approval, standard design certification, or manufacturing license.

Decommission or decommissioning means to remove a plant or site safely from service and reduce residual radioactivity to a level that permits—

- (1) Release of the property for unrestricted use and termination of the license; or
- (2) Release of the property under restricted conditions and termination of the license.

Defense in depth means inclusion of two or more independent and redundant layers of defense in the design of a facility and its operating procedures to compensate for uncertainties such that no single layer of defense, no matter how robust, is exclusively relied upon. Defense in depth includes, but is not limited to, the use of access controls, physical barriers, redundant and diverse safety functions, and emergency response measures.

Design-basis accidents (DBAs) means postulated event sequences that are used to set functional design criteria and performance objectives for the design of SR SSCs through deterministic analyses. Design-basis accidents are a type of licensing-basis event and are based on the capabilities and reliabilities of SR SSCs needed to mitigate and prevent event sequences, respectively.

Design-basis external hazard level means the level of severity or intensity of an external hazard for which the SR SSCs are protected against or designed to withstand without losing their capability to perform their safety functions.

Design features means the active and passive SSCs and the inherent characteristics of those SSCs that contribute to limiting the total effective dose equivalent to individual members of the public during normal operations and prevent or mitigate the consequences of event sequences.

Early site permit (ESP) means a Commission approval, issued under subpart H of this part, for a site for one or more commercial nuclear plants. An early site permit is a partial construction permit.

Electric utility means any entity that generates or distributes electricity and that recovers the cost of this electricity, either directly or indirectly, through rates established by the entity itself or by a separate regulatory authority. Investor-owned utilities, including generation or distribution subsidiaries, public utility districts, municipalities, rural electric cooperatives, and State and Federal agencies, including associations of any of the foregoing, are included within the meaning of “electric utility.”

Event sequence means a postulated initiating event defined for a set of initial plant conditions followed by system, safety function, and operator successes or failures, and terminating in a specified end state depending on the system, safety function, and operator successes and failures (e.g., prevention of release of radioactive material or release in one of the reactor-specific release categories). An event sequence may include many unique variations of events that are similar in terms of results or end states.

Exclusion area means that area surrounding the reactor, in which the reactor licensee has the authority to determine all activities including exclusion or removal of personnel and property from the area. This area may be traversed by a highway, railroad, or waterway, provided these are not so close to the facility as to interfere with normal operations of the facility and provided appropriate and effective arrangements are made to control traffic on the highway, railroad, or waterway, in case of emergency, to protect the public health and safety. Residence within the exclusion area must normally be prohibited. In any event, residents must be subject to ready removal in case of necessity. Activities unrelated to operation of the reactor may be permitted in an exclusion area under appropriate limitations, provided that no significant hazards to the public health and safety will result.

Fission product release means the amount and composition of radioactive material released to the environment, after accounting for any retention of radionuclides provided by reactor design features.

Fuel means special nuclear material (SNM) or source material, discrete elements that physically contain SNM or source material, and homogeneous mixtures that contain SNM or source material, intended to or used to create power in a commercial nuclear plant.

Functional design criteria means metrics for the performance of SSCs. For SR SSCs, these criteria define performance metrics necessary to demonstrate compliance with the safety criteria in § 53.210. For NSRSS SSCs, these criteria define performance metrics necessary to demonstrate compliance with the safety criteria in § 53.220.

License, when used in the context of a facility, means a limited work authorization, CP, OL, early site permit, COL, or ML under this part, or a renewed license issued by the Commission under this part. When used in the context of a license authorizing an individual to manipulate the controls of a facility, *license* means a license issued by the Commission to perform the function of an operator, senior operator, or generally licensed reactor operator as defined in this part.

Licensee means a person who is authorized to conduct activities under a license issued under this part by the Commission.

Licensing-basis events means a collection of event sequences considered in the design and licensing of the commercial nuclear plant. Licensing-basis events are unplanned events and include anticipated event sequences, unlikely event sequences, very unlikely event sequences, and DBAs.

Licensing-basis information means the information contained in regulations, orders, licenses, certifications, or approvals issued by the NRC for a commercial nuclear plant licensed under this part and that information submitted to the NRC by an applicant or licensee in a Safety Analysis Report, program description, or other licensing-related document required under this part.

Low-population zone means the area immediately surrounding the exclusion area which contains residents, the total number and density of which are such that there is a reasonable probability that appropriate protective measures could be taken on their behalf in the event of a serious accident. A permissible population density or total population within this zone is not included in this definition because the situation may vary from case to case. Whether a specific number of people can, for example, be evacuated from a specific area or instructed to take shelter on a timely basis, will depend on many factors such as location, number and size of highways, scope and extent of advance planning, and actual distribution of residents within the area.

Major decommissioning activity means, for a commercial nuclear plant, any activity that results in permanent removal of major radioactive components, permanently modifies the structure of the containment, if applicable, or results in dismantling components for shipment containing greater than class C waste in accordance with § 61.55 of this chapter.

Major feature of the emergency plans means an aspect of those plans necessary to:

- (1) Address in whole or part either one or more of the 16 standards in 10 CFR 50.47(b) or the requirements of 10 CFR 50.160(b), as applicable; or
- (2) Describe the emergency planning zones as required in 10 CFR 53.1109(g).

Manufactured reactor means the essential portions of a nuclear reactor that are manufactured under an ML and subsequently transported and incorporated into a commercial nuclear plant under a COL or CP.

Manufacturing license means a license issued under this part that authorizes the manufacture of manufactured reactors but not its construction, installation, or operation.

Non-Safety-Related but Safety-Significant (NSRSS) SSCs means those SSCs which are not SR but are relied on to achieve adequate defense in depth or perform risk-significant functions and warrant special treatment.

Non-Safety-Significant SSCs means those SSCs that are not SR or NSRSS, are not relied on to achieve adequate defense in depth or to perform risk-significant functions, and do not warrant special treatment.

Person means—

- (1) Any individual, corporation, partnership, firm, association, trust, estate, public or private institution, group, government agency other than the Commission or the Department of Energy, except that the Department of Energy shall be considered a person to the extent that its facilities are subject to the licensing and related regulatory authority of the Commission pursuant to section 202 of the Energy Reorganization Act of 1974, any State or any political subdivision of, or any political entity within a State, any foreign government or nation or any political subdivision of any such government or nation, or other entity; and
- (2) Any legal successor, representative, agent, or agency of the foregoing.

Population center distance means the distance from the reactor to the nearest boundary of a densely populated center containing more than about 25,000 residents.

Programmatic controls means administrative measures that govern human action in implementing programs and operating, monitoring, and maintaining SSCs and equipment of a commercial nuclear plant. Programmatic controls considered to be licensing basis information are addressed by programs under § 53.845 and are specified in an application for a requested activity of the Commission.

Quality assurance (QA) means all those planned and systematic actions necessary to ensure that a structure, system, or component will perform satisfactorily in service. Quality assurance includes quality control, which comprises those QA actions related to the physical characteristics of a material, structure, component, or system which provide a means to control the quality of the material, structure, component, or system to predetermined requirements.

Safety criteria means performance-based metrics that establish a level of safety provided in requirements in §§ 53.210 and 53.220.

Safety-related structures, systems, or components means those SSCs that are relied upon to demonstrate compliance with the safety criteria in § 53.210 and warrant special treatment.

Small modular reactor means a power reactor, which may be of modular design as defined in § 52.1 of this chapter, licensed under this part to produce heat energy up to 1,000 megawatts thermal per module.

Site characteristics means the actual physical, environmental, and demographic features of a site. Site characteristics are specified in an early site permit or in a Preliminary or Final Safety Analysis Report for a limited work authorization, CP, or COL, as applicable.

Site parameters are the postulated physical, environmental, and demographic features of an assumed site. Site parameters are specified in a standard design approval, standard design certification, or ML.

Source material means source material as defined in subsection 11z. of the Atomic Energy Act of 1954, as amended, (the Act) and in the regulations contained in part 40 of this chapter.

Special nuclear material (SNM) means:

- (1) Plutonium, uranium-233, uranium enriched in the isotope-233 or in the isotope-235, and any other material which the Commission, pursuant to the provisions of section 51 of the Act, determines to be SNM, but does not include source material; or
- (2) Any material artificially enriched by any of the foregoing, but does not include source material.

Special treatment means those requirements, such as QA, design criteria, and programmatic controls, that are taken beyond the procurement, installation, and maintenance of commercial grade products to ensure that SR and NSRSS SSCs will provide defense in depth or perform risk-significant functions. The requirements also ensure that the SSCs will perform under the service conditions and with the reliability assumed in the analysis performed under § 53.450 to demonstrate compliance with the safety criteria in §§ 53.210 for SR SSCs and 53.220 for SR and NSRSS SSCs.

Standard design means a design which is sufficiently detailed and complete to support certification or approval in accordance with subpart H of this part, and which is usable under of this part for a multiple number of units or at a multiple number of sites without reopening or repeating the review.

Standard design approval or design approval means an NRC staff approval, issued under subpart H of this part, of a final standard design for a commercial nuclear plant. The approval may be for either the final design for the entire reactor facility or the final design of major portions thereof.

Standard design certification or design certification means a Commission approval, issued under subpart H of this part, of a final standard design for a nuclear power facility. This design may be referred to as a certified standard design.

Total effective dose equivalent means the sum of the effective dose equivalent (for external exposures) and the committed effective dose equivalent (for internal exposures).

Utilization facility means any commercial nuclear reactor other than one designed or used primarily for the formation of plutonium or uranium-233.

Unlikely event sequences means event sequences that are not expected to occur in the life of a commercial nuclear plant and are less likely than anticipated event sequences, but are infrequent rather than rare. Unlikely event sequences take into account the expected response of all SSCs within the plant regardless of safety classification.

Very unlikely event sequences means event sequences that are not expected to occur in the life of a commercial nuclear plant, are less likely than an unlikely event sequence, and are rare. Very unlikely event sequences take into account the expected response of all SSCs within the plant regardless of safety classification.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 18773, Apr. 13, 2026]

§ 53.030 [Reserved]

§ 53.040 Written communications.

- (a) **General requirements.** All correspondence, reports, applications, and other written communications from the applicant or licensee to the NRC concerning the regulations in this part or individual license conditions must be sent either by mail addressed: ATTN: Document Control Desk, U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001; by hand delivery to the NRC's offices at 11555 Rockville Pike, Rockville, Maryland, between the hours of 8:15 a.m. and 4 p.m. eastern time; or, where practicable, by electronic submission, for example, via Electronic Information Exchange, email, or CD-ROM. Electronic submissions must be made in a manner that enables the NRC to receive, read, authenticate, distribute, and archive the submission, and process and retrieve it a single page at a time. Detailed guidance on making electronic submissions can be obtained by visiting the NRC's website at <https://www.nrc.gov/site-help/e-submittals.html>; by email to MSHD.Resource@nrc.gov; or by writing the Office of the Chief Information Officer, U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001. The guidance discusses, among other topics, the formats the NRC can accept, the use of electronic signatures, and the

treatment of nonpublic information. If the communication is on paper, the signed original must be sent. If a submission due date falls on a Saturday, Sunday, or Federal holiday, the next Federal working day becomes the official due date.

- (b) **Distribution requirements.** Copies of all correspondence, reports, and other written communications concerning the regulations in this part or individual license conditions, or the terms and conditions of an early site permit or standard design approval, must be submitted to the persons listed below (addresses for the NRC Regional Offices are listed in appendix D to 10 CFR part 20).
- (1) **Applications for amendment of permits and licenses, reports, and other communications.** All written communications (including responses to generic letters, bulletins, information notices, regulatory information summaries, inspection reports, and miscellaneous requests for additional information) that are required of holders of licenses, permits, and design approvals issued pursuant to this part, must be submitted as follows, except as otherwise specified in paragraphs (b)(2) through (7) of this section: to the NRC's Document Control Desk (if on paper, the signed original), with a copy to the appropriate Regional Office, and a copy to the appropriate NRC Resident Inspector if one has been assigned to the site of the facility or the place of manufacture of a reactor licensed under this part.
- (2) **Applications for permits and licenses, and amendments to applications.** Applications for licenses, permits, and design approvals and amendments to any of these types of applications must be submitted to the NRC's Document Control Desk, with a copy to the appropriate Regional Office, and a copy to the appropriate NRC Resident Inspector if one has been assigned to the facility or the place of manufacture of a reactor licensed under this part, except as otherwise specified in paragraphs (b)(3) through (9) of this section. If the application or amendment is on paper, the submission to the Document Control Desk must be the signed original.
- (3) **Acceptance review application.** Written communications required for an application for determination of suitability for docketing must be submitted to the NRC's Document Control Desk, with a copy to the appropriate Regional Office. If the communication is on paper, the submission to the Document Control Desk must be the signed original.
- (4) **Security plan and related submissions.** Written communications, as defined in paragraphs (b)(4)(i) through (v) of this section, must be submitted to the NRC's Document Control Desk, with a copy to the appropriate Regional Office. If the communication is on paper, the submission to the Document Control Desk must be the signed original. Submissions should include the following as appropriate:
- (i) Physical security plan;
 - (ii) Safeguards contingency plan;
 - (iii) Cybersecurity plan;
 - (iv) Change to security plan, guard training and qualification plan, safeguards contingency plan, or cybersecurity plan made without prior Commission approval under § 53.1565; and
 - (v) Application for amendment of physical security plan, guard training and qualification plan, safeguards contingency plan, or cybersecurity plan under § 53.1510.

- (5) **Emergency plan and related submissions.** Written communications as defined in paragraphs (b)(5)(i) through (iii) of this section must be submitted to the NRC's Document Control Desk, with a copy to the appropriate Regional Office, and a copy to the appropriate NRC Resident Inspector if one has been assigned to the site of the facility. If the communication is on paper, the submission to the Document Control Desk must be the signed original. Submissions should include the following as appropriate:
- (i) Emergency plan;
 - (ii) Change to an emergency plan under § 53.1565; and
 - (iii) Emergency implementing procedures under § 53.855.
- (6) **Updated Final Safety Analysis Report.** An updated Final Safety Analysis Report or replacement pages under § 53.1545 must be submitted to the NRC's Document Control Desk, with a copy to the appropriate Regional Office, and a copy to the appropriate NRC Resident Inspector if one has been assigned to the site of the facility or the place of manufacture of a reactor licensed under this part. Paper copy submissions may be made using replacement pages; however, if a licensee chooses to use electronic submission, all subsequent updates or submissions must be performed electronically on a total replacement basis. If the communication is on paper, the submission to the Document Control Desk must be the signed original. If the communications are submitted electronically, see Guidance for Electronic Submissions to the Commission.
- (7) **Quality assurance related submissions.**
- (i) A change to the Safety Analysis Report QA program description under § 53.1565, or a change to a licensee's NRC-accepted QA topical report under § 53.1565, must be submitted to the NRC's Document Control Desk, with a copy to the appropriate Regional Office, and a copy to the appropriate NRC Resident Inspector if one has been assigned to the site of the facility or the place of manufacture of a reactor licensed under this part. If the communication is on paper, the submission to the Document Control Desk must be the signed original.
 - (ii) A change to an NRC-accepted QA topical report from non-licensees (*i.e.*, architect/engineers, nuclear steam supply system suppliers, fuel suppliers, constructors, etc.) must be submitted to the NRC's Document Control Desk. If the communication is on paper, the signed original must be sent.
- (8) **Certification of permanent cessation of operations.** The licensee's certification of permanent cessation of operations, under subpart G of this part, must state the date on which operations have ceased or will cease, and must be submitted to the NRC's Document Control Desk. This submission must be under oath or affirmation.
- (9) **Certification of permanent fuel removal.** The licensee's certification of permanent fuel removal, under subpart G of this part, must state the date on which the fuel was removed from the reactor vessel and the disposition of the fuel, and must be submitted to the NRC's Document Control Desk. This submission must be under oath or affirmation.
- (c) **Form of communications.** All paper copies submitted to demonstrate compliance with the requirements set forth in paragraph (b) of this section must be typewritten, printed, or otherwise reproduced in permanent form on unglazed paper. Exceptions to these requirements imposed on paper submissions may be granted for the submission of micrographic, photographic, or similar forms.

- (d) **Regulation governing submission.** Licensees, applicants, and holders of standard design approvals submitting correspondence, reports, and other written communications under the regulations of this part are requested but not required to cite whenever practical, in the upper right corner of the first page of the submission, the specific regulation or other basis requiring submission.

§ 53.050 Deliberate misconduct.

- (a) Any licensee or applicant for a license; holder of or applicant for a standard design approval; applicant for a standard design certification; employee of a licensee, holder of a standard design approval, or applicant for a license, standard design approval, or standard design certification; or any contractor (including a supplier or consultant), subcontractor, employee of a contractor or subcontractor of any licensee or applicant for a license, holder of or applicant for a standard design approval, or applicant for a standard design certification, who knowingly provides to any licensee, applicant, contractor, or subcontractor, any components, equipment, materials, or other goods or services that relate to a licensee's or applicant's activities in this part, may not—
 - (1) Engage in deliberate misconduct that causes or would have caused, if not detected, a licensee or applicant to be in violation of any rule, regulation, or order; or any term, condition, or limitation of any license issued by the Commission; or
 - (2) Deliberately submit to the NRC, a licensee, an applicant, or a licensee's or applicant's contractor or subcontractor, information that the person submitting the information knows to be incomplete or inaccurate in some respect material to the NRC.
- (b) A person who violates paragraph (a)(1) or (2) of this section may be subject to enforcement action in accordance with the procedures in subpart B of 10 CFR part 2.
- (c) For the purposes of paragraph (a)(1) of this section, deliberate misconduct by a person means an intentional act or omission that the person knows—
 - (1) Would cause a licensee or applicant to be in violation of any rule, regulation, or order; or any term, condition, or limitation, of any license issued by the Commission; or
 - (2) Constitutes a violation of a requirement, procedure, instruction, contract, purchase order, or policy of a licensee, applicant, contractor, or subcontractor.

§ 53.060 Employee protection.

- (a) Discrimination by a Commission licensee, holder of a standard design approval, an applicant for a license, standard design certification, or standard design approval, a contractor or subcontractor of a Commission licensee, holder of a standard design approval, applicant for a license, standard design certification, or standard design approval, against an employee for engaging in certain protected activities is prohibited. Discrimination includes discharge and other actions that relate to compensation, terms, conditions, or privileges of employment. The protected activities are established in section 211 of the Energy Reorganization Act of 1974, as amended, and in general are related to the administration or enforcement of a requirement imposed under the Act or the Energy Reorganization Act of 1974, as amended.
 - (1) The protected activities include but are not limited to—
 - (i) Providing the Commission or his or her employer information about alleged violations of either of the statutes named in paragraph (a) of this section or possible violations of requirements imposed under either of those statutes;

- (ii) Refusing to engage in any practice made unlawful under either of the statutes named in paragraph (a) of this section or under these requirements if the employee has identified the alleged illegality to the employer;
 - (iii) Requesting the NRC to institute action against his or her employer for the administration or enforcement of these requirements;
 - (iv) Testifying in any Commission proceeding, or before Congress, or at any Federal or State proceeding regarding any provision (or proposed provision) of either of the statutes named in paragraph (a) of this section; and
 - (v) Assisting or participating in, or being about to assist or participate in, these activities.
- (2) These activities are protected even if no formal proceeding is actually initiated as a result of the employee assistance or participation.
- (3) This section has no application to any employee alleging discrimination prohibited by this section who, acting without direction from his or her employer (or the employer's agent), deliberately causes a violation of any requirement of the Energy Reorganization Act of 1974, as amended, or the Act.
- (b) Any employee who believes that they have been discharged or otherwise discriminated against by any person for engaging in protected activities specified in paragraph (a)(1) of this section may seek a remedy for the discharge or discrimination through an administrative proceeding in the Department of Labor. The administrative proceeding must be initiated within 180 days after an alleged violation occurs. The employee may do this by filing a complaint alleging the violation with the Department of Labor, Wage and Hour Division. The Department of Labor may order reinstatement, back pay, and compensatory damages.
- (c) A violation of paragraph (a), (e), or (f) of this section by a Commission licensee, a holder of a standard design approval, an applicant for a Commission license, standard design certification, or a standard design approval, or a contractor or subcontractor of a Commission licensee, holder of a standard design approval, or any applicant may be grounds for—
 - (1) Denial, revocation, or suspension of the license or standard design approval;
 - (2) Withdrawal or revocation of a proposed or final standard design certification;
 - (3) Imposition of a civil penalty on the licensee, holder of a standard design approval, or applicant (including an applicant for a standard design certification under this part following Commission adoption of final design certification rule) or a contractor or subcontractor of the licensee, holder of a standard design approval, or applicant; or
 - (4) Other enforcement action.
- (d) Actions taken by an employer, or others, which adversely affect an employee may be predicated upon nondiscriminatory grounds. The prohibition applies when the adverse action occurs because the employee has engaged in protected activities. An employee's engagement in protected activities does not automatically render him or her immune from discharge or discipline for legitimate reasons or from adverse action dictated by nonprohibited considerations.
- (e)

- (1) Each holder or applicant for a license or design approval, must prominently post the revision of NRC Form 3, "Notice to Employees," referenced in § 19.11(e)(1) of this chapter. This form must be posted at locations sufficient to permit employees protected by this section to observe a copy on the way to or from their place of work. Premises must be posted no later than 30 days after an application is docketed and remain posted while the application is pending before the Commission, during the term of the license, and for 30 days following license termination.
- (2) Copies of NRC Form 3 may be obtained by writing to the Regional Administrator of the appropriate NRC Regional Office listed in appendix D to 10 CFR part 20, via email to Forms.Resource@nrc.gov, or by visiting the NRC's online library at <https://www.nrc.gov/reading-rm/doc-collections/forms/>.
- (f) No agreement affecting the compensation, terms, conditions, or privileges of employment, including an agreement to settle a complaint filed by an employee with the Department of Labor pursuant to section 211 of the Energy Reorganization Act of 1974, as amended, may contain any provision which would prohibit, restrict, or otherwise discourage an employee from participating in protected activity as defined in paragraph (a)(1) of this section, including, but not limited to, providing information to the NRC or to his or her employer on potential violations or other matters within NRC's regulatory responsibilities.
- (g) Part 19 of 10 CFR sets forth requirements and regulatory provisions applicable to licensees, holders of a standard design approval, applicants for a license, standard design certification, or standard design approval, and contractors or subcontractors of a Commission licensee, or holder of a standard design approval, and are in addition to the requirements in this section.

§ 53.070 Completeness and accuracy of information.

- (a) Information provided to the Commission by a holder of a license, permit, design certification, or standard design approval under this part or an applicant for a license, permit, design certification, or standard design approval under this part, and information required by statute or by the Commission's regulations, orders, license conditions, or terms and conditions of a standard design approval to be maintained by the applicant or the licensee must be complete and accurate in all material respects.
- (b) Each applicant or licensee, each holder of a standard design approval under this part, and each applicant for a standard design certification under this part following Commission adoption of a final design certification regulation, must notify the Commission of information identified by the applicant or licensee as having for the regulated activity a significant implication for public health and safety or common defense and security. An applicant, licensee, or holder violates this paragraph (b) only if the applicant, licensee, or holder fails to notify the Commission of information that the applicant, licensee, or holder has identified as having a significant implication for public health and safety or common defense and security. Notification must be provided to the Administrator of the appropriate Regional Office within 2 working days of identifying the information. This requirement is not applicable to information which is already required to be provided to the Commission by other reporting or updating requirements.

§ 53.080 Specific exemptions.

- (a) The Commission may, upon application by any interested person or upon its own initiative, grant exemptions from the requirements of the regulations of this part, which are authorized by law, will not present an undue risk to the public health and safety, and are consistent with the common defense and security.
- (b) The Commission will not consider granting an exemption unless special circumstances are present. Special circumstances are present whenever—

- (1) Application of the regulation in the particular circumstances conflicts with other rules or requirements of the Commission;
 - (2) Application of the regulation in the particular circumstances would not serve the underlying purpose of the rule or is not necessary to achieve the underlying purpose of the rule;
 - (3) Compliance would result in undue hardship or other costs that are significantly in excess of those contemplated when the regulation was adopted, or that are significantly in excess of those incurred by others similarly situated;
 - (4) The exemption would result in benefit to the public health and safety that compensates for any decrease in safety that may result from the grant of the exemption;
 - (5) The exemption would provide only temporary relief from the applicable regulation and the licensee or applicant has made good faith efforts to comply with the regulation; or
 - (6) There is present any other material circumstance not considered when the regulation was adopted for which it would be in the public interest to grant an exemption. If such condition is relied on exclusively for demonstrating compliance with paragraph (b) of this section, the exemption may not be granted until the Executive Director for Operations has consulted with the Commission.
- (c) Any person may request an exemption permitting the conduct of construction activities prior to the issuance of a CP. The Commission may grant such an exemption upon considering and balancing the following factors:
- (1) Whether conduct of the proposed activities will give rise to a significant adverse impact on the environment and the nature and extent of such impact, if any;
 - (2) Whether redress of any adverse environment impact from conduct of the proposed activities can reasonably be effective should such redress be necessary;
 - (3) Whether conduct of the proposed activities would foreclose subsequent adoption of alternatives; and
 - (4) The effect of delay in conducting such activities on the public interest, including whether the power needs to be used by the proposed facility, the availability of alternative sources, if any, to meet those needs on a timely basis, and delay costs to the applicant and to consumers.
- (d) Issuance of such an exemption must not be deemed to constitute a commitment to issue a CP. During the period of any exemption granted pursuant to paragraph (c) of this section, any activities conducted must be carried out in such a manner as will minimize or reduce their environmental impact.
- (e) The Commission's consideration of requests for exemptions from requirements of the regulations of other parts in this chapter that are applicable by virtue of this part must be governed by the exemption requirements of those parts.

§ 53.090 Standards for review.

- (a) **Common standards.** In determining that a CP, OL, early site permit, COL, or ML under this part will be issued to an applicant, the Commission will be guided by the following considerations:

- (1) Except for an early site permit or ML, the processes to be performed, the operating procedures, the facility and equipment, the use of the facility, and other technical specifications, or the proposals, in regard to any of the foregoing, collectively provide reasonable assurance that the applicant will comply with the regulations in this chapter, including the regulations in 10 CFR part 20, and that the health and safety of the public will not be endangered.
 - (2) The applicant for a CP, OL, COL, or ML is technically and financially qualified to engage in the proposed activities in accordance with the regulations in this chapter. However, no consideration of financial qualification is necessary for an electric utility applicant for an OL for a utilization facility of the type described in paragraph (d) of this section or for an applicant for an ML.
 - (3) The issuance of a CP, OL, early site permit, COL, or ML to the applicant will not, in the opinion of the Commission, be inimical to the common defense and security or to the health and safety of the public.
 - (4) Any applicable requirements of 10 CFR part 51 have been satisfied.
- (b) **Additional standards for licenses.** In determining whether a license will be issued to an applicant, the Commission will, in addition to applying the standards set forth in paragraph (a) of this section, consider whether the proposed activities will serve a useful purpose proportionate to the quantities of SNM or source material to be utilized.
- (c) **Additional standards and provisions affecting licenses for commercial power.** In addition to applying the standards set forth in paragraphs (a) and (b) of this section, paragraphs (c)(1) through (c)(4) of this section apply in the case of a license for a facility for the generation of commercial power.
- (1) The NRC will—
 - (i) Give notice in writing of each application to the regulatory agency or State as may have jurisdiction over the rates and services incident to the proposed activity;
 - (ii) Publish notice of the application in trade or news publications as it deems appropriate to give reasonable notice to municipalities, private utilities, public bodies, and cooperatives which might have a potential interest in the utilization or production facility; and
 - (iii) Publish notice of the application once each week for four consecutive weeks in the FEDERAL REGISTER. No license will be issued by the NRC prior to the giving of these notices and until four weeks after the last notice is published in the FEDERAL REGISTER.
 - (2) If there are conflicting applications for a limited opportunity for such license, the Commission will give preferred consideration in the following order: first, to applications submitted by public or cooperative bodies for facilities to be located in high cost power areas in the United States; second, to applications submitted by others for facilities to be located in such areas; third, to applications submitted by public or cooperative bodies for facilities to be located in areas other than high cost power areas; and, fourth, to all other applicants.
 - (3) The licensee who transmits electric energy in interstate commerce, or sells it at wholesale in interstate commerce, must be subject to the regulatory provisions of the Federal Power Act.
 - (4) Nothing will preclude any government agency, now or hereafter authorized by law to engage in the production, marketing, or distribution of electric energy, if otherwise qualified, from obtaining a CP, OL, or COL under this part for a utilization facility for the primary purpose of producing electric energy for disposition for ultimate public consumption.

- (d) *Licenses for commercial nuclear plants.* A license will be issued, to an applicant who qualifies, for any one or more of the following: to transfer or receive in interstate commerce, or manufacture, produce, transfer, acquire, possess, or use a utilization facility for industrial or commercial purposes.

§ 53.100 Jurisdictional limits.

No permit, license, standard design approval, or standard design certification under this part shall be deemed to have been issued for activities that are not under or within the jurisdiction of the United States.

§ 53.110 Attacks and destructive acts.

Licensees, applicants for licenses, permits, certifications, and design approvals, and applicants for an amendment to any license, permit, certification, or design approval under this part are not required to provide for design features or other measures for the specific purpose of protection against the effects of—

- (a) Attacks and destructive acts, including sabotage, directed against the facility by an enemy of the United States, whether a foreign government or other person; or
- (b) Use or deployment of weapons incident to U.S. defense activities.

§ 53.115 Rights related to special nuclear material.

- (a) No right to the SNM will be conferred by a license issued under this part except as may be defined by the license.
- (b) Neither a license issued under this part, nor any right thereunder, nor any right to utilize or produce SNM may be transferred, assigned, or disposed of in any manner, either voluntarily or involuntarily, directly or indirectly, through transfer of control of the license to any person, unless the Commission, after securing full information, finds that the transfer is in accordance with the provisions of the Act and gives its consent in writing.

§ 53.117 License suspension and rights of recapture.

Any license issued under this part must be subject to suspension and to the rights of recapture of the material or control of the facility reserved to the Commission under section 108 of the Act in a state of war or national emergency declared by Congress.

§ 53.120 Information collection requirements: OMB approval.

- (a) The NRC has submitted the information collection requirements contained in this part to the Office of Management and Budget (OMB) for approval as required by the Paperwork Reduction Act (44 U.S.C. 3501 *et seq.*). The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number. OMB has approved the information collection requirements contained in this part under control number 3150-0274.
- (b) The approved information collection requirements contained in this part appear in §§ 53.070, 53.080, 53.240, 53.410, 53.420, 53.425, 53.430, 53.440, 53.450, 53.480, 53.500, 53.540, 53.605, 53.610, 53.620, 53.700, 53.710, 53.715, 53.720, 53.730, 53.780, 53.785, 53.805, 53.810, 53.815, 53.830, 53.850, 53.855, 53.865, 53.870, 53.875, 53.880, 53.910, 53.1010, 53.1020, 53.1030, 53.1045, 53.1060, 53.1070, 53.1075, 53.1080, 53.1100, 53.1109, 53.1115, 53.1130, 53.1140, 53.1144, 53.1146, 53.1173, 53.1182, 53.1188, 53.1200, 53.1206, 53.1209, 53.1210, 53.1221, 53.1230, 53.1236, 53.1239, 53.1241, 53.1254, 53.1257, 53.1263, 53.1270, 53.1276, 53.1279, 53.1282, 53.1288, 53.1295, 53.1300, 53.1306, 53.1309, 53.1312,

53.1327, 53.1330, 53.1333, 53.1336, 53.1348, 53.1360, 53.1366, 53.1369, 53.1372, 53.1384, 53.1410, 53.1413, 53.1416, 53.1419, 53.1437, 53.1449, 53.1452, 53.1458, 53.1470, 53.1505, 53.1510, 53.1515, 53.1525, 53.1530, 53.1535, 53.1540, 53.1545, 53.1550, 53.1560, 53.1565, 53.1570, 53.1575, 53.1580, 53.1620, 53.1630, 53.1645, 53.1690, 53.1720.

- (c) This part contains information collection requirements in addition to those approved under the control number specified in paragraph (a) of this section. The information collection requirement and the control numbers under which it is approved are as follows:
- (1) In §§ 53.765, 53.770, 53.780, and 53.795, NRC Form 396 is approved under control number 3150-0024.
 - (2) In §§ 53.775 and 53.795, NRC Form 398 is approved under control number 3150-0090.
 - (3) In § 53.1640, NRC Form 366 is approved under control number 3150-0104.
 - (4) In § 53.1630, NRC Form 361S is approved under control number 3150-0238.
 - (5) In § 53.1650, International Atomic Energy Agency Design Information Questionnaire forms are approved under control number 3150-0056.
 - (6) In § 53.1650, DOC/NRC Form AP-A and associated forms are approved under control numbers 0694-0135.

Subpart B—Technology-Inclusive Safety Requirements

§ 53.210 Safety criteria for design-basis accidents.

Design features and programmatic controls must be provided for each commercial nuclear plant such that identification and analyses of design-basis accidents (DBAs) in accordance with § 53.240 demonstrate the following:

- (a) An individual located at any point on the boundary of the exclusion area for any 2-hour period following the onset of the postulated fission product release would not receive a radiation dose in excess of 25 rem (250 millisieverts) total effective dose equivalent (TEDE); and
- (b) An individual located at any point on the outer boundary of the low-population zone who is exposed to the radioactive cloud resulting from the postulated fission product release (during the entire period of its passage) would not receive a radiation dose in excess of 25 rem (250 millisieverts) TEDE.^[1]

§ 53.220 Safety criteria for licensing-basis events other than design-basis accidents.

Design features and programmatic controls must be provided for each commercial nuclear plant such that identification and analysis of licensing-basis events (LBEs) other than DBAs in accordance with § 53.240 demonstrate the following:

^[1] The use of 25 rem TEDE is not intended to imply that this number constitutes an acceptable limit for an emergency dose to the public under accident conditions. Rather, this dose value has been set forth in this section as a reference value, which can be used in the evaluation of plant design features with respect to postulated reactor accidents, to assure that these designs provide assurance of low risk of public exposure to radiation, in the event of an accident.

- (a) Plant structures, systems, and components (SSCs), personnel, and programs provide the necessary capabilities and maintain the necessary reliability to address LBEs other than DBAs in accordance with §§ 53.240 and 53.450(e), and provide measures for defense in depth in accordance with § 53.250; and
- (b) The analysis of risks to public health and safety resulting from LBEs other than DBAs under § 53.450(e) includes comprehensive risk metrics that satisfy associated risk performance objectives that are acceptable to the U.S. Nuclear Regulatory Commission (NRC) and provide an appropriate level of safety.

§ 53.230 Safety functions.

- (a) The primary safety function is limiting the release of radioactive materials from the facility and must be maintained during normal operation and for LBEs over the life of the plant.
- (b) Additional safety functions needed to support the retention of radioactive materials during LBEs—such as controlling reactivity, heat generation, heat removal, and chemical interactions—must be identified for each commercial nuclear plant.
- (c) The primary and additional safety functions are required to satisfy the safety criteria defined in §§ 53.210 and 53.220 and must be fulfilled by the design features, human actions, and programmatic controls specified throughout this part.

§ 53.240 Licensing-basis events.

- (a) Licensing-basis events must be identified for each commercial nuclear plant and analyzed under § 53.450 to demonstrate that the safety requirements in this subpart have been satisfied.
- (b) The identified LBEs, ranging from anticipated event sequences to very unlikely event sequences, must collectively address appropriate risk-informed combinations of malfunctions of plant SSCs, human errors, facility hazards, and the effects of external hazards.
- (c) The analysis of LBEs must—
 - (1) Include analysis of one or more DBAs under § 53.450(f);
 - (2) Confirm the adequacy of design features and programmatic controls needed to satisfy the safety criteria defined in §§ 53.210 and 53.220, and
 - (3) Establish related functional requirements for plant SSCs, personnel, and programs.

§ 53.250 Defense in depth.

- (a) Measures must be taken for each commercial nuclear plant to ensure appropriate defense in depth is provided to compensate for uncertainties in the analysis of the safety criteria such that there is reasonable assurance that the safety criteria in this subpart are met over the life of the plant.
- (b) The uncertainties that must be addressed under paragraph (a) of this section include those related to the state of knowledge and modeling capabilities, the ability of barriers to limit the release of radioactive materials from the facility during LBEs other than DBAs, the reliability and performance of plant SSCs and personnel, and the effectiveness of programmatic controls.
- (c) The safety analysis may not exclusively rely upon a single engineered design feature, human action, or programmatic control, no matter how robust, to address the range of LBEs other than DBAs.

§ 53.260 Normal operations.

Holders of licenses to operate commercial nuclear plants under this part must control public doses and dose rates in unrestricted areas from normal plant operations to meet the requirements in 10 CFR part 20.

§ 53.270 Protection of plant workers.

Holders of licenses to operate commercial nuclear plants under this part must control occupational doses to meet the requirements in 10 CFR part 20.

Subpart C—Design and Analysis Requirements

§ 53.400 Design features for licensing-basis events.

- (a) Design features must be provided for each commercial nuclear plant such that, when combined with corresponding human actions and programmatic controls, the plant will satisfy the safety criteria defined in §§ 53.210 and 53.220.
- (b) Design features must ensure that the safety functions identified in § 53.230 are fulfilled during licensing-basis events (LBEs).

§ 53.410 Functional design criteria for design-basis accidents.

- (a) Functional design criteria must be defined for each design feature classified as safety-related (SR) in terms of its role in demonstrating compliance with the safety criteria defined in § 53.210.
- (b) The identification of special treatments associated with the design of SR structures, systems, and components (SSCs) must consider human actions and programmatic controls identified and implemented in accordance with this and other subparts to achieve and maintain the reliability and capability of SSCs relied upon to satisfy the defined functional design criteria and the safety criteria required in § 53.210, and to maintain consistency with analyses required by § 53.450(f).

§ 53.415 Protection against external hazards.

Safety-related SSCs must be protected against or must be designed to withstand the effects of natural phenomena (e.g., earthquakes, tornadoes, hurricanes, floods, tsunami, and seiches) and constructed hazards (e.g., dams, transportation routes, military and industrial facilities) considering an event severity up to the design-basis external hazard levels as determined under § 53.510 without losing the capability to perform the safety functions identified under § 53.230. Specific requirements for earthquake engineering are included in § 53.480.

§ 53.420 Functional design criteria for licensing-basis events other than design-basis accidents.

- (a) Functional design criteria must be defined for each design feature classified as SR or non-safety-related but safety-significant (NSRSS) in terms of its role in demonstrating compliance with—
 - (1) The safety criteria in § 53.220; and
 - (2) The evaluation criteria in § 53.450(e).
- (b) The identification of special treatments associated with the design of SR and NSRSS SSCs must consider human actions and programmatic controls identified and implemented in accordance with this and other subparts to achieve and maintain the reliability and capability of SSCs relied upon to satisfy—

- (1) The safety criteria in § 53.220; and
- (2) The evaluation criteria in § 53.450(e).

§ 53.425 Design features and functional design criteria for normal operations.

- (a) Design features must be provided for each commercial nuclear plant to support the Radiation Protection Program required in § 53.850.
- (b) Functional design criteria must be defined for each design feature relied upon to demonstrate compliance with § 53.850.
- (c) Functional design criteria, including design objectives for dose to the maximally exposed member of the public, must be defined for design features to show that plant design features and corresponding programmatic controls, including monitoring programs, control liquid, gaseous, and solid wastes, as required under part 20 of this chapter.

§ 53.430 Design features and functional design criteria for protection of plant workers.

- (a) Design features must be provided for each commercial nuclear plant such that, when combined with corresponding programmatic controls, the requirements in § 53.270 can be met.
- (b) Functional design criteria must be defined for each design feature relied upon to demonstrate compliance with § 53.270.

§ 53.440 Design requirements.

- (a)
 - (1) Analysis, appropriate test programs, prototype testing, operating experience, or a combination thereof must demonstrate that each design feature required by § 53.400 meets the defined functional design criteria required by §§ 53.410 and 53.420. This demonstration must consider interdependent effects throughout the commercial nuclear plant and the range of conditions under which the design features required by § 53.400 must function throughout the plant's lifetime.
 - (2) The design processes for SR and NSRSS SSCs under this part must include administrative procedures for evaluating operating, design, and construction experience and for considering applicable important industry experiences in the design of those SSCs.
- (b) The design features classified as SR must, wherever applicable, be designed using generally accepted consensus codes and standards that have been endorsed or otherwise found acceptable by the U.S. Nuclear Regulatory Commission (NRC).
- (c) The materials used for each SR and NSRSS SSC must be qualified for their service conditions over the design life of the SSC as appropriate to satisfy the special treatments established for the SSC under § 53.460.
- (d) Possible degradation mechanisms related to aging, fatigue, chemical interactions, operating temperatures, effects of irradiation, and other environmental factors that may affect the performance of SR and NSRSS SSCs must be evaluated and used to inform the design and the development of integrity assessment programs under § 53.870.
- (e)

- (1) Safety-related SSCs and, where appropriate, NSRSS SSCs must be designed and located to minimize, consistent with other safety requirements in this part, the probability and effect of fires and explosions.
- (2) Noncombustible and fire-resistant materials must be used wherever practical throughout the facility, particularly in locations with SR and NSRSS SSCs.
- (3) Fire detection and fire suppression systems of appropriate capacity and capability must be provided and designed to minimize the adverse effects of fires on SR and NSRSS SSCs.
- (4) Fire suppression systems must be designed to ensure that their rupture or inadvertent operation does not significantly impair the ability of SR and NSRSS SSCs to perform their safety functions to satisfy § 53.230.
- (f) Safety and security must be considered together in the design process such that, where possible, security issues are effectively resolved through design and engineered security features.
- (g) The reactor system and waste stores for each commercial nuclear plant must be capable of achieving and maintaining a subcritical condition during normal operations and following any LBE identified in accordance with § 53.240.
- (h) Each commercial nuclear plant must have a capability to provide long-term cooling of the reactor fuel and waste stores during normal operations and following any LBE identified in accordance with § 53.240.
- (i) The design, analysis, staffing, and programmatic controls for each commercial nuclear plant must consider the number of reactors, waste stores, and other significant inventories of radioactive materials and the associated operating configurations, common systems, system interfaces, and system interactions.
- (j) [Reserved]
- (k) Design features, related functional design criteria, programmatic controls, or a combination thereof must be defined such that analyses demonstrate a low risk of permanent injury to the public due to the health effects of the chemical hazards of licensed material.
- (l) Measures must be taken during the design of commercial nuclear plants to minimize, to the extent practicable, contamination of the facility and the environment, facilitate eventual decommissioning, and minimize, to the extent practicable, the generation of radioactive waste in accordance with § 20.1406 of this chapter.
- (m)
 - (1) Each commercial nuclear plant must include criticality monitoring capabilities meeting the requirements of either § 70.24 of this chapter or paragraph (m)(2) of this section.
 - (2) In lieu of maintaining a monitoring system capable of detecting criticality as described in § 70.24 of this chapter, criticality accident requirements may be satisfied by—
 - (i) Demonstrating the sub-criticality of special nuclear material, except when it is inside the reactor and the reactor is being operated, by maintaining k-effective below 0.95 at a 95 percent probability, 95 percent confidence level, under conditions that maximize reactivity for the applicable storage and handling configurations, and
 - (ii) Providing radiation monitors for fuel storage and associated handling areas when fuel is present to detect excessive radiation levels and to support initiating appropriate safety actions.

- (3) While a spent fuel transportation package approved under 10 CFR part 71 of this chapter or spent fuel storage cask approved under 10 CFR part 72 is in the special nuclear material handling or storage area, the requirements in 10 CFR parts 71 or 72, as applicable, and the requirements of the certificate of compliance for that package or cask, are the applicable requirements for the fuel within that package or cask.

(n)

- (1) The design of each commercial nuclear plant must reflect state-of-the-art human factors principles for safe and reliable performance in all locations that human activities are expected for performing or supporting the continued availability of plant safety or emergency response functions.
- (2) The design must provide for the capabilities described in § 53.730(b) to ensure the plant staff are able to monitor plant conditions and respond to events.
- (3) The means by which the design and human actions together will achieve the safety requirements of subpart B of this part must be evaluated and used to inform the design and the development of the concept of operations required by § 53.730(c).
- (4) A functional requirements analysis and function allocation must be used to ensure that plant design features address how safety functions and functional safety criteria are satisfied, and how the safety functions will be assigned to appropriate combinations of human action, automation, active safety features, passive safety features, or inherent safety characteristics.

§ 53.450 Analysis requirements.

- (a) **Requirement to have a probabilistic risk assessment (PRA), or other systematic risk evaluations (SREs), or a combination thereof.** A PRA, other SREs, or a combination thereof for each commercial nuclear plant must be performed and used together with other generally accepted approaches for systematically evaluating engineered systems to identify potential failures, susceptibility to internal and external hazards, and other contributing factors to event sequences that might challenge the safety functions identified in § 53.230 and to support demonstrating that each commercial nuclear plant meets the safety criteria of § 53.220.
- (b) **Specific uses of analyses.** The PRA, other SREs, or a combination thereof, together with other generally accepted approaches for systematically evaluating engineered systems must be used to—
 - (1) Inform the selection of the LBEs, as described in § 53.240, which must be considered in the design to determine compliance with the safety criteria in subpart B of this part.
 - (2) Inform the classification of SSCs according to their safety significance in accordance with § 53.460 and to identify the environmental conditions under which the SSCs and operating staff must perform their safety functions.
 - (3) Evaluate the adequacy of defense-in-depth measures required in accordance with § 53.250.
 - (4) Identify and assess all plant operating states where there is the potential for the uncontrolled release of radioactive material to the environment.
 - (5) Identify and assess events that challenge plant control and safety systems whose failure could lead to the uncontrolled release of radioactive material to the environment. These include internal events, such as human errors and equipment failures, and external events identified in accordance with subpart D of this part.

- (6) Inform the establishment and updating of appropriate measures for plant operations, including availability controls, to ensure that the configurations and special treatments for SR SSCs and NSRSS SSCs provide the capabilities, availability, and reliability consistent with satisfying the safety criteria under § 53.220 and the analyses of licensing-basis events other than design-basis accidents (DBAs) under § 53.450(e).
- (c) **Maintenance and upgrade of analyses.** The PRA, other SREs, or a combination thereof must be maintained (e.g., updated to reflect plant changes such as modifications, procedure changes, or plant performance data) at least every 5 years until the permanent cessation of operations under § 53.1070 and upgraded (e.g., changed in scope or use of new methods) in conformance with generally accepted methods, standards, and practices that have been endorsed or otherwise found acceptable by the NRC.
- (d) **Qualification of analytical codes.** The analytical codes used in modeling the physical behavior of plant systems in the analyses of licensing-basis events (including but not limited to thermodynamics, reactor physics, fuel performance, and mechanistic source term codes) must be qualified for the range of conditions for which they are to be used.
- (e) **Analyses of licensing-basis events other than design-basis accidents.**
 - (1) Analyses must be performed for LBEs other than design-basis accidents (DBAs). These LBEs must be identified using insights from a PRA, other SREs, or a combination thereof with other generally accepted approaches for systematically evaluating engineered systems to identify and analyze equipment failures and human errors.
 - (2) The analysis of LBEs other than DBAs must include definitions of evaluation criteria for each event or specific categories of LBEs to determine the acceptability of the plant response to the challenges posed by internal and external hazards to provide an appropriate level of safety.
 - (3) The analyses of LBEs other than DBAs must address event sequences from initiation to a defined end state and be used in combination with other engineering analyses to demonstrate that the functional design criteria required by § 53.420 provide sufficient barriers to the unplanned release of radionuclides to satisfy the evaluation criteria defined for each LBE other than DBAs, to satisfy the safety criteria specified in accordance with § 53.220 and provide defense in depth as required by § 53.250.
 - (4) The methodology used to identify, categorize, and analyze LBEs must include a means to identify event sequences deemed significant for controlling the risks posed to public health and safety.
- (f) **Analysis of design-basis accidents.**
 - (1) The analysis of LBEs required by § 53.240 must include analysis of DBAs that address possible challenges to the safety functions identified under § 53.230. The events selected as DBAs must be those that, if not terminated, have the potential for exceeding the safety criteria in § 53.210.
 - (2) The DBAs selected must be analyzed using deterministic methods that address event sequences from initiation to a safe stable end state and assume only the SR SSCs identified under § 53.460 and human actions addressed by the requirements of subpart F of this part are available to perform the safety functions identified in accordance with § 53.230.
 - (3) The analysis must conservatively demonstrate compliance with the safety criteria in § 53.210.
- (g) **Other required analyses.** Analyses must be performed to assess—

- (1) **Fire protection.** Fire protection measures to demonstrate, through inclusion of fires in the analysis of LBEs or by separate analyses, that a fire or explosion in any plant area would not—
 - (i) Prevent equipment from fulfilling the safety functions identified in accordance with § 53.230; or
 - (ii) Challenge the safety criteria in §§ 53.210 and 53.220.
- (2) [Reserved]
- (3) **Dose to members of the public.** Measures taken under § 53.425, including estimating—
 - (i) The quantity of each of the principal radionuclides expected to be released annually to unrestricted areas in liquid effluents produced during normal reactor operations and the dose to the maximally exposed member of the public in unrestricted areas.
 - (ii) The quantities of each of the principal radionuclides of the gases, halides, and particulates expected to be released annually to unrestricted areas in gaseous effluents produced during normal reactor operations and the dose to the maximally exposed member of the public in unrestricted areas.
 - (iii) The annual external radiation dose in unrestricted areas and the maximally exposed member of the public in unrestricted areas due to direct radiation from contained radiation sources from the commercial nuclear plant during normal reactor operations.

§ 53.460 Safety categorization and special treatments.

- (a) Structures, systems, and components must be classified according to their safety significance. The SSC categories must include “Safety-Related,” “Non-Safety-Related but Safety-Significant,” and “Non-Safety-Significant,” as defined in subpart A of this part.
- (b) For SR and NSRSS SSCs, the conditions under which they must perform their safety function in § 53.230 must be identified. Special treatments must be established in accordance with this and other subparts to provide confidence that the SSCs will perform under the service conditions and with reliability consistent with the analysis performed under § 53.450 to demonstrate meeting the safety criteria in §§ 53.210 and 53.220.
 - (1) The special treatments for SR SSCs must include meeting the applicable quality assurance requirements from appendix B of part 50 of this chapter.
 - (2) The special treatments for NSRSS SSCs and special treatments for SR SSCs beyond those required under paragraph (b)(1) of this section may include meeting selected quality assurance requirements from appendix B of part 50 of this chapter when such treatment is needed to address performance requirements, equipment reliability, or uncertainties.
- (c) The identification of special treatments for SR and NSRSS SSCs must account for human actions needed to prevent or mitigate LBEs, the need to perform such actions reliably under the postulated environmental conditions, and the role of programs established in accordance with subpart F of this part to provide confidence that those actions will be performed as assumed in the analysis performed in accordance with § 53.450 to demonstrate meeting the applicable criteria in §§ 53.210, 53.220, and 53.450(e).

§ 53.470 [Reserved]

§ 53.480 Earthquake engineering.

- (a) **Effects of earthquakes.** Structures, systems, and components classified as SR or NSRSS must be able to withstand the effects of earthquakes, commensurate with the safety significance of the SSC, without loss of capability to perform their role in fulfilling the safety functions required by § 53.230.

- (b) **Definitions.** As used in this section—

Design-Basis Ground Motions (DBGMs) are the vibratory ground motions for which certain SSCs must be designed to remain functional.

Operating basis earthquake (OBE) ground motion is the vibratory ground motion for which those features of the commercial nuclear plant necessary for continued operation without undue risk to the health and safety of the public are designed to remain functional. The OBE ground motion is used in § 53.720.

Response spectrum is a plot of the maximum responses (acceleration, velocity, or displacement) of idealized single-degree-of-freedom oscillators as a function of the natural frequencies of the oscillators for a given damping value. The response spectrum is calculated for a specified vibratory motion input at the oscillators' supports.

Surface deformation is the distortion of geologic strata on or near the ground surface that occurs because of tectonic forces that result from earthquakes.

- (c) **Design considerations —**

(1) **Design-Basis Ground Motions.**

- (i) The DBGMs must be derived from the Site Ground Motion Response Spectra developed in accordance with § 53.510(c), by taking into consideration the functional design criteria of SSCs in accordance with §§ 53.410 and 53.420. The horizontal component of the DBGM(s) in the free-field at the foundation level of the structures must be an appropriate response spectrum that is determined based on the risk-significance of SSCs and their safety functions. In view of the limited data available on vibratory ground motion of strong earthquakes, it is acceptable that the design response spectra be smoothed spectra.
- (ii) The commercial nuclear plant must be designed so that, if the DBGMs occur, the following SSCs remain functional and within applicable stress, strain, and deformation limits:
- (A) Structures, systems, and components for which functional design criteria are established in accordance with § 53.410 or § 53.420; and
- (B) Structures, systems, and components classified as SR or NSRSS commensurate with safety significance in accordance with § 53.460.
- (iii) In addition to seismic loads, applicable concurrent normal operating, functional, and accident-induced loads must be taken into account in the design of the SR SSCs and, commensurate with safety significance, NSRSS SSCs.
- (iv) The design of the commercial nuclear plant must take into account the possible effects of seismic-induced ground disruption, such as fissuring, lateral spreads, differential settlement, liquefaction, and landsliding, on the facility foundations.
- (v) The SSCs fulfilling the safety functions required by § 53.230 must be demonstrated through design, testing, or qualification methods to be able to fulfill those safety functions during and after the vibratory ground motion associated with the DBGMs.

- (vi) The evaluation of SSCs required by this section to show they are able to function during and after earthquake ground motion should consider, if applicable, soil-structure interaction effects and the expected duration of vibratory motion. It is permissible to design for inelastic behavior in some of these SSCs during the DBGMs and under the postulated concurrent loads, provided the necessary safety functions are maintained.
 - (2) **OBE Ground Motion.** The OBE Ground Motion must be characterized by response spectra. The value of the OBE Ground Motion must be set to one-third or less of the DBGMs response spectra.
 - (3) [Reserved]
 - (4) **Required seismic instrumentation.** Suitable instrumentation must be provided so that the seismic response of commercial nuclear plant SR SSCs or NSRSS SSCs can be evaluated promptly after an earthquake.
- (d) **Surface deformation.**
- (1) The potential for surface deformation must be taken into account in the design of the commercial nuclear plant by providing reasonable assurance that in the event of deformation, SSCs classified as SR or NSRSS in accordance with § 53.460 will remain functional.
 - (2) In addition to surface deformation induced loads, the design of SSCs must take into account, commensurate with safety significance, seismic loads and applicable concurrent functional and accident-induced loads.
 - (3) The design provisions for surface deformation must be based on its postulated occurrence in any direction and azimuth and under any part of the commercial nuclear plant, unless evidence indicates this assumption is not appropriate, and must take into account the estimated rate at which the surface deformation may occur.
- (e) **Seismically induced floods and water waves and other design conditions.** Seismically induced floods and water waves from either locally or distantly generated seismic activity and other design conditions determined pursuant to subpart D of this part must be taken into account in the design of the commercial nuclear plant so as to prevent undue risk to the health and safety of the public.
- (f) **Analysis.** The analyses required by § 53.450 must address seismic hazards and related SSC responses in determining that the safety criteria defined in § 53.220 will be met.
- (g) **Design criteria, human actions, and programmatic controls.** Functional design criteria, human actions, and programmatic controls needed to address seismic events must be identified and implemented in accordance with this and other subparts to achieve and maintain the performance of SSCs relied upon to satisfy the safety criteria in § 53.220 and to maintain consistency with analyses required by § 53.450 when accounting for the site-specific frequencies and magnitudes of earthquakes for a commercial nuclear plant.

Subpart D—Siting Requirements

§ 53.500 General siting and siting assessment.

The purpose of this subpart and the specific requirements therein is to ensure that:

- (a) The siting of each commercial nuclear plant is supported by assessments of proposed sites such that the design, including design features and programmatic controls corresponding to the site characteristics, satisfies the safety criteria defined in §§ 53.210 and 53.220. The siting assessment addresses the site characteristics that might contribute to the initiation, progression, or consequences of licensing-basis events (LBEs) analyzed under §§ 53.450 and 53.480 that are identified and mitigated by design features or programmatic controls. The siting assessment takes into consideration the potential adverse impacts that a commercial nuclear plant may have on nearby populations as a result of normal operations or LBEs.
- (b) Activities performed to identify site characteristics or otherwise needed to determine site-specific contributors to functional design criteria or analysis assumptions under subpart C of this part satisfy the applicable special treatment requirements of § 53.460, including, where applicable, the quality assurance requirements from appendix B of part 50 of this chapter.

§ 53.510 External hazards.

- (a) **General external hazard requirements.** The design-basis external hazard level for the relevant external hazards for a site must be identified and characterized based on site-specific assessments of natural and constructed hazards with the potential to adversely affect plant functions. The external hazard frequencies and magnitudes determined from the site-specific assessments must take into account uncertainties and variabilities in data, models, and methods relied on to characterize the external hazards.
- (b) **Definitions.** As used in this section, the following terms mean:

Geological Siting Factors are geological and seismic factors that may affect the design and operation of the proposed commercial nuclear plant.

Ground Motion Response Spectra (GMRS) are the site-specific GMRS resulting from the geologic investigations and evaluations of the site vicinity and region and used to determine design-basis ground motions for structures, systems, and components under § 53.480.

Probabilistic Seismic Hazard Analysis is an analytical methodology that incorporates uncertainty into estimates of an annual frequency of exceedance for a certain ground motion parameter (e.g., peak ground acceleration, peak ground velocity, response spectral values) at a site.

- (c) **Geological investigations.** The GMRS for the site must be determined based on the results of investigations of the geological, seismological, and engineering characteristics of the site and its environs and must be characterized by both horizontal and vertical free-field GMRS at the free ground surface. The size of the region to be investigated and the type of data pertinent to the investigations must be determined based on the nature of the region surrounding the site. Data on vibratory ground motion, earthquake recurrence rates, fault geometry and slip rates, and site subsurface material properties must be obtained by reviewing pertinent literature and carrying out field investigations. Uncertainties are inherent in the parameters and models used to estimate the GMRS for the site. The site assessment must reflect these uncertainties through an appropriate analysis, such as a probabilistic seismic hazard analysis.
- (d) **Geologic and seismic siting factors.** The geologic and seismic siting factors considered for design under §§ 53.415 and 53.480 must include, but are not limited to, determination of the potential for surface tectonic and nontectonic deformations, the size and character of seismically induced floods and water waves that could affect a site from either locally or distantly generated seismic activity, soil and rock stability, liquefaction potential, and natural and artificial slope stability.

§ 53.520 Site characteristics.

Site characteristics that might contribute to the initiation, progression, or consequences of LBEs analyzed under § 53.450 must be identified, assessed, and considered in the design and analyses required by subpart C of this part.

§ 53.530 Population-related considerations.

Every site must have an exclusion area, a low-population zone, and a population center distance as defined in § 53.020.

- (a) The offsite radiological consequences estimated by the analyses required by § 53.450(f) must be used to confirm that—
 - (1) An individual located at any point on the boundary of the exclusion area for any 2-hour period following onset of the postulated fission product release would not receive a radiation dose in excess of 25 rem (250 millisieverts) total effective dose equivalent.
 - (2) An individual located at any point on the outer boundary of the low-population zone who is exposed to the radioactive cloud resulting from the postulated fission product release (during the entire period of its passage) would not receive a radiation dose in excess of 25 rem (250 millisieverts) total effective dose equivalent.
- (b) The reactor site must either:
 - (1) Provide a population center distance of at least one and one-third times the distance from the reactor to the outer boundary of the low-population zone; or
 - (2) Be found acceptable to the U.S. Nuclear Regulatory Commission (NRC) based on assessments of societal risks in comparison to societal benefits for the specific site. The boundary of the population center or the alternate area assessed considering societal risks and benefits must be determined upon consideration of population distribution. Political boundaries are not controlling in the calculation of population center distance or the alternate area assessed considering societal risks and benefits.
- (c) Reactor sites should be located away from very densely populated centers or otherwise be shown to be acceptable by assessments of societal risks in comparison to societal benefits for the specific site. Areas of low-population density are, generally, preferred. However, in determining the acceptability of a particular site located away from a very densely populated center but not in an area of low-population density or when assessing a site considering societal risks and benefits, consideration will be given to safety, environmental, economic, or other factors, which may result in the site being found acceptable.

§ 53.540 Siting interfaces.

Site characteristics must be addressed by the design features, programmatic controls, and supporting analyses used to demonstrate that the safety criteria in §§ 53.210 and 53.220 are met for each commercial nuclear plant. Site characteristics must be such that adequate emergency plans and security plans can be developed and maintained.

Subpart E—Construction and Manufacturing Requirements

§ 53.600 Construction and manufacturing—scope and purpose.

This subpart applies to those construction and manufacturing activities authorized by a construction permit (CP), combined license (COL), manufacturing license (ML), or limited work authorization (LWA) issued under this part.

§ 53.605 Reporting of defects and noncompliance.

Each CP and ML issued under this part is subject to the terms and conditions in this section, and each COL issued under this part is subject to the terms and conditions in this section until the date that the Commission makes the finding under § 53.1452(g).

- (a) **Definitions.** The definitions in § 21.3 of this chapter apply to this section.
- (b) **Posting requirements.**
 - (1) Each individual, partnership, corporation, dedicating entity, or other entity subject to the regulations in this section must post current copies of this section and the regulations in 10 CFR part 21; section 206 of the Energy Reorganization Act of 1974, as amended; and procedures adopted under these regulations. These documents must be posted in a conspicuous position on any premises within the United States where the activities subject to the license are conducted.
 - (2) If posting of these regulations or the procedures adopted under them is not practical, the licensee may, in addition to posting section 206 of the Energy Reorganization Act of 1974, as amended, post a notice that describes the regulations/procedures, including the name of the individual to whom reports may be made, and states where they may be examined.
- (c) **Procedures.** The holder of a CP, COL, or ML subject to this section must adopt appropriate procedures to—
 - (1) Evaluate deviations and failures to comply to identify defects and failures to comply associated with substantial safety hazards as soon as practicable, and, except as provided in paragraph (c)(2) of this section, in all cases within 60 days of discovery, to identify a reportable defect or failure to comply that could create a substantial safety hazard, were it to remain uncorrected.
 - (2) Ensure that if an evaluation of an identified deviation or failure to comply potentially associated with a substantial safety hazard cannot be completed within 60 days from the discovery of the deviation or failure to comply, an interim report is prepared and submitted to the Commission through a director or responsible officer, or designated person as discussed in paragraph (d)(5) of this section. The interim report should describe the deviation or failure to comply that is being evaluated and should also state when the evaluation will be completed. This interim report must be submitted in writing within 60 days of discovery of the deviation or failure to comply.
 - (3) Ensure that a director or responsible officer of the holder of a CP, COL, or ML subject to this section is informed as soon as practicable, and, in all cases, within the 5 working days after completion of the evaluation described in paragraph (c)(1) or (c)(2) of this section, if the construction or manufacture of a facility or activity, or a basic component supplied for such a facility or activity—
 - (i) Fails to comply with the Atomic Energy Act of 1954, as amended, or any applicable regulation, order, or license of the Commission relating to a substantial safety hazard;
 - (ii) Contains a defect; or

- (iii) Underwent any significant breakdown in any portion of the quality assurance program (QAP) conducted under the requirements of appendix B to part 50 of this chapter that could have produced a defect in a basic component. These breakdowns in the QAP are reportable whether or not the breakdown actually resulted in a defect in a design approved and released for construction, installation, or manufacture.

(d) ***Reporting defects and noncompliance.***

- (1) The holder of a CP, COL, or ML subject to this section that obtains information reasonably indicating that the facility or manufactured reactors fails to comply with the Atomic Energy Act of 1954, as amended, or any applicable regulation, order, or license of the Commission relating to a substantial safety hazard must notify the Commission of the failure to comply through a director, responsible officer, or designated person as discussed in paragraph (d)(5) of this section.
- (2) The holder of a CP, COL, or ML subject to this section that obtains information reasonably indicating the existence of any defect found in the construction or manufacture, or any defect found in the final design of a facility as approved and released for construction or manufacture, must notify the Commission of the defect through a director, responsible officer, or designated person as discussed in paragraph (d)(5) of this section.
- (3) The holder of a CP, COL, or ML subject to this part, who obtains information reasonably indicating that the QAP has undergone any significant breakdown discussed in paragraph (c)(3)(iii) of this section must notify the Commission of the breakdown in the QAP through a director, responsible officer, or designated person as discussed in paragraph (d)(5) of this section.
- (4) When acting as a dedicating entity, the holder of a CP, COL, or ML subject to this section is responsible for identifying and evaluating deviations; reporting defects and failures to comply associated with substantial safety hazards for dedicated items; and maintaining auditable records for the dedication process.
- (5) The notification requirements of this paragraph (d) apply to all defects and failures to comply associated with a substantial safety hazard regardless of whether extensive evaluation, redesign, or repair is required to conform to the criteria and bases stated in the Safety Analysis Report, CP, COL, or ML. Evaluation of potential defects and failures to comply and reporting of defects and failures to comply under this section satisfies the CP holder's, COL holder's, and ML holder's evaluation and notification obligations under 10 CFR part 21, and satisfies the responsibility of individual directors or responsible officers or holders of a CP, COL, or ML subject to this section to report defects, and failures to comply associated with substantial safety hazards under section 206 of the Energy Reorganization Act of 1974, as amended. The director or responsible officer may authorize an individual to provide the notification required by this section. However, this does not relieve the director or responsible officer of his or her responsibility under this section.

(e) ***Notification—timing and where sent.*** The notification required by paragraph (d) of this section must consist of—

- (1) Initial notification by telephone, facsimile, or email identified in appendix A to 10 CFR part 73 to the U.S. Nuclear Regulatory Commission (NRC) Operations Center within 2 days following receipt of information by the director or responsible corporate officer under paragraph (c)(3) of this section, on the identification of a defect or a failure to comply. If the CP, COL, or ML holder elects to use facsimile, verification that the facsimile has been received should be made by calling the NRC Operations Center. This paragraph (e)(1) does not apply to interim reports described in paragraph (c)(2) of this section.

- (2) Written notification submitted to the NRC Document Control Desk by an appropriate method listed in § 53.040, with a copy to the appropriate NRC Regional Administrator at the address specified in appendix D to 10 CFR part 20 and a copy to the appropriate NRC resident inspector, if applicable, within 30 days following receipt of information by the director or responsible corporate officer under paragraph (c)(3) of this section, on the identification of a defect or failure to comply.
- (f) **Content of notification.** The written notification required by paragraph (e)(2) of this section must clearly indicate that the written notification is being submitted under this section and include the following information, to the extent known.
 - (1) Name and address of the individual or individuals informing the Commission.
 - (2) Identification of the facility, the activity, or the basic component supplied for the facility or the activity within the United States which contains a defect or fails to comply.
 - (3) Identification of the firm constructing or manufacturing the facility or supplying the basic component which fails to comply or contains a defect.
 - (4) Nature of the defect or failure to comply and the safety hazard which is created or could be created by the defect or failure to comply.
 - (5) The date on which the information of a defect or failure to comply was obtained.
 - (6) In the case of a basic component that contains a defect or failure to comply, the number and location of these components in use at the facility subject to the regulations in this part.
 - (7) In the case of a completed reactor manufactured under this part, the entities to which the reactor was supplied.
 - (8) The corrective action which has been, is being, or will be taken; the name of the individual or organization responsible for the action; and the length of time that has been or will be taken to complete the action.
 - (9) Any advice related to the defect or failure to comply about the facility, activity, or basic component that has been, is being, or will be given to other entities.
- (g) **Procurement documents.** Each holder of a CP, COL, or ML subject to this section must ensure that each procurement document for a facility or a basic component specifies the provisions of 10 CFR part 21 or this section that apply, as applicable.
- (h) **Coordination with 10 CFR part 21.** The requirements of this section are satisfied when the defect or failure to comply associated with a substantial safety hazard has been previously reported under 10 CFR part 21, under § 73.1205 of this chapter, under this section, or under § 53.1640.
- (i) **Records retention.** The holder of a CP, COL, or ML subject to this section must prepare and maintain records necessary to accomplish the purposes of this section, specifically—
 - (1) Retain procurement documents, which define the requirements that facilities or basic components must satisfy in order to be considered acceptable, for the lifetime of the facility or basic component.
 - (2) Retain records of evaluations of all deviations and failures to comply under paragraph (c)(1) of this section for the longest of—
 - (i) Ten years from the date of the evaluation;
 - (ii) Five years from the date that an early site permit is referenced in an application for a COL; or

- (iii) Five years from the date of delivery of a manufactured reactor.
- (3) Retain records of all interim reports to the Commission made under paragraph (c)(2) of this section, or notifications to the Commission made under paragraph (d) of this section for the minimum time periods stated in paragraph (i)(2) of this section;
- (4) Suppliers of basic components must retain records of—
 - (i) All notifications sent to affected licensees or purchasers under paragraph (d)(4) of this section for a minimum of 10 years following the date of the notification;
 - (ii) The facilities or other purchasers to whom the basic components or associated services were supplied for a minimum of 15 years from the delivery of the basic component or associated services.
- (5) Maintaining reports in accordance with this section satisfies the recordkeeping obligations under 10 CFR part 21 of the entities, including directors or responsible officers thereof, subject to this section.

§ 53.610 Construction.

- (a) **Management and control.** Licensees must ensure that the following plans, programs, and organizational units are developed and implemented to manage and control the construction activities:
 - (1) Programs to ensure that the construction of a commercial nuclear plant supports the eventual compliance with the design and analysis requirements in subpart C of this part.
 - (2) An organization, headed by qualified personnel, responsible for managing, controlling, and evaluating the adequacy of the construction activities.
 - (3) Procedures describing the qualifications for personnel in key positions in the licensee's management and control organization and the organizational responsibilities, authority, and interfaces with other parts of the licensee's organization.
 - (4) Procedures to evaluate the applicability of other national and international construction experience to the planned and ongoing construction activities and to ensure the applicable experience will be provided to those constructing the plant.
 - (5) A fitness-for-duty program, under 10 CFR part 26.
 - (6)
 - (i) A QAP meeting the requirements of appendix B of part 50 of this chapter as required by § 53.460(b).
 - (ii) Appropriate programmatic controls to provide special treatment for non-safety-related but safety-significant structures, systems, and components (SSCs).
 - (7) A radiation protection program, in accordance with 10 CFR part 20, that includes measures for monitoring the dose to individuals working with radioactive materials brought onto the site, as applicable.
 - (8) An information security program in accordance with §§ 73.21, 73.22, and 73.23 of this chapter, as applicable.

(b) **Construction activities.** No person may begin the construction of a commercial nuclear plant on a site on which the facility is to be operated under this part until that person has been issued either a CP or COL, an early site permit authorizing activities under § 53.1130, or an LWA under this part.

(1) Licensees must satisfy the following requirements:

- (i) As appropriate, considering the types and quantities of radioactive materials being brought onto the site—
 - (A) The licensee must maintain and follow a special nuclear material (SNM) material control and accounting program, a measurement control program, and other material control procedures that include corresponding record management requirements as required by the provisions of § 70.32 of this chapter. Prior to initial receipt of SNM onsite, the licensee must implement an SNM material control and accounting program in accordance with 10 CFR part 74.
 - (B) Procedures must be in place to receive, possess, use, and store source, byproduct, and SNM in accordance with applicable portions of 10 CFR parts 30, 40, and 70.
 - (C) A plant staff training program associated with the receipt of radioactive material must be approved and implemented prior to initial receipt of byproduct, source or SNM (excluding exempt quantities as described in § 30.18 of this chapter).
- (ii) For construction of a commercial nuclear plant involving multiple reactor units, plans and procedures must be in place to prevent or mitigate potential hazards to the SSCs of operating units resulting from construction activities, including the managerial and administrative controls to be used to provide assurance that the limiting conditions for operation of the operating units are not exceeded as a result of construction activities.
- (iii) Procedures must be in place prior to the start of construction activities that describe how construction will be controlled so as not to impact other features important to the design, such as dewatering, slope stability, backfill, compaction, and seepage.
- (iv) For LWA holders, a plan must be developed for redress of activities performed under the LWA should one of the following situations arise:
 - (A) LWA work activities are terminated by the holder of the LWA;
 - (B) The LWA is revoked by the NRC; or
 - (C) The Commission denies the associated CP or COL application.

(2)

- (i) Onsite fresh fuel must be protected and stored in compliance with § 73.67 of this chapter.
- (ii) Before initial fuel load into the reactor (or, for a fueled manufactured reactor, before initiating the removal of the features to prevent criticality required under § 53.620(d)(1)), a cybersecurity program that meets the requirements of § 73.54 or § 73.110 of this chapter, a physical security program that meets the requirements of § 73.55 or § 73.100 of this chapter, and an access authorization program that meets the requirements of § 73.56 or § 73.120 of this chapter must be established, as applicable.

- (iii) Fire protection measures must be implemented for work and storage areas (including adjacent fire areas that could affect the work or storage area) before initial receipt of byproduct, source, or non-fuel SNM (excluding exempt quantities as described in § 30.18 of this chapter). The fire protection measures for areas associated with new fuel (including all fuel handling, fuel storage, and adjacent fire areas that could affect the new fuel) must be implemented before receipt of fuel. Prior to the receipt of fuel, a formal letter of agreement must be in place with the local fire department specifying the nature of arrangements in support of the fire protection program.

(c) **Inspection and acceptance.**

- (1) The licensee must have a process for accepting individual or groups of SSCs upon completion of construction and protecting them from damage or tampering as other construction activities continue.
- (2) The post-construction acceptance process must address the inspections, tests, analyses, and acceptance criteria specified in the COL under § 53.1440 or the equivalent verifications needed to support the issuance of an operating license under § 53.1387.

§ 53.620 Manufacturing.

- (a) **Management and control.** Holders of MLs must ensure that the following plans, programs, and organizational units are developed and implemented to manage and control the manufacturing activities within the scope of the ML:
 - (1) Programs to ensure that the manufacturing of a manufactured reactor or portions of a manufactured reactor complies with the design and analysis requirements in subpart C of this part. The entity with design authority for the manufactured reactor covered by the ML must be identified in the license.
 - (2) An organizational and management structure responsible for managing, controlling, and evaluating the adequacy of the reactor design and manufacturing activities.
 - (3) Procedures describing the qualifications for personnel in key positions in the licensee's management and control organization and the organizational responsibilities, authority, and interfaces with other parts of the licensee's organization.
 - (4) A program to evaluate the applicability of other national and international design and manufacturing experience to the planned and ongoing manufacturing activities.
 - (5) A fitness-for-duty program, in accordance with 10 CFR part 26.
 - (6)
 - (i) A QAP meeting the requirements of appendix B to part 50 of this chapter, to be applied to the design, fabrication, construction, and testing of the SSCs of the manufactured reactor.
 - (ii) Appropriate programmatic controls to provide special treatment measures for non-safety-related but safety-significant SSCs.
 - (7) A radiation protection program, in accordance with 10 CFR part 20, that includes measures for monitoring the dose to individuals if the manufacturing activities include working with radioactive materials.
 - (8) An information security program in accordance with §§ 73.21, 73.22 and 73.23 of this chapter, as applicable.

(b) **Manufacturing activities.** Holders of MLs must satisfy the following requirements:

- (1) The manufacturing process must be conducted within facilities for which the ML holder has the authority to establish controls on any activity that might affect manufacturing. The licensee must establish access controls to the portions of each facility involved in the manufacturing processes governed by the ML.
- (2) Manufacturing processes must be performed in accordance with the ML and the referenced codes and standards that have been endorsed or otherwise found acceptable by the NRC.
- (3) A post-manufacturing inspection and acceptance process must be established and implemented before transporting a manufactured reactor or portions of a manufactured reactor for installation at a commercial nuclear plant. The process must consider the results of inspections, tests, and analyses that have been performed and the acceptance criteria that are necessary and sufficient to conclude that manufacturing activities have been completed in accordance with the ML.

(c) **Control of radioactive materials.** As appropriate considering the types and quantities of radioactive materials being brought into the manufacturing facility—

- (1) Procedures must be in place to receive, transfer, possess, and use source, byproduct, and SNM in accordance with the applicable portions of 10 CFR parts 30, 40 and 70.
- (2) A fire protection program must be established and implemented before the initial receipt of byproduct, source, or non-fuel SNM (excluding exempt quantities as described in § 30.18 of this chapter).
- (3) An emergency plan appropriate for responding to the facility-specific hazards of an accidental release of radioactive material and to limit the health effects of the associated chemical hazards of licensed material must be approved and implemented prior to the receipt of byproduct, source, or SNM (excluding exempt quantities as described in § 30.18 of this chapter).
- (4) A plant staff training program associated with the receipt of radioactive material must be approved and implemented before initial receipt of byproduct, source, or SNM (excluding exempt quantities as described in § 30.18 of this chapter).
- (5) Security requirements must be implemented for the protection of SNM based on the type, enrichment, and quantity in accordance with 10 CFR part 73, as applicable, and for the protection of Category 1 and Category 2 quantities of radioactive material in accordance with 10 CFR part 37, as applicable.

(d) **Fuel loading.**

- (1)
 - (i) An ML may authorize possession of a manufactured reactor into which the licensee has loaded fresh (unirradiated) fuel pursuant to a license issued under part 70 of this chapter only if the manufactured reactor is configured during its loading, storage, and transport with features to prevent criticality that are specified in the ML.
 - (ii) The ML applicant may file a separate, subsequent application for the 10 CFR part 70 license or combine the application for the 10 CFR part 70 license with the application for an ML.
 - (iii) The Commission has determined that any such fueled manufactured reactor in which the features to prevent criticality are in place is not in operation.

- (iv) Upon installation of the fueled manufactured reactor in its place of operation and a Commission finding that the acceptance criteria in the COL that authorized reactor construction are met under § 53.1452(g), or that any conditions in the CP that authorized reactor construction are met and the associated operating license (OL) issued, the features to prevent criticality may be removed. Upon initiating the removal of the features to prevent criticality, the fueled manufactured reactor has commenced operation.
- (2) Holders of part 70 licenses authorizing the possession and loading of fresh fuel into manufactured reactors must comply with the requirements of part 70 for the facilities and activities related to the storage, movement, and loading of fresh fuel in the manufactured reactor. Holders of these part 70 licenses must comply with the requirements of Subpart H to part 70, regardless of whether their proposed activities meet the applicability criteria found in 10 CFR 70.60. Procedures, equipment, and personnel required by the 10 CFR part 70 license, must be in place before the receipt of SNM at the manufacturing facility.
 - (i) Before the receipt of SNM, the licensee must have security programs in place that meet the performance objectives of 10 CFR 73.67, with the following additions and exceptions:
 - (A) A physical security plan describing the physical security program must be maintained and a cybersecurity program must be established for the possession and loading of fresh fuel into a manufactured reactor authorized by a 10 CFR part 70 license, regardless of fuel type, enrichment, and quantity.
 - (B) The physical security program must be designed to prevent unintended and uncontrolled criticality events.
 - (C) The cybersecurity program must provide reasonable assurance that a cyberattack does not adversely impact the functions performed by digital assets necessary for implementing the physical security requirements of this section, or the radiation monitoring and criticality requirements in this section or in 10 CFR part 70.
 - (D) All holders of a part 70 license that authorizes loading of fresh fuel into a manufactured reactor must perform the screening required in § 73.67(d)(4) of this chapter to confirm the identity, trustworthiness, and reliability of individuals prior to granting unescorted access to special nuclear material; these determinations must be documented.
 - (ii) [Reserved]
- (3) The loading or unloading of fresh fuel into or from a manufactured reactor and any changes to the configuration of reactivity control and prevention systems for the fueled manufactured reactor must be performed by a certified fuel handler meeting the requirements in subpart F of this part.
- (e) **Transportation.**
 - (1) A holder of an ML may not transport or allow to be removed from the places of manufacture the manufactured reactor or portions thereof as defined in the ML except for either transport to a site for which the Commission has issued a COL or CP that references the subject ML or export in accordance with 10 CFR part 110.
 - (2) A holder of an ML must include in any contract governing the transport of a manufactured reactor or portions thereof as defined in the ML from the places of manufacture to any other location, a provision requiring that the person transporting the manufactured reactor comply with all shipping requirements in applicable NRC regulations, certificates of compliance, and NRC-issued licenses.

- (3) Procedures governing the preparation of the manufactured reactor or portions thereof as defined in the ML for transport and the conduct of the transport must be issued prior to transport. The procedures must implement the protective measures and restrictions described in NRC regulations and NRC-issued licenses to protect the reactor from potential conditions that would adversely affect the safe operation of a commercial nuclear plant.
- (4) For a manufactured reactor that is to be loaded with fresh fuel before transport to the place of operation, the ML must specify that transportation will be in accordance with parts 71 and 73 of this chapter.
- (f) *Acceptance and installation at the site for which the Commission has issued a COL or CP that references the subject ML.*
 - (1) Installation at the site for which the Commission has issued a COL or CP that references the subject ML must follow the regulations in § 53.610.
 - (2) Upon arrival at the site, the manufactured reactor or portions of a manufactured reactor may not be installed in its place of operation unless the COL or CP holder performs inspections sufficient to verify the reactor is in compliance with the ML and has not been damaged in transit. The COL or CP holder must perform these inspections in accordance with documented procedures subject to quality assurance measures commensurate with their importance to safety. In addition, inspections must confirm that the interface requirements between the manufactured reactor or portions of a manufactured reactor and the remaining portions of the commercial nuclear plant are met.

Subpart F—Requirements for Operation

§ 53.700 Operational objectives.

The purpose of this subpart and the specific requirements herein is to ensure that:

- (a) Each holder of an operating license (OL) or combined license (COL) under this part develops, implements, and maintains controls for plant structures, systems, and components (SSCs), responsibilities of personnel, and plant programs during the operating life of each commercial nuclear plant such that the requirements defined in subpart B are satisfied. More specifically:
 - (1) Under § 53.710 through § 53.730, each holder of an OL or COL under this part must maintain the capabilities, availability, and reliability of plant SSCs to ensure that the safety functions identified in § 53.230 will be performed if called upon during licensing-basis events (LBEs).
 - (2) Under § 53.725 through § 53.830, each holder of an OL or COL under this part must ensure that personnel have adequate knowledge and skills to perform their assigned duties that support the performance of the safety functions identified in § 53.230.
 - (3) Under § 53.845 through § 53.910, each holder of an OL or COL under this part must implement plant programs sufficient to ensure that the safety functions identified in § 53.230 will be performed if called upon during normal operations and LBEs.
- (b) [Reserved]

§ 53.710 Maintaining capabilities and availability of structures, systems, and components.

Measures must be provided for each commercial nuclear plant licensed under this part such that the capabilities, availability, and reliability of plant SSCs, when combined with corresponding programmatic controls and human actions, provide that the safety criteria defined in §§ 53.210 and 53.220 will be met.

- (a) Technical specifications must be developed, implemented, and maintained that define conditions or limitations on plant operations that are necessary to ensure that safety-related (SR) SSCs can fulfill the safety functions identified under § 53.230 and support meeting the safety criteria of § 53.210. The technical specifications must describe the following requirements:
 - (1) Limits on the inventory of radioactive materials within the reactor system and supporting systems with the potential, individually or collectively, to cause a release exceeding the safety criteria in § 53.210 as a result of a design-basis accident analyzed in accordance with § 53.450(f).
 - (2) Operating limits for the facility that if exceeded could lead to a failure to perform a required safety function necessary to demonstrate compliance with the safety criteria in § 53.210.
 - (3) For each SSC classified as SR in accordance with § 53.460, technical specifications must define—
 - (i) **Limiting conditions for operation.** Limiting conditions for operation are the lowest functional capability or performance levels of SR SSCs required to ensure that the design-basis accidents analyzed in accordance with § 53.450(f) satisfy the safety criteria of § 53.210. When a limiting condition for operation is not met, the licensee must shut down the plant or follow any remedial action permitted by the technical specifications until the condition can be met.
 - (ii) **Surveillance requirements.** Surveillance requirements are requirements relating to test, calibration, or inspection to assure that the necessary quality of systems and components is maintained and that the limiting conditions for operation will be met.
 - (4) Design elements to be included are those elements of the plant such as materials of construction and geometric arrangements, which, if altered or modified, would have a significant effect on safety and are not covered in categories described in paragraphs (a)(1) through (3) of this section.
 - (5) Administrative controls are the provisions relating to organization and management, procedures, recordkeeping, review and audit, and reporting necessary to assure operation of the plant in a safe manner. Each licensee must submit any reports to the Commission pursuant to approved technical specifications under § 53.040.
- (b) Control measures on plant operations, including availability controls, must be developed and implemented to ensure that the configurations and special treatments for SR SSCs and non-safety-related but safety-significant (NSRSS) SSCs provide the capabilities, availability, and reliability required to demonstrate compliance with the criteria of §§ 53.220 and 53.450(e).^[1] The control measures must—
 - (1)
 - (i) Identify who within the licensee's organization has authority to make configuration changes;

^[1] The comprehensive risk metrics and related risk performance objectives established under § 53.220 involve assessing and averaging the risks over a defined period (e.g., plant year) and do not constitute a real-time requirement that must be continuously demonstrated by the licensee.

- (ii) Establish processes to make configuration changes to NSRSS SSCs; and
 - (iii) Establish processes to ensure that all organizations of the commercial nuclear plant affected by the configuration changes are formally notified and approve of the change.
- (2) Describe how the special treatments for each NSRSS SSC and special treatments for SR SSCs beyond those under paragraph (a) of this section will be established and maintained over the operating life of the commercial nuclear plant.

§ 53.715 Maintenance, repair, and inspection programs.

- (a) A program to control maintenance activities and monitor the performance or condition of SR and NSRSS SSCs must be developed, implemented, and maintained.
- (b) Whenever a licensee determines through activities related to maintenance, repair, and inspection of SSCs, the activities under § 53.710, or otherwise that the performance or condition of an SR or NSRSS SSC does not demonstrate compliance with established special treatments or performance goals related to capabilities, availability, or reliability, the licensee must take appropriate corrective action.
- (c) Performance and condition monitoring activities and associated goals and preventive maintenance activities must be evaluated at least every 24 months. The evaluations must take into account, where practical, industry-wide operating experience. Adjustments must be made where necessary to ensure that the objective of preventing failures of SSCs through maintenance is appropriately balanced against the objective of minimizing unavailability of SSCs due to monitoring or preventive maintenance.
- (d) Before performing maintenance activities (including but not limited to surveillance, post-maintenance testing, and corrective and preventive maintenance), the licensee must assess and manage the increase in risk that may result from the proposed maintenance activities.

§ 53.720 Response to seismic events.

If vibratory ground motion exceeding that of the operating basis earthquake Ground Motion or significant plant damage due to vibratory ground motion occurs, the licensee must shut down the commercial nuclear plant. If structures, systems, or components necessary for the safe shutdown of the commercial nuclear plant are not available after the occurrence of this vibratory ground motion, the licensee must consult with the Commission and must propose a plan for the timely, safe shutdown of the commercial nuclear plant. Prior to resuming operations, the licensee must demonstrate to the Commission that those features necessary for continued operation without undue risk to the health and safety of the public or necessary to maintain the licensing basis of the commercial nuclear plant were either not functionally damaged or have been repaired.

§ 53.725 General staffing, training, personnel qualifications, and human factors requirements.

- (a) *Two classes of commercial nuclear plants.* Commercial nuclear plants licensed under this part are either of the class of self-reliant-mitigation facilities or of interaction-dependent-mitigation facilities, based upon the similarity of operating and technical characteristics of the plants in the class. A commercial nuclear plant is a self-reliant-mitigation facility if the U.S. Nuclear Regulatory Commission (NRC) determined as part of its approval of the OL or COL for that plant that its design demonstrates compliance with the criteria of § 53.800(a)(1) through (a)(5). Otherwise, the commercial nuclear plant is an interaction-dependent-mitigation facility.

- (b) **Purpose and applicability.** The regulations in §§ 53.725 through 53.830 address areas related to staffing, training, personnel qualifications, and human factors engineering for applicants for or holders of OLs or COLs under this part. These regulations are organized as follows:
- (1) Sections 53.725 through 53.745 address general requirements for staffing, training, personnel qualifications, and human factors engineering. The regulations within these sections are applicable to all applicants for or holders of OLs or COLs under this part, except where specifically stated otherwise.
 - (2) Sections 53.760 through 53.795 address operator and senior operator licensing requirements. The regulations within these sections are applicable to those applicants for or holders of OLs or COLs under this part for interaction-dependent-mitigation facilities that have not yet certified the permanent cessation of operations and permanent removal of fuel from the reactor vessel as described under § 53.1070.
 - (3) Sections 53.800 through 53.820 address generally licensed reactor operator requirements. The regulations within these sections are in lieu of §§ 53.760 through 53.795 for those applicants for or holders of OLs or COLs under this part for self-reliant-mitigation facilities that have not yet certified the permanent cessation of operations and permanent removal of fuel from the reactor vessel as described under § 53.1070.
 - (4) Section 53.830 provides general personnel training requirements. The regulations within this section are applicable to all applicants for or holders of OLs or COLs under this part.
- (c) **Definitions.** When used in §§ 53.725 through 53.830, *applicant* refers to an applicant for an operator or senior operator license; *licensee* refers to the holder of an operator, senior operator, or generally licensed reactor operator license; and *facility licensee* refers to the licensee for the commercial nuclear plant where the applicant would be licensed or the licensee is licensed. As also used in §§ 53.725 through 53.830—
- Automation** means a device or system that accomplishes (partially or fully) a function or task.
- Auxiliary operator** means any individual who operates components of a commercial nuclear plant but does not manipulate controls or direct the manipulation of controls of the plant and is not required to be licensed under the provisions of this part.
- Controls** when used with respect to a nuclear reactor means apparatus and mechanisms, the manipulation of which directly affects the reactivity or power level of the reactor.
- Generally licensed reactor operator** means any individual licensed under the provisions of § 53.810 to manipulate controls of a self-reliant-mitigation facility and to direct the licensed activities of generally licensed reactor operators.
- Interaction-dependent-mitigation facility** means a commercial nuclear plant design other than one that demonstrates compliance with the operating and technical characteristics defined under § 53.800.
- Load following** means a commercial nuclear plant automatically changing its output to match expected demand in response to externally originated instructions or signals.
- Operator** means any individual licensed under the provisions of §§ 53.760 through 53.795 to manipulate controls of an interaction-dependent-mitigation facility.
- Performance testing** means testing conducted to verify a simulation facility's performance as compared to actual or predicted reference plant performance.

Reference plant means the specific commercial nuclear plant, or plant design for facilities which are not yet constructed, on which a simulation facility's configuration, system control arrangement, and design data are based.

Self-reliant-mitigation facility means a commercial nuclear plant design that demonstrates compliance with the operating and technical characteristics defined under § 53.800.

Senior operator means any individual licensed under the provisions of §§ 53.760 through 53.795 to manipulate controls of an interaction-dependent-mitigation facility and to direct the licensed activities of operators.

Simulation facility means an interface designed to provide a realistic imitation of the operation of a commercial nuclear plant used for the administration of examinations, for training, and/or to demonstrate compliance with experience requirements for applicants or licensees. A simulation facility may rely, in whole or part, upon the physical utilization of the reference plant itself.

Systems approach to training means a training program that includes the following five elements:

- (i) Systematic analysis of the jobs to be performed.
- (ii) Learning objectives derived from the analysis which describe desired performance after training.
- (iii) Training design and implementation based on the learning objectives.
- (iv) Evaluation of trainee mastery of the objectives during training.
- (v) Evaluation and revision of the training based on the performance of trained personnel in the job setting.

§ 53.726 Communications.

- (a) An applicant or licensee or facility licensee must submit any communication or report required by the regulations contained within §§ 53.725 through 53.830 and must submit any application filed under these regulations to the Commission.
- (b) Each facility licensee that is required to comply with the requirements of §§ 53.760 through 53.795 (i.e., interaction-dependent-mitigation facilities) must notify the appropriate NRC contact within 30 days of the following in regard to a licensed operator or senior operator:
 - (1) Permanent reassignment from the position for which the facility licensee has certified the need for a licensed operator or senior operator under § 53.775(a)(1);
 - (2) Termination of any operator or senior operator; or
 - (3) Permanent disability or illness as required under § 53.770.

§ 53.728 Completeness and accuracy of information.

Information provided to the Commission by an applicant for an operator or senior operator license or by a licensee or information required by statute or by the Commission's regulations, orders, or license conditions to be maintained by the applicant or the licensee must be complete and accurate in all material respects.

§ 53.730 Defining, fulfilling, and maintaining the role of personnel in ensuring safe operations.

Each applicant for or holder of an OL or COL for a commercial nuclear plant under this part must comply with the following:

- (a) **Human factors engineering design requirements.** The plant design must reflect state-of-the-art human factors engineering principles for safe and reliable performance in all locations that human activities are expected for performing or supporting the continued availability of plant safety or emergency response functions.
- (b) **Human system interface design requirements.** The plant design must provide for the following to support operating personnel in monitoring plant conditions and responding to plant events:
 - (1) Features for displaying to operating personnel a minimum set of parameters that define the safety status of the plant and are capable of displaying both the full range of important plant parameters and data trends on demand, as well as indicating when process limits are being approached or exceeded;
 - (2) Automatic indication of the bypassed and operable status of safety systems;
 - (3) Direct indication of SSC status that relates to the ability of the SSC to perform its safety function, such as relief and safety valve position (*i.e.*, open or closed) for barriers important to fulfilling safety functions with such devices, and ultimate heat sink and cooling system status and availability;
 - (4) Instrumentation to measure, record, and display key plant parameters related to the performance of SSCs and the integrity of barriers important to fulfilling safety functions to support operators in monitoring plant conditions and responding to plant events. Examples include temperatures and pressures within important systems or structures, core or fuel system conditions (including possible damage states), temperatures and levels associated with cooling functions, combustible gas concentrations, radiation levels in systems and within structures, and radioactive effluent releases;
 - (5) Leakage control and detection in the design of systems that pass through barriers important to fulfilling safety functions for the release of radionuclides. An example is an SSC that penetrates a containment structure that might contain radioactive materials that could contribute to the source term during an accident;
 - (6) Monitoring of in-plant radiation and airborne radioactivity as appropriate for a broad range of normal operating and accident conditions; and
 - (7) For self-reliant-mitigation facilities, the plant design must also provide the generally licensed reactor operators with the capability to do the following:
 - (i) Receive plant operating data, including reactor parameters and information needed for the evaluation of emergency conditions.
 - (ii) Promptly dispatch operations and maintenance personnel.
 - (iii) Immediately implement responsibilities under the facility emergency plan, as applicable.
 - (8) For both interaction-dependent and self-reliant mitigation facilities, the plant design must provide licensed operators with the capability of immediately initiating a reactor shutdown from their location.

- (c) **Concept of operations.** A concept of operations that is of sufficient scope and detail to address the following must be provided:
 - (1) Plant goals;
 - (2) The roles and responsibilities of operating personnel and automation (or any combination thereof) that are responsible for completing plant functions;
 - (3) Staffing, qualifications, and training;
 - (4) The management of normal operations;
 - (5) The management of off-normal conditions and emergencies;
 - (6) The management of maintenance and modifications; and
 - (7) The management of tests, inspections, and surveillances.
- (d) **Functional requirements analysis and function allocation.** A functional requirements analysis and a function allocation must be provided that are sufficient to demonstrate compliance with the following:
 - (1) The functional requirements analysis must address how safety functions and functional safety criteria are satisfied; and
 - (2) The function allocation must describe how the safety functions will be assigned to human action, automation, active safety features, passive safety features, and/or inherent safety characteristics.
- (e) **Operating experience.** A program, during construction and during operation, as applicable, for evaluating and applying operating experience must be developed, implemented, and maintained.
- (f) **Staffing plan.** A staffing plan must be developed and comply with the following:
 - (1) The staffing plan must include a description of how engineering expertise will be available to the on-shift operating personnel during all plant conditions, to assist if they encounter a situation not covered by procedures or training. Engineering expertise includes familiarity with the operation of the plant for which the expertise is provided and one of the following:
 - (i) A bachelor's degree in engineering, engineering technology, or physical science from an institution accredited by a U.S. Government recognized accrediting body or equivalent; or
 - (ii) A Professional Engineer's license from a U.S. State or territory.
 - (2) Applicants for or holders of OLs or COLs for interaction-dependent-mitigation facilities must include within their staffing plans a description of how the proposed numbers, positions, and qualifications of operators and senior operators across all modes of plant operations will be sufficient to ensure that plant safety functions will be maintained. This description must be supported by human factors engineering analyses and assessments.
 - (3) Applicants for or holders of OLs or COLs for self-reliant-mitigation facilities must include within their staffing plans a description of how generally licensed reactor operator staffing that is both sufficient to continually monitor the operations of fueled reactors and to provide for a continuity of responsibility for facility operations at all times during the operating phase will be maintained.

- (4) Applicants for or holders of OLs or COLs under this part must include within their staffing plans a description of how the positions and responsibilities of personnel contained within those plans will adequately satisfy necessary support functions within areas such as plant operations, equipment surveillance and maintenance, radiological protection, chemistry control, fire brigades, engineering, security, and emergency response.
- (5) The staffing plan must be approved by the NRC as part of its approval of the OL or COL for the plant. The approved staffing plan is subject to the requirements of § 53.1565.
- (g) **Training, examination, and proficiency programs.** Develop, implement, and maintain programs that comply with the following requirements. These programs must be approved by the NRC as part of its approval of the OL or COL for the plant:
 - (1) For those applicants for or holders of OLs or COLs for interaction-dependent-mitigation facilities:
 - (i) The operator licensing initial training program required under § 53.780(a);
 - (ii) The operator licensing initial examination program required under § 53.780(b);
 - (iii) The operator licensing requalification program required under § 53.780(c); and
 - (iv) The operator proficiency program required under § 53.780(g).
 - (2) For those applicants for or holders of OLs or COLs for self-reliant-mitigation facilities, the generally licensed reactor operator training, examination, and proficiency programs required under § 53.815.
 - (3) The operator licensing requalification programs required under § 53.780(c) or § 53.815(b) must be implemented upon commencing the administration of initial examinations under the operator licensing examination program required under § 53.780(b) or § 53.815(b), respectively.

§ 53.735 General exemptions.

The regulations in §§ 53.725 through 53.830 do not require a license for an individual who—

- (a) Under the direction and in the presence of an operator or senior operator or a generally licensed reactor operator, as appropriate, manipulates the controls of a commercial nuclear plant as a part of the individual's training in a facility licensee's training program as approved by the Commission to qualify for an operator or senior operator license or a generally licensed reactor operator license there, as appropriate, under these regulations; or
- (b) Under the direction and in the presence of a senior operator or generally licensed reactor operator, as appropriate, manipulates the controls of a commercial nuclear plant to load or unload the fuel into, out of, or within the reactor vessel while the reactor is not operating.

§ 53.740 Facility licensee requirements—general.

- (a) Facility licensees must demonstrate compliance with the requirements of either §§ 53.760 through 53.795 for interaction-dependent-mitigation facilities or §§ 53.800 through 53.820 for self-reliant-mitigation facilities.
- (b) The facility licensee must maintain the staffing complement described under its approved facility staffing plan until such time as the permanent cessation of operations and permanent removal of fuel from the reactor vessel has been certified as described under § 53.1070. The approved staffing plan is subject to the requirements of § 53.1565.

- (c) Except as provided under § 53.735, the facility licensee may not permit the manipulation of the controls of a commercial nuclear plant by anyone who is not an operator or senior operator or generally licensed reactor operator, as appropriate.
- (d) Facility licensees for interaction-dependent-mitigation facilities that have not yet certified the permanent cessation of operations and permanent removal of fuel from the reactor vessel as described under § 53.1070 must designate senior operators to be responsible for supervising the licensed activities of operators.
- (e) Apparatus and mechanisms other than controls, the operation of which may affect the reactivity or power level of a reactor, must be manipulated only while plant conditions are being monitored by an individual who is an operator or senior operator or a generally licensed reactor operator, as appropriate.
- (f)
 - (1) Load following is permitted if at least one of the following is immediately capable of refusing demands when they could challenge the safe operation of the plant or when precluded by the plant equipment conditions:
 - (i) The actuation of an automatic protection system that utilizes setpoints more conservative than those otherwise credited for the purposes of reactor protection; or
 - (ii) An automated control system; or
 - (iii) An operator or senior operator or a generally licensed reactor operator, as appropriate.
 - (2) The provisions of paragraph (e) of this section do not apply during load following operations.
- (g)
 - (1) Facility licensees for interaction-dependent-mitigation facilities must have present during alteration of the core (including fuel loading or transfer) an individual holding a senior operator license, or a senior operator license limited to fuel handling to directly supervise the activity and, during this time, the facility licensee must not assign other duties to this person.
 - (2) Facility licensees for self-reliant-mitigation facilities must have present during alteration of the core (including fuel loading or transfer) an individual holding a generally licensed reactor operator license to directly supervise the activity and, during this time, the facility licensee must not assign other duties to this person.
 - (3) The provisions of paragraphs (g)(1) and (2) of this section do not apply to core alterations performed as part of refueling operations while a facility that is capable of online refueling is operating at power.
- (h) Facility licensees may take reasonable action that departs from a license condition or a technical specification (contained in a license issued under this part) in an emergency when this action is immediately needed to protect the public health and safety and no action consistent with license conditions and technical specifications that can provide adequate or equivalent protection is immediately apparent. Such facility licensee action must be approved, as a minimum, by a senior operator or a generally licensed reactor operator, as applicable, or, after certifying the permanent cessation of operations and permanent removal of fuel from the reactor vessel as described under § 53.1070 by a certified fuel handler, senior operator, or generally licensed reactor operator, as applicable, prior to taking the action.

§ 53.745 Operator license requirements.

A person must be authorized by a license issued by the Commission to perform the function of an operator, senior operator, or generally licensed reactor operator as defined in this part.

§ 53.760 Operator licensing.

- (a) **Applicability.** Sections 53.760 through 53.795 address operator and senior operator licensing requirements. The regulations within these sections are applicable to those applicants for or holders of OLs or COLs under this part for interaction-dependent-mitigation facilities that have not yet certified the permanent cessation of operations and permanent removal of fuel from the reactor vessel as described under § 53.1070.
- (b) [Reserved]

§ 53.765 Medical requirements.

- (a) An applicant for an operator or senior operator license must have a medical examination by a physician. An operator or senior operator must have a medical examination by a physician every 2 years.
- (b) To certify the medical fitness of an applicant for an operator or senior operator license, an authorized representative of the facility licensee must complete and sign NRC Form 396, "Certification of Medical Examination by Facility Licensee," which can be obtained by writing the Office of the Chief Information Officer, U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, by calling 301-415-7232, or by visiting the NRC's website at <https://www.nrc.gov> and selecting forms from the index found on the home page, or by other means provided by the NRC.
 - (1) NRC Form 396 must certify that a physician has conducted the medical examination of the applicant as required in paragraph (a) of this section.
 - (2) When the medical certification requests a conditional license based on medical evidence, the medical evidence must be submitted on NRC Form 396 to the Commission to enable the Commission to make a determination in accordance with § 53.775(b).
- (c) The facility licensee must document and maintain the results of medical qualifications data, test results, and each operator's or senior operator's medical history for the current license period and provide the documentation to the Commission upon request. The facility licensee must retain this documentation while an individual performs the functions of an operator or senior operator.

§ 53.770 Incapacitation because of disability or illness.

If, during the term of the operator or senior operator license, the licensee develops a permanent physical or mental condition that causes the licensee to fail to demonstrate compliance with the requirements of § 53.775(b)(1)(i), the facility licensee must notify the Commission within 30 days of learning of the diagnosis. For conditions for which a conditional license (as described in § 53.775(b)) is requested, the facility licensee must provide medical certification on NRC Form 396 to the Commission (as described in § 53.765(b)).

§ 53.775 Applications for operators and senior operators.

- (a) **How to apply.**
 - (1) The applicant for an operator or senior operator license must—

- (i) Complete NRC Form 398, "Personal Qualification Statement—Licensee," which can be obtained by writing the Office of the Chief Information Officer, U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, by calling 301-415-5877, or by visiting the NRC's website at <https://www.nrc.gov> and selecting forms from the index found on the home page, or by other means provided by the NRC;
- (ii) File an original of NRC Form 398, or an equivalent electronic submittal, together with the information required in paragraphs (a)(1)(iii) and (a)(1)(iv) of this section, with the appropriate Regional Administrator.
- (iii) Provide evidence that the applicant, as a trainee, has successfully demonstrated competence in manipulating the controls of either the facility for which a license is sought or a simulation facility that demonstrates compliance with the requirements of § 53.780(e). For operators applying for a senior operator license, certification that the operator has successfully operated the controls of the facility as an operator will be accepted; and
- (iv) Provide certification by the facility licensee of medical condition and general health on NRC Form 396, to comply with § 53.765.

- (2) The Commission may at any time after the application has been filed, and before the license has expired, require further information under oath or affirmation to enable it to determine whether to grant or deny the application or whether to revoke, modify, or suspend the license.
- (3) An applicant whose application has been denied because of a medical condition or their general health may submit a further medical report at any time as a supplement to the application.
- (4) Each application and statement must contain complete and accurate disclosure as to all matters required to be disclosed. The applicant must sign statements required by paragraphs (a)(1)(i) and (a)(1)(ii) of this section.

(b) *Disposition of an initial application* —

- (1) **License approval.** The Commission will approve an initial application if it finds that the following criteria are met:
 - (i) **Health.** The applicant's medical condition and general health will not adversely affect the performance of assigned operator or senior operator job duties or cause operational errors endangering public health and safety. The Commission will base its finding upon the certification by the facility licensee as detailed in § 53.765(b).
 - (ii) **Examination.** The applicant has passed the requisite examination in accordance with § 53.780(b). The examination determines whether the applicant for an operator's or senior operator's license has learned to operate a facility competently and safely, and additionally, in the case of a senior operator, whether the applicant has learned to supervise the licensed activities of operators competently and safely.
- (2) **Conditional license.** If an applicant's general medical condition does not demonstrate compliance with the minimum standards under § 53.775(b)(1)(i), the Commission may approve the application and include conditions in the license to accommodate the medical condition. The Commission will consider the recommendations and supporting evidence of the facility licensee and of the examining physician (provided on NRC Form 396) in arriving at its decision.

(c) *Re-applications.*

- (1) An applicant whose application for a license has been denied because of failure to pass the examination may file a new application. The application must be submitted on NRC Form 398 and include a statement signed by an authorized representative of the facility licensee by whom the applicant will be employed that states in detail the extent of the applicant's additional training and remediation since the denial and certifies that the applicant is ready for re-examination.
- (2) An applicant who has passed a portion of the examination and failed another may request in a new application on NRC Form 398 to be excused from re-examination on the portions of the examination that the applicant has passed. The Commission may in its discretion grant the request if it determines that sufficient justification is presented.

§ 53.780 Training, examination, and proficiency program.

(a) *Operator licensing initial training program.*

- (1) A program that is based upon a systems approach to training, as defined by § 53.725(b), must be utilized for the training of applicants for operator and senior operator licenses. The program must ensure that applicants at the facility will possess the knowledge, skills, and abilities necessary to protect the public health and maintain those plant safety functions specific to the facility design. The program must be approved by the Commission prior to its use for training applicants, as described under § 53.730(g). The approved operator licensing initial training program is subject to the requirements of § 53.1565.
- (2) The facility licensee must maintain operator licensing initial training program records documenting the initial operator licensing training administered and completed by each applicant. The facility licensee must retain these records during the period in which any trainees subsequently remain licensed as operators or senior operators at the facility.

(b) *Operator licensing initial examination program.*

- (1) The facility licensee must establish and implement an examination program for testing a representative sample of the knowledge, skills, and abilities needed to safely perform operator and senior operator duties, to include both the examination methods and criteria to be used to assess passing performance. The program must provide for valid and reliable examinations and be approved by the Commission prior to its use for examining applicants, as described under § 53.730(g). The approved operator licensing initial examination program is subject to the requirements of § 53.1565.
- (2) The facility licensee must submit prepared examinations to the Commission for review and approval in advance of their administration.
- (3) The Commission will either administer an approved examination or allow the facility licensee to administer the examination. The facility licensee must ensure that sufficient advance notification is provided to the Commission to either administer the examination or allow for a representative of the Commission to be afforded the opportunity to be present when the facility licensee administers the examination.
- (4) Graded examination documentation for each applicant must be provided to the Commission for review in making operator licensing decisions.
- (5) The facility licensee must maintain operator licensing initial examination program records documenting the participation of each operator and senior operator applicant in the initial examination. The records must contain copies of examinations administered, the answers given by

the applicant, documentation of the grading of examinations, and documentation of any additional training administered in areas in which an applicant exhibited deficiencies. The facility licensee must retain these records during the period in which the associated operators or senior operators remain licensed at the facility.

(c) *Operator licensing requalification program.*

- (1) A program based upon a systems approach to training, as defined by § 53.725(b), must be utilized for the continuing training of operators and senior operators.
 - (i) The program must ensure that operators and senior operators at the facility maintain the knowledge, skills, and abilities necessary to protect the public health and maintain those plant safety functions specific to the facility design. The program must be conducted for a continuous period not to exceed 24 months in duration.
 - (ii) The program must be approved by the Commission prior to its use for continuing training, as described under § 53.730(g). The approved operator licensing requalification program is subject to the requirements of § 53.1565.
- (2) The following requirements apply to operator licensing requalification programs:
 - (i) The facility licensee must propose a requalification examination program for testing, for each requalification period, a sample of the topics included under the systems approach to training, to include both the examination methods and criteria to be used to assess passing performance. The program must provide for valid and reliable examinations and be approved by the Commission prior to its use for examining operators and senior operators, as described under § 53.730(g). The approved requalification examination program is subject to the requirements of § 53.1565.
 - (ii) The following requirements apply to the requalification examination program:
 - (A) The facility licensee must make prepared requalification examinations available to the Commission for review.
 - (B) The facility licensee must ensure that a representative of the Commission is afforded the opportunity to be present during requalification examination administration.
 - (C) The facility licensee must ensure that each operator and senior operator is administered a complete requalification examination on a periodicity not to exceed 24 months. Additionally, the facility licensee must ensure that any licensed operator or senior licensed operator who either demonstrates unsatisfactory performance on, or fails to complete, this biennial requalification examination is removed from the performance of licensed operator and senior licensed operator duties until any necessary remedial training has been completed and a retake examination has been passed.
 - (D) The facility licensee must promptly provide a summary of examination results to the NRC for each operator and senior operator following the completion of the requalification examination.
- (3) The facility licensee must maintain operator licensing requalification program records documenting the participation of each operator and senior operator in the requalification program. The records must contain copies of examinations administered, the answers given by the operator or senior

operator, documentation of the grading of examinations, and documentation of any additional training administered in areas in which an operator or senior operator exhibited deficiencies. The facility licensee must retain these records until the operator's or senior operator's license is renewed.

(d) **Examination integrity.** Applicants, operators and senior operators, and facility licensees must not engage in any activity that compromises the integrity of any application or examination required by §§ 53.760 through 53.795. The integrity of an examination is considered compromised if any activity, regardless of intent, affected, or, but for detection, could have affected the consistent administration of the examination. This includes activities related to the preparation and certification of applications and all activities related to the preparation, administration, and grading of examinations required by §§ 53.760 through 53.795.

(e) **Simulation facilities.**

- (1) This section addresses the use of a simulation facility for the administration of examinations, for training, or to demonstrate compliance with experience requirements for applicants for operator and senior operator licenses.
- (2) Simulation facilities used for training purposes, for demonstrating compliance with experience requirements, or for the conduct of examinations under § 53.780(b) and (c) must demonstrate compliance with the following criteria as they relate to the facility licensee's reference plant:
 - (i) The simulation facility must be of sufficient scope and fidelity for individuals to acquire and demonstrate the necessary knowledge, skills, and abilities to safely perform operator and senior operator duties.
 - (ii) The simulation facility must utilize models relating to nuclear, thermal-hydraulic, and other applicable design-specific characteristics that either replicate the most recent fuel load in the reference commercial nuclear plant or, prior to initial fuel load (or, for a fueled manufactured reactor, prior to initiating the removal of the features to prevent criticality required under § 53.620(d)(1)), replicate the intended initial fuel load for the reference commercial nuclear plant, with the exception of those portions of the simulation facility that utilize the reference plant itself.
 - (iii) Simulation facility fidelity must be demonstrated so that significant control manipulations are completed without procedural exceptions, simulator performance exceptions, or deviation from the approved training scenario sequence.
- (3) Facility licensees that maintain a simulation facility that has been approved by the Commission for training purposes, demonstrating compliance with experience requirements, or the conduct of examinations under § 53.780(b) and (c) for the facility licensee's reference plant must:
 - (i) Conduct performance testing throughout the life of the simulation facility in a manner sufficient to ensure that paragraph (e)(2) of this section is met;
 - (ii) Retain the results of performance testing for 4 years after the completion of each performance test or until superseded by updated test results;
 - (iii) Promptly correct modeling and hardware discrepancies and discrepancies identified from scenario validation and from performance testing or provide justification as to why the presence of such discrepancies will not adversely affect simulator performance with respect to the criteria of paragraph (e)(2) of this section;

- (iv) Make the results of any uncorrected performance test failures that may exist at the time of the initial license examination or requalification examination available for NRC review, prior to or concurrent with preparations for each initial license examination or requalification examination; and
 - (v) Maintain the provisions for license application and examination integrity consistent with § 53.780(d).
- (4) A simulation facility must demonstrate compliance with the requirements of paragraphs (e)(2) and (e)(3) of this section for the Commission to accept the simulation facility for conducting initial examinations as described in § 53.780(b), requalification training as described in § 53.780(c), or performing control manipulations that affect reactivity to establish eligibility for an operator or senior operator license as described in § 53.775(a).
- (f) **Waiver of examination requirement.** On application, the Commission may waive any or all of the requirements for an initial licensing examination if it finds that the applicant has demonstrated the required knowledge, skills, and abilities to safely operate the plant, and is capable of continuing to do so. The Commission may make such a finding based on demonstration of the following:
- (1) Recent operating experience at a comparable facility;
 - (2) Proof of the applicant's past competent and safe performance; and
 - (3) Proof of the applicant's current qualifications.
- (g) **Proficiency.** The facility licensee must develop, implement, and maintain a proficiency program to ensure that operators and senior operators will actively perform the functions of an operator or senior operator, respectively, as needed to maintain proficiency with on-shift duties and familiarity with plant status. This program must include those steps that will be taken to re-establish proficiency when it cannot be maintained. This program must be approved by the Commission as part of its approval of the OL or COL for the plant. The approved proficiency program is subject to the requirements of § 53.1565.
- (h) **Records.** Each record required by this section must be legible throughout the retention period specified by each Commission regulation. The record may be the original, a reproduced copy, or an electronic copy provided that the copy is authenticated by authorized personnel.

§ 53.785 Conditions of operator and senior operator licenses.

Each operator and senior operator license contains and is subject to the following conditions whether stated in the license or not:

- (a) Neither the license nor any right under the license may be assigned or otherwise transferred.
- (b) The license is limited to the facility or facilities for which it is issued.
- (c) The license is limited to those controls of the facility or facilities specified in the license.
- (d) The license is subject to, and the licensee must observe, all applicable rules, regulations, and orders of the Commission.
- (e) The licensee must maintain or re-establish proficiency in accordance with the facility licensee's Commission-approved proficiency program required under § 53.780(g).
- (f) The licensee must be subject to the facility's Commission-approved operator licensing requalification and requalification examination programs required under § 53.780(c).

- (g) The licensee must have a biennial medical examination as described by § 53.765.
- (h) The licensee must notify the Commission within 30 days about a conviction for a felony.
- (i) The licensee must not consume or ingest alcoholic beverages within the protected area of commercial nuclear plants. The licensee must not use, possess, or sell any illegal drugs. The licensee must not perform activities authorized by a license issued under this part while under the influence of alcohol or any prescription, over-the-counter, or illegal substance that could adversely affect his or her ability to safely and competently perform his or her licensed duties. For the purpose of this paragraph (i), with respect to alcoholic beverages and drugs, the term “under the influence” means the licensee exceeded, as evidenced by a confirmed test result, the lower of the cutoff levels for drugs or alcohol contained in 10 CFR part 26, or as established by the facility licensee. The term “under the influence” also means the licensee could be mentally or physically impaired as a result of substance use including prescription and over-the-counter drugs, as determined under the provisions, policies, and procedures established by the facility licensee for its fitness-for-duty program, in such a manner as to adversely affect his or her ability to safely and competently perform licensed duties.
- (j) Each licensee must participate in the drug and alcohol testing programs as required under 10 CFR part 26.
- (k) The licensee must comply with any other conditions that the Commission may impose to protect health or to minimize danger to life or property.

§ 53.790 Issuance, modification, and revocation of operator and senior operator licenses.

- (a) **Issuance of operator and senior operator licenses.** If the Commission determines that an applicant for an operator license or a senior operator license demonstrates compliance with the requirements of the Atomic Energy Act of 1954, as amended, (the Act) and its regulations, it will issue a license in the form and containing any conditions and limitations it considers appropriate and necessary.
- (b) **Modification and revocation of operator and senior operator licenses.**
 - (1) The terms and conditions of all operator and senior operator licenses are subject to amendment, revision, or modification by reason of rules, regulations, or orders issued in accordance with the Act or any amendments thereto.
 - (2) Any license may be revoked, suspended, or modified, in whole or in part—
 - (i) For any material false statement in the application or in any statement of fact required under section 182 of the Act;
 - (ii) Because of conditions revealed by the application or statement of fact or any report, record, inspection, or other means that would warrant the Commission to refuse to grant a license on an original application;
 - (iii) For willful violation of, or failure to observe, any of the terms and conditions of the Act or the license, or of any rule, regulation, or order of the Commission;
 - (iv) For any conduct determined by the Commission to be a hazard to safe operation of the facility; or

- (v) For the sale, use, or possession of illegal drugs, or refusal to participate in the facility drug and alcohol testing program, or a confirmed positive test for drugs, drug metabolites, or alcohol in violation of the conditions and cutoff levels established by § 53.785(i) or the consumption of alcoholic beverages within the protected area of commercial nuclear plants, or a determination of unfitness for scheduled work as a result of the consumption of alcoholic beverages.

§ 53.795 Expiration and renewal of operator and senior operator licenses.

(a) *Expiration.*

- (1) Each operator license and senior operator license expires 6 years after the date of issuance, upon termination of employment with the facility licensee, or upon determination by the facility licensee that the licensed individual no longer needs to maintain a license.
- (2) If a licensee files an application for renewal or an upgrade of an existing license on NRC Form 398 at least 30 days before the expiration of the existing license, it does not expire until disposition of the application for renewal or for an upgraded license has been finally determined by the Commission. Filing by mail will be deemed to be complete at the time the application is postmarked

(b) *Renewal.*

- (1) The applicant for renewal of an operator license or senior operator license must—
 - (i) Complete and sign NRC Form 398 and include the number of the license for which renewal is sought.
 - (ii) File an original of NRC Form 398 as specified in § 53.775.
 - (iii) Provide written evidence of the applicant's experience under the existing license and the approximate number of hours that the licensee has operated the facility.
 - (iv) Provide a statement by an authorized representative of the facility licensee that during the effective term of the current license the applicant has satisfactorily completed the requalification program for the facility for which operator or senior operator license renewal is sought.
 - (v) Provide evidence that the applicant has discharged the license responsibilities competently and safely. The Commission may accept as evidence of the applicant's having met this requirement a certificate of an authorized representative of the facility licensee or holder of an authorization by which the licensee has been employed.
 - (vi) Provide certification by the facility licensee of medical condition and general health on NRC Form 396, to comply with § 53.765.
- (2) The license will be renewed if the Commission finds that—
 - (i) The medical condition and the general health of the licensee continue to be such as not to cause operational errors that endanger public health and safety. The Commission will base this finding upon the certification by the facility licensee as described in § 53.765(b).
 - (ii) The licensee—
 - (A) Is capable of continuing to competently and safely assume licensed duties;
 - (B) Has successfully completed a requalification program that has been approved by the Commission as required by § 53.780(c); and

- (C) Has passed the requalification examinations as required by § 53.780(c).
- (iii) There is a continued need for an operator to operate or for a senior operator to supervise operators at the facility designated in the application.
- (iv) The past performance of the licensee has been satisfactory to the Commission. In making its finding, the Commission will include in its evaluation information such as notices of violations or letters of reprimand in the licensee's docket.

§ 53.800 Facility licensees for self-reliant-mitigation facilities.

- (a) A commercial nuclear plant is a self-reliant-mitigation facility if the NRC determined as part of its approval of the OL or COL for that plant that its design demonstrates compliance with the criteria in paragraphs (a)(1) through (a)(5) of this section. A self-reliant-mitigation facility is of a class, based upon the similarity of operating and technical characteristics of the plants in the class, such that its licensee must comply with the requirements of §§ 53.800 through 53.820 in lieu of those in §§ 53.760 through 53.795.
 - (1) The safety performance criteria of §§ 53.210 and 53.220 must be met without reliance upon human action for credited event mitigation.
 - (2) The results of the probabilistic risk assessment (PRA), other systematic risk evaluations, or a combination thereof required by § 53.450(a) must demonstrate that the evaluation criteria for the events analyzed in accordance with § 53.450 will be met without reliance on human actions to achieve acceptable event mitigation.
 - (3) The functional requirements analysis and function allocation performed under § 53.730(d) must demonstrate that functions required for safety are not reliant upon credited human action.
 - (4) The plant response to events analyzed under § 53.450 must rely exclusively on safety features and characteristics that will neither be rendered unavailable by credible human errors of commission or omission nor credibly require manual human operation in response to equipment failures. Compliance with this paragraph (a)(4) may be achieved through the use of SSCs that function through inherent characteristics or that have engineered protections against human failures.
 - (5) Assessments of credited human actions within the analysis of design-basis accidents (DBAs) and across the range of LBEs other than DBAs do not identify important human actions needed to ensure appropriate defense in depth is provided, as required by § 53.250.
- (b) [Reserved]

§ 53.805 Facility licensee requirements related to generally licensed reactor operators.

- (a) Licensees for self-reliant-mitigation facilities that have not yet certified the permanent cessation of operations and permanent removal of fuel from the reactor vessel as described under § 53.1070 must demonstrate compliance with the following requirements:
 - (1) Ensure that, in addition to being qualified to perform those items identified by the facility-specific systems approach to training conducted under § 53.815, generally licensed reactor operators are qualified to safely and competently—
 - (i) Perform administrative tasks, including compliance with technical specifications, and perform operability determinations;
 - (ii) Implement maintenance and configuration controls;

- (iii) Comply with radioactive release limitations;
 - (iv) Understand plant operating data, including reactor parameters, and evaluate emergency conditions;
 - (v) Initiate a reactor shutdown from necessary locations;
 - (vi) Dispatch and direct operations and maintenance personnel;
 - (vii) Implement any applicable responsibilities under the facility emergency plan; and
 - (viii) Make required notifications to local, State, participating Tribal, and Federal authorities.
- (2) Develop, implement, and maintain facility technical specifications that provide the necessary administrative controls to ensure the implementation of the requirements in this section.
 - (3) Develop, implement, and maintain the generally licensed reactor operator training, examination, and proficiency programs required under § 53.815.
 - (4) Ensure that generally licensed reactor operators are subject to the facility's generally licensed reactor operator training, examination, and proficiency programs required under § 53.815. Ensure that generally licensed reactor operators are subject to and comply with the applicable programmatic requirements for personnel required under 10 CFR parts 26 and 73. An individual that is not in compliance with any of these programs is not qualified to be in a position that may involve the manipulation of the controls of the commercial nuclear plant.
 - (5) Report annually to the NRC the identity of all generally licensed reactor operators at the commercial nuclear plant, including all additions and deletions since the previous report.
 - (6) Ensure that the facility design continues to meet the criteria of § 53.800.
- (b) [Reserved]

§ 53.810 Generally licensed reactor operators.

- (a) A general license to manipulate the controls of a self-reliant-mitigation facility and to direct the licensed activities of generally licensed reactor operators is hereby issued to any individual employed in a position that may involve the manipulation of the controls of that self-reliant-mitigation facility and who observes the restrictions of this section.
- (b) A generally licensed reactor operator must comply with the operating procedures and other conditions specified in the license authorizing operation of the facility.
- (c) The general license is limited to the facility or facilities at which the operator is employed.
- (d) The Commission will suspend the general license on an individual operator basis for violations of any provision of the Act or any rule or regulation issued thereunder whenever the Commission deems such suspension desirable, including—
 - (1) For willful violation of, or failure to observe, any of the terms and conditions of the Act or the general license, or of any rule, regulation, or order of the Commission;
 - (2) For any conduct determined by the Commission to be a hazard to safe operation of the facility; or

- (3) For the sale, use, or possession of illegal drugs, or refusal to participate in the facility drug and alcohol testing program, or a confirmed positive test for drugs, drug metabolites, or alcohol in violation of the conditions and cutoff levels established by § 53.810(f) or the consumption of alcoholic beverages within the protected area of commercial nuclear plants, or a determination of unfitness for scheduled work as a result of the consumption of alcoholic beverages.
- (e) The Commission may require information from a generally licensed reactor operator to determine whether a general license should be revoked or suspended with respect to that operator.
- (f) The generally licensed reactor operator must not consume or ingest alcoholic beverages within the protected area of commercial nuclear plants. The generally licensed reactor operator must not use, possess, or sell any illegal drugs. The generally licensed reactor operator must not perform activities requiring a general license while under the influence of alcohol or any prescription, over-the-counter, or illegal substance that could adversely affect his or her ability to safely and competently perform these activities. For the purpose of this paragraph (f), with respect to alcoholic beverages and drugs, the term “under the influence” means the generally licensed reactor operator exceeded, as evidenced by a confirmed test result, the lower of the cutoff levels for drugs or alcohol contained in 10 CFR part 26, or as established by the facility licensee. The term “under the influence” also means the generally licensed reactor operator could be mentally or physically impaired as a result of substance use including prescription and over-the-counter drugs, as determined under the provisions, policies, and procedures established by the facility licensee for its fitness-for-duty program, in such a manner as to adversely affect his or her ability to safely and competently perform generally licensed reactor operator duties.
- (g) The generally licensed reactor operator must notify the Commission within 30 days about a conviction for a felony.

§ 53.815 Generally licensed reactor operator training, examination, and proficiency programs.

- (a) **Applicability.** The requirements of this section apply to each licensee of a self-reliant-mitigation facility that has not yet certified the permanent cessation of operations and permanent removal of fuel from the reactor vessel as described under § 53.1070.
- (b) **Requirements.**
 - (1) The facility licensee must develop, implement, and maintain training and examination programs that demonstrate compliance with the requirements of paragraphs (b)(2) through (b)(3) of this section.
 - (2) The training program must provide for both the initial and continuing training of generally licensed reactor operators and be derived from a systems approach to training as defined in this part.
 - (3)
 - (i) The training program must incorporate the instructional requirements necessary to provide qualified generally licensed reactor operators to operate and maintain the facility in a safe manner in all modes of operation. The training program must comply with the facility license, including all technical specifications and applicable regulations. The facility licensee must periodically evaluate and revise the training program as appropriate to reflect industry experience and relevant changes, including changes to the facility, procedures, regulations, and quality assurance (QA) requirements. Facility licensee management must periodically review the training program for effectiveness.
 - (ii) The training program must ensure that generally licensed reactor operators have and maintain the necessary knowledge, skills, and abilities.

- (iii) The training program must include the generally licensed reactor operator manipulating the controls of either the facility or a simulation facility that demonstrates compliance with the requirements of § 53.815(e).
 - (iv) The training program must include an initial examination program for testing a representative sample of the knowledge, skills, and abilities needed to safely perform generally licensed reactor operator duties, to include both the examination methods and criteria to be used to assess passing performance. The facility licensee must provide the opportunity for a representative of the Commission to be present during initial examination administration.
 - (v) The training program must include a requalification examination program for testing a sample of the topics included under the systems approach to training, to include the examination methods and criteria to be used to assess passing performance. The requalification examination program must specify an appropriate periodicity for administering a complete requalification examination to each generally licensed reactor operator, and the facility licensee must provide the opportunity for a representative of the Commission to be present during requalification examination administration.
 - (A) The facility licensee must ensure that any generally licensed reactor operator who either demonstrates unsatisfactory performance on, or fails to complete, the requalification examination is removed from the performance of generally licensed reactor operator duties until such time that any necessary remedial training has been completed and a retake examination has been passed.
 - (B) [Reserved]
 - (vi) The training program must be approved by the Commission prior to its use, as described under § 53.730(g). The examination program must provide for valid and reliable examinations and must be approved by the Commission prior to their use, as described under § 53.730(g). The approved programs are subject to the requirements of § 53.1565.
- (c) **Records.** The following is required regarding the documentation of the generally licensed reactor operator training and examination programs:
- (1) Sufficient records must be maintained by the facility licensee to maintain the integrity of the programs and kept available for NRC inspection to verify the adequacy of the programs.
 - (2) The facility licensee must maintain records documenting the participation of each generally licensed reactor operator in the training and examination programs. The records must contain copies of examinations administered, the answers given by the generally licensed reactor operator, documentation of the grading of examinations, and documentation of any additional training administered in areas in which a generally licensed reactor operator exhibited deficiencies. The facility licensee must retain these records while the associated generally licensed reactor operators remain employed at the facility.
 - (3) Each record required by this part must be legible throughout the retention period. The record may be the original, a reproduced copy, or an electronic copy provided that the copy is authenticated by authorized personnel.
- (d) **Examination integrity.** Generally licensed reactor operators and facility licensees must not engage in any activity that compromises the integrity of any examination conducted under the generally licensed reactor operator training and examination programs. The integrity of an examination is considered compromised

if any activity, regardless of intent, affected, or, but for detection, could have affected the consistent administration of the examination. This includes all activities related to the preparation, administration, and grading of examinations.

(e) **Simulation facilities.**

- (1) Simulation facilities used for training purposes, for maintaining proficiency, or for the conduct of examinations must demonstrate compliance with the following criteria as they relate to the facility licensee's reference plant:
 - (i) The simulation facility must be of sufficient scope and fidelity for individuals to acquire and demonstrate the necessary knowledge, skills, and abilities to safely perform generally licensed reactor operator duties.
 - (ii) The simulation facility must utilize models relating to nuclear, thermal-hydraulic, and other applicable design-specific characteristics that either replicate the most recent fuel load in the reference commercial nuclear plant or, prior to initial fuel load (or, for a fueled manufactured reactor, prior to initiating the removal of the features to prevent criticality required under § 53.620(d)(1)), replicate the intended initial fuel load for the reference commercial nuclear plant, with the exception of those portions of the simulation facility that utilize the reference plant itself.
 - (iii) Simulator fidelity must be demonstrated so that significant control manipulations are completed without procedural exceptions, simulator performance exceptions, or deviation from the approved training scenario sequence.
- (2) Facility licensees that maintain a simulation facility for training purposes, for maintaining proficiency, or for the conduct of examinations must—
 - (i) Conduct performance testing throughout the life of the simulation facility in a manner sufficient to ensure that paragraph (e)(1) of this section is met;
 - (ii) Retain the results of performance testing for 4 years after the completion of each performance test or until superseded by updated test results;
 - (iii) Promptly correct modeling and hardware discrepancies and discrepancies identified from scenario validation and from performance testing or provide justification for why the presence of such discrepancies will not adversely affect the criteria of paragraph (e)(1) of this section;
 - (iv) Make the results of any uncorrected performance test failures that may exist at the time of an inspection available for NRC review; and
 - (v) Maintain the provisions for examination integrity consistent with § 53.815(d).
- (f) **Waiver of examination requirement.** The facility licensee may waive any or all of the requirements for an examination in accordance with the facility licensee's Commission-approved generally licensed reactor operator training and examination programs.
- (g) **Proficiency.** The facility licensee must develop, implement, and maintain a proficiency program to allow generally licensed reactor operators to maintain proficiency regarding position functions and familiarity with plant status. This program must include those steps that will be taken in order to re-establish proficiency when it cannot be maintained.

§ 53.820 Cessation of individual applicability.

The general license ceases to be applicable on an individual basis once a generally licensed reactor operator is no longer being employed in a position that may involve the manipulation of the controls of the self-reliant mitigation facility.

§ 53.830 Training and qualification of commercial nuclear personnel.

- (a) This section addresses personnel training requirements. The regulations within this section are applicable to all applicants for or holders of OLs or COLs under this part.
- (b) Prior to initial fuel load (or, for a fueled manufactured reactor, prior to initiating the removal of the features to prevent criticality required under § 53.620(d)(1)), each holder of an operating or COL under this part must, with sufficient time to provide trained and qualified personnel to operate the facility, establish, implement, and maintain a training program that demonstrates compliance with the requirements of paragraphs (c) and (d) of this section.
- (c) The training program must be derived from a systems approach to training as defined in this part and must provide, at a minimum, for the training and qualification of the following categories of commercial nuclear personnel:
 - (1) Supervisors (e.g., shift supervisors);
 - (2) Technicians (e.g., maintenance, chemistry, and radiological); and
 - (3) Other appropriate operating personnel (e.g., auxiliary operators, certified fuel handlers, and individuals who provide engineering expertise to on-shift operating personnel).
- (d) The training program must incorporate the instructional requirements necessary to provide qualified personnel to operate components of a commercial nuclear plant and maintain the facility in a safe manner in all modes of operation. The training program must be developed to be in compliance with the facility license, including all technical specifications and applicable regulations.
 - (1) The training program must be periodically evaluated and revised as appropriate to reflect industry experience and relevant changes, including changes to the facility, procedures, regulations, and QA requirements. The training program must be periodically reviewed by facility licensee management for effectiveness.
 - (2) Sufficient records must be maintained by the facility licensee to maintain program integrity and kept available for NRC inspection to verify the adequacy of the training program.

§ 53.845 Programs.

- (a) The required plant programs under this part must include but are not necessarily limited to the programs described in the following sections of this subpart. Licensees may combine, separate, and otherwise organize programs and related documents as appropriate for the technologies and organizations associated with the commercial nuclear plant.
- (b) In addition to the programs described in the following sections, programs must be provided for each commercial nuclear plant, if necessary, to ensure that the performance of design features and human actions are consistent with the analyses performed under §§ 53.450 and 53.730 and that the plant will demonstrate compliance with the safety criteria defined in §§ 53.210 and 53.220.

§ 53.850 Radiation protection.

- (a) Each holder of an OL or COL under this part must develop, implement, and maintain a Radiation Protection Program for operations that is commensurate with the scope and extent of licensed activities under this part and includes measures for limiting and monitoring radioactive plant effluents and limiting and monitoring the dose to individuals working with radioactive materials in accordance with 10 CFR part 20.
- (b) Each holder of an OL or COL under this part must develop, implement, and maintain a program for the control of radioactive effluents and for environmental monitoring. The program must be contained in an Offsite Dose Calculations Manual, must be implemented by procedures, and must include remedial actions to be taken whenever the program limits are exceeded. The Offsite Dose Calculations Manual must—
 - (1) Contain the methodology and parameters used in the calculation of offsite doses resulting from radioactive gaseous and liquid effluents, in the calculation of gaseous and liquid effluent monitoring alarm and trip setpoints, and in the conduct of the radiological environmental monitoring program; and
 - (2) Contain the radioactive effluent controls and radiological environmental monitoring activities, and descriptions of the information that should be included in the Annual Radiological Environmental Operating and Radioactive Effluent Release Reports required by § 53.1645.
- (c) Each holder of an OL or COL under this part must develop, implement, and maintain a Process Control Program that identifies the administrative and operational controls for solid radioactive waste processing, process parameters, and surveillance requirements sufficient to ensure compliance with the requirements of 10 CFR part 20, 10 CFR part 61, and 10 CFR part 71.

§ 53.855 Emergency preparedness.

- (a) Each holder of an OL or COL under this part must have an emergency response plan that must contain information needed to demonstrate compliance with either the requirements in § 50.160 of this chapter or the requirements in appendix E to part 50 and the planning standards of § 50.47(b) of this chapter.
- (b) No initial OL, initial COL, or early site permit that includes complete and integrated emergency plans will be issued under this part unless a finding is made by the NRC, in accordance with § 50.47 of this chapter, that there is reasonable assurance that adequate protective measures can and will be taken in the event of a radiological emergency.

§ 53.860 Security programs.

- (a) **Physical protection program.** Each holder of an OL or COL under this part must develop, implement, and maintain a physical protection program under the following requirements:
 - (1) The licensee must implement security requirements for the protection of special nuclear material based on the type, enrichment, and quantity in accordance with 10 CFR part 73, as applicable, and implement security requirements for the protection of Category 1 and Category 2 quantities of radioactive material in accordance with 10 CFR part 37, as applicable; and
 - (2) The licensee must demonstrate compliance with the provisions set forth in either § 73.55 or § 73.100 of this chapter.

- (b) ***Fitness-for-duty.*** Each holder of an OL or COL under this part must develop, implement, and maintain a fitness-for-duty program under 10 CFR part 26.
- (c) ***Access authorization.*** Each holder of an OL or COL under this part must develop, implement, and maintain an access authorization program under § 73.56 or § 73.120 of this chapter, as applicable.
- (d) ***Cybersecurity.*** Each holder of an OL or COL under this part must develop, implement, and maintain a cybersecurity program under § 73.54 or § 73.110 of this chapter.
- (e) ***Information security.*** Each holder of an OL or COL under this part must develop, implement, and maintain an information protection system under §§ 73.21, 73.22, and 73.23 of this chapter, as applicable.

§ 53.865 Quality assurance.

Each holder of an OL or COL under this part must develop, implement, and maintain a quality assurance program in accordance with appendix B of part 50 of this chapter. A written quality assurance program manual must be developed and used to guide the conduct of the program.

§ 53.870 Integrity assessment programs.

Each holder of an OL or COL under this part must develop, implement, and maintain an integrity assessment program to monitor, evaluate, and manage—

- (a) The effects of plant aging on SR and NSRSS SSCs. The program may refer to surveillances, tests, and inspections conducted for specific SSCs in accordance with other requirements in this part or conducted in accordance with applicable consensus codes and standards endorsed or otherwise found acceptable by the NRC;
- (b) Cyclic or transient load limits to ensure that SR and NSRSS SSCs are maintained within the applicable design limits; and
- (c) Degradation mechanisms related to chemical interactions, operating temperatures, effects of irradiation, and other environmental factors to ensure that the capabilities, availability, and reliability of SR and NSRSS SSCs demonstrate compliance with the functional design criteria of §§ 53.410 and 53.420.

§ 53.875 Fire protection.

- (a)
 - (1) Each holder of an OL or COL under this part must have a fire protection plan that describes the overall fire protection program for the facility; identifies the various positions within the licensee's organization that are responsible for the program; states the authorities that are delegated to each of these positions to implement those responsibilities; and outlines the plans for fire protection, fire detection and suppression capability; and limitation of fire damage.
 - (2) The fire protection plan must also describe specific features necessary to implement the program described in paragraph (a)(1) of this section such as the following: administrative controls and personnel requirements for fire prevention and manual fire suppression activities; automatic and manually operated fire detection and suppression systems; and the means to limit fire damage to SSCs so that the capability to demonstrate compliance with the requirements of § 53.210 is ensured.
- (b)

- (1) Each holder of an OL or COL under this part must develop a performance-based or deterministic fire protection program that demonstrates compliance with the safety criteria outlined in §§ 53.210 and 53.220, related safety functions outlined in § 53.230, and defense in depth as outlined in § 53.250 with specific fire protection measures related to fire prevention, fire detection, and fire suppression.
- (2) The fire protection program must comply with the following:
 - (i) Safety-related and, where appropriate, NSRSS SSCs must be designed, located, and maintained to minimize, consistent with other safety requirements, the probability and effect of fires and explosions.
 - (ii) Noncombustible and fire-resistant materials must be used wherever practical throughout the facility, particularly in locations with SR and NSRSS SSCs.
 - (iii) Fire detection and fire suppression systems of appropriate capacity and capability must be provided and designed and maintained to minimize the adverse effects of fires on SR and NSRSS SSCs.
 - (iv) Fire suppression systems must be designed and maintained to ensure that their rupture or inadvertent operation does not significantly impair the ability of SR and NSRSS SSCs to perform their safety functions to satisfy § 53.230.

§ 53.880 Inservice inspection and inservice testing.

- (a) Each holder of an OL or COL under this part must develop, implement, and maintain a program for inservice inspection (ISI) and inservice testing (IST) prior to receiving an OL or COL. The ISI/IST programs must, wherever applicable, be in accordance with generally accepted consensus codes and standards that have been endorsed or otherwise found acceptable by the NRC. The ISI/IST program must include all inspections and tests required by the codes and standards used in the design and be supplemented by risk insights that identify the most important SSCs to plant safety. The types of testing and inspections and their frequency should be informed by risk insights to maintain the reliability and performance of SSCs consistent with the associated design and analyses activities involving those SSCs. Risk insights must also be used to determine when to conduct the inspections and tests (e.g., full power, shutdown, refueling) to minimize risk to the plant workers and the public. The ISI/IST program must be documented in a written manual and managed by qualified personnel reporting to the director, responsible officer, or designated person.
- (b) Prior to plant operation, baseline inspections and testing must be performed using the same techniques as will be used for future inspections and testing. The results of these inspections and testing must be used as benchmarks for evaluating the results of future inspections and testing. Sufficient room and support must be provided to accommodate the personnel, ISI/IST equipment, and shielding necessary to perform the inspections and testing. Acceptance criteria for determining whether corrective action is needed must be developed (or taken from the codes and standards used in the design) for evaluating the results of the inspections and testing. The results of the inspections and testing must be provided to the director, responsible officer, or designated person who is responsible for determining what, if any, corrective action is needed and when it should be taken. The ISI/IST results and corrective actions must be documented and the documentation retained for the life of the plant.

§ 53.910 Procedures and guidelines.

- (a) Each holder of an OL or COL under this part must have a program for developing, implementing, and maintaining an integrated set of procedures, guidelines, and related supporting activities to support normal operations and respond to possible unplanned events.
- (b) The program required by paragraph (a) of this section must include but is not limited to development, implementation, maintenance, and supporting activities of procedures and guidelines for the following:
 - (1) Plant operations;
 - (2) Maintenance activities under § 53.715;
 - (3) Program requirements under this subpart;
 - (4) Emergency operating procedures, if developed to address the role of human actions in responding to LBEs;
 - (5) Accident management guidelines, if developed to address the role of human actions in responding to LBEs;
 - (6) Procedures for each area in which licensed special nuclear material is handled, used, or stored to protect personnel upon the sounding of a criticality alarm required by § 53.440(m); and
 - (7) Procedures that describe how the licensee will address the following areas if the licensee is notified of a potential aircraft threat:
 - (i) Verification of the authenticity of threat notifications;
 - (ii) Maintenance of continuous communication with threat notification sources;
 - (iii) Contacting all onsite personnel and applicable offsite response organizations;
 - (iv) Onsite actions necessary to enhance the capability of the facility to mitigate the consequences of an aircraft impact;
 - (v) Measures to reduce visual discrimination of the site relative to its surroundings or individual buildings within the protected area;
 - (vi) Dispersal of equipment and personnel, as well as rapid entry into site protected areas for essential onsite personnel and offsite responders who are necessary to mitigate the event; and
 - (vii) Recall of site personnel.

Subpart G—Decommissioning Requirements

§ 53.1000 Scope and purpose.

This subpart defines the requirements related to decommissioning for applicants for, or holders of, an operating license (OL) or combined license (COL). The requirements related to maintaining financial assurance for decommissioning are in §§ 53.1010 through 53.1060. The requirements for transitioning from operations to decommissioning and for the release of property and termination of the license are in §§ 53.1070 through 53.1080.

§ 53.1010 Financial assurance for decommissioning.

- (a) This section establishes requirements for indicating to the U.S. Nuclear Regulatory Commission (NRC) how an applicant for or holder of an OL or COL under this part will provide reasonable assurance that funds will be available for the decommissioning process. Reasonable assurance consists of a series of steps as provided in paragraph (b) of this section and §§ 53.1020, 53.1030 and 53.1040. Funding for the decommissioning of commercial nuclear plants may also be subject to the regulation of Federal or State government agencies (e.g., Federal Energy Regulatory Commission (FERC) and State Public Utility Commissions) that have jurisdiction over rate regulation. The requirements of this subpart, in particular § 53.1020, are in addition to, and not a substitution for, other requirements, and are not intended to be used by themselves or by other agencies to establish rates.
- (b) Each applicant for an OL or COL under this part must prepare a plan and an associated decommissioning report that ensures and documents that adequate funding will be available to decommission the facility. Each holder of an OL or COL must implement and maintain the plan.
 - (1)
 - (i) Before the Commission issues an OL under this part, the applicant must update the decommissioning report to certify that it has provided financial assurance for decommissioning in the amount proposed in the application and approved by the NRC under § 53.1020.
 - (ii) No later than 30 days after the Commission issues the notice of intended operation under § 53.1452 for a COL under this part, the licensee must update the decommissioning report to certify that it has provided financial assurance for decommissioning in the amount proposed in the application and approved by the NRC under § 53.1020.
 - (2) The amount of financial assurance for decommissioning to be provided must be based on a site-specific cost estimate for decommissioning the facility under § 53.1020.
 - (3) The amount of financial assurance for decommissioning to be provided must be adjusted annually using a rate at least equal to that stated in § 53.1030.
 - (4) The amount of financial assurance for decommissioning to be provided must be covered by one or more of the methods described in § 53.1040 as acceptable to the NRC. A copy of the financial instrument obtained to satisfy the requirements of § 53.1040 must be submitted to the NRC as part of the application for an OL under this part; however, an applicant for or holder of a COL need not obtain such financial instrument or submit a copy to the Commission except as provided in § 53.1060(b).

§ 53.1020 Cost estimates for decommissioning.

Cost estimates for decommissioning must be site-specific. Site-specific decommissioning cost estimates (DCEs) must account for the engineering, labor, equipment, transportation, disposal, and related charges needed to support termination of the license. They must include the costs for decontaminating structures, systems, and components and the site environs; removal of contaminated components and materials from the plant and the site environs; disposal of removed components and materials in appropriate facilities; and any other activities supporting the release of the property and termination of the license. They must also address the approach to annual adjustments required by § 53.1030. Finally, site-specific DCEs must include plans for adjusting levels of funds assured for decommissioning to demonstrate that a reasonable level of assurance will be provided that funds will be available when needed to cover the cost of decommissioning.

§ 53.1030 Annual adjustments to cost estimates for decommissioning.

Each holder of an OL or COL under this part must annually adjust the cost estimate for decommissioning to account for escalation in labor, energy, and waste burial costs. Licensees may elect to use either a site-specific adjustment factor, approved as part of the plan and associated decommissioning report required by § 53.1010, in paragraph (a) of this section or the generic adjustment factor in paragraph (b) of this section.

- (a) A site-specific adjustment factor must address the estimated contributions and escalation of costs for the following aspects of decommissioning:
 - (1) Labor, materials, and services;
 - (2) Energy and waste transportation; and
 - (3) Radioactive waste burial or other disposition.
- (b) A generic adjustment factor must be at least equal to $0.65 L + 0.13 E + 0.22 B$, where L and E are escalation factors for labor and energy, respectively, and are to be taken from regional data of U.S. Department of Labor Bureau of Labor Statistics and B is an escalation factor for waste burial and is to be taken from NRC report NUREG-1307, "Report on Waste Burial Charges."

§ 53.1040 Methods for providing financial assurance for decommissioning.

Financial assurance for decommissioning is to be provided by the following methods.

- (a) **Prepayment.** Prepayment is the deposit made preceding the start of operation or the transfer of a license under § 53.1570 into an account segregated from licensee assets and outside the administrative control of the licensee and its subsidiaries or affiliates of cash or liquid assets such that the amount of funds would be sufficient to pay decommissioning costs. Prepayment may be in the form of a trust, escrow account, or Government fund with payment by certificate of deposit, deposit of government or other securities, or other method acceptable to the NRC. This trust, escrow account, Government fund, or other type of agreement must be established in writing and maintained at all times in the United States with an entity that is an appropriate State or Federal government agency, or an entity whose operations in which the prepayment deposit is managed are regulated and examined by a Federal or State agency. A licensee that has prepaid funds based on a site-specific cost estimate under § 53.1020 may take credit for projected earnings on the prepaid decommissioning trust funds, using up to a 2 percent annual real rate of return through the time of termination of the license. A licensee may use a credit of greater than 2 percent if the licensee's rate-setting authority has specifically authorized a higher rate. Actual earnings on existing funds may be used to calculate future fund needs.
- (b) **External sinking fund.** An external sinking fund is a fund established and maintained by setting funds aside periodically in an account segregated from licensee assets and outside the administrative control of the licensee and its subsidiaries or affiliates in which the total amount of funds would be sufficient to pay decommissioning costs. An external sinking fund may be in the form of a trust, escrow account, or Government fund, with payment by certificate of deposit, deposit of government or other securities, or other method acceptable to the NRC. This trust, escrow account, Government fund, or other type of agreement must be established in writing and maintained at all times in the United States with an entity that is an appropriate State or Federal government agency, or an entity whose operations in which the external sinking fund is managed are regulated and examined by a Federal or State agency. A licensee that has collected funds based on a site-specific cost estimate under § 53.1020 may take credit for projected earnings on the external sinking funds using up to a 2 percent annual real rate of return from the

time of future funds' collection through the time of termination of the license. A licensee may use a credit of greater than 2 percent if the licensee's rate-setting authority has specifically authorized a higher rate. Actual earnings on existing funds may be used to calculate future fund needs. A licensee whose rates for decommissioning costs cover only a portion of these costs may make use of this method only for the portion of these costs that are collected in one of the manners described in this paragraph (b). This method may be used as the exclusive mechanism relied upon for providing financial assurance for decommissioning in the following circumstances:

- (1) By a licensee that recovers, either directly or indirectly, the estimated total cost of decommissioning through rates established by "cost of service" or similar ratemaking regulation. Public utility districts, municipalities, rural electric cooperatives, and State and Federal agencies, including associations of any of the foregoing, that establish their own rates and are able to recover their cost of service allocable to decommissioning, are deemed to satisfy this condition.
- (2) By a licensee whose source of revenues for its external sinking fund is a "non-bypassable charge," the total amount of which will provide funds estimated to be needed for decommissioning pursuant to § 53.1020, § 53.1060, or § 53.1575.

(c) ***A surety method, insurance, or other guarantee method.***

- (1) These methods guarantee that decommissioning costs will be paid. A surety method may be in the form of a surety bond, or letter of credit. Any surety method or insurance used to provide financial assurance for decommissioning must contain the following conditions:
 - (i) The surety method or insurance must be open-ended, or, if written for a specified term, such as 5 years, must be renewed automatically, unless 90 days or more prior to the renewal day the issuer notifies the NRC, the beneficiary, and the licensee of its intention not to renew. The surety or insurance must also provide that the full-face amount be paid to the beneficiary automatically prior to the expiration without proof of forfeiture if the licensee fails to provide a replacement acceptable to the NRC within 30 days after receipt of notification of cancellation.
 - (ii) The surety or insurance must be payable to a trust established for decommissioning costs. The trustee and trust must be acceptable to the NRC. An acceptable trustee includes an appropriate State or Federal government agency or an entity that has the authority to act as a trustee and whose trust operations are regulated and examined by a Federal or State agency.
- (2) A parent company guarantee of funds for decommissioning costs based on a financial test may be used if the guarantee and test are as contained in appendix A to 10 CFR part 30.
- (3) For commercial companies that issue bonds, a guarantee of funds by the applicant or licensee for decommissioning costs based on a financial test may be used if the guarantee and test are as contained in appendix C to 10 CFR part 30. For commercial companies that do not issue bonds, a guarantee of funds by the applicant or licensee for decommissioning costs may be used if the guarantee and test are as contained in appendix D to 10 CFR part 30. A guarantee by the applicant or licensee may not be used in any situation in which the applicant or licensee has a parent company holding majority control of voting stock of the company.

- (d) ***Funding method for Federal licensees.*** For a Federal licensee, a statement of intent containing a cost estimate for decommissioning and indicating that funds for decommissioning will be obtained when necessary.

- (e) **Contractual funding method.** Contractual obligation(s) on the part of a licensee's customer(s), the total amount of which over the duration of the contract(s) will provide the licensee's total share of uncollected funds estimated to be needed for decommissioning pursuant to § 53.1020, § 53.1060, or § 53.1575. To be acceptable to the NRC as a method of decommissioning funding assurance, the terms of the contract(s) must include provisions that the buyer(s) of electricity or other products will pay for the decommissioning obligations specified in the contract(s), notwithstanding the operational status either of the licensed plant to which the contract(s) pertains or force majeure provisions. All proceeds from the contract(s) for decommissioning funding will be deposited to the external sinking fund. The NRC reserves the right to evaluate the terms of any contract(s) and the financial qualifications of the contracting entity or entities offered as assurance for decommissioning funding.
- (f) **Other funding mechanisms.** Any other mechanism, or combination of mechanisms, that provides, as determined by the NRC upon its evaluation of the specific circumstances of each licensee submittal, assurance of decommissioning funding equivalent to that provided by the mechanisms specified in paragraphs (a) through (e) of this section. Licensees who do not have sources of funding described in paragraph (b) of this section may use an external sinking fund in combination with a guarantee mechanism, as specified in paragraph (c) of this section, provided that the total amount of funds estimated to be necessary for decommissioning is assured.

§ 53.1045 Limitations on the use of decommissioning trust funds.

- (a)
 - (1) Decommissioning trust funds may be used by licensees if—
 - (i) The withdrawals are for expenses for decommissioning activities consistent with the definition of decommission or decommissioning in § 53.020;
 - (ii) The expenditure would not reduce the value of the decommissioning trust below an amount necessary to place and maintain the reactor in a safe storage condition if unforeseen conditions or expenses arise; and
 - (iii) The withdrawals would not inhibit the ability of the licensee to complete funding of any shortfalls in the decommissioning trust needed to ensure the availability of funds to ultimately release the site and terminate the license.
 - (2) Initially, 3 percent of the amount determined in accordance with § 53.1020 may be used for decommissioning planning. For licensees that have submitted the certifications required under § 53.1070 and commencing 90 days after the NRC has received the post-shutdown decommissioning activities report (PSDAR) required by § 53.1060, an additional 20 percent may be used. An updated site-specific DCE must be submitted to the NRC prior to the licensee using any funding in excess of these amounts.
- (b) Licensees that are not “electric utilities” as defined in § 53.020 that use prepayment or an external sinking fund to provide financial assurance must provide in the terms of the arrangements governing the trust, escrow account, or Government fund, used to segregate and manage the funds that—
 - (1) The trustee, manager, investment advisor, or other person directing investment of the funds—
 - (i) Is prohibited from investing the funds in securities or other obligations of the licensee or any other owner or operator of any commercial nuclear plant or their affiliates, subsidiaries, successors or assigns, or in a mutual fund in which at least 50 percent of the fund is invested in the securities of a licensee or parent company whose subsidiary is an owner or operator of a

foreign or domestic commercial nuclear plant. However, the funds may be invested in securities tied to market indices or other non-nuclear sector collective, commingled, or mutual funds, provided that no more than 10 percent of trust assets may be indirectly invested in securities of any entity owning or operating one or more commercial nuclear plants.

- (ii) Is obligated at all times to adhere to a standard of care set forth in the trust, which either shall be the standard of care, whether in investing or otherwise, required by State or Federal law or one or more State or Federal regulatory agencies with jurisdiction over the trust funds, or, in the absence of any such standard of care, whether in investing or otherwise, that a prudent investor would use in the same circumstances. The term "prudent investor," shall have the same meaning as set forth in FERC's "Regulations Governing Nuclear Plant Decommissioning Trust Funds" at 18 CFR 35.32(a)(3), or any successor regulation.
- (2) The licensee, its affiliates, and its subsidiaries are prohibited from being engaged as investment manager for the funds or from giving day-to-day management direction of the funds' investments or direction on individual investments by the funds, except in the case of passive fund management of trust funds where management is limited to investments tracking market indices.
 - (3) The trust, escrow account, Government fund, or other account used to segregate and manage the funds may not be amended in any material respect without written notification to the Director, Office of Nuclear Reactor Regulation, or Director, Office of Nuclear Material Safety and Safeguards, as applicable, at least 30 working days before the proposed effective date of the amendment. The licensee must provide the text of the proposed amendment and a statement of the reason for the proposed amendment. The trust, escrow account, Government fund, or other account may not be amended if the person responsible for managing the trust, escrow account, Government fund, or other account receives written notice of objection from the Director, Office of Nuclear Reactor Regulation, or Director, Office of Nuclear Material Safety and Safeguards, as applicable, within the notice period.
 - (4) Except for withdrawals being made under paragraph (a) of this section or for payments of ordinary administrative costs (including taxes) and other incidental expenses of the fund (including legal, accounting, actuarial, and trustee expenses) in connection with the operation of the fund, no disbursement or payment may be made from the trust, escrow account, Government fund, or other account used to segregate and manage the funds until written notice of the intention to make a disbursement or payment has been given to the Director, Office of Nuclear Reactor Regulation, or Director, Office of Nuclear Material Safety and Safeguards, as applicable, at least 30 working days before the date of the intended disbursement or payment. The disbursement or payment from the trust, escrow account, Government fund or other account may be made following the 30 working day notice period if the person responsible for managing the trust, escrow account, Government fund, or other account does not receive written notice of objection from the Director, Office of Nuclear Reactor Regulation, or Director, Office of Nuclear Material Safety and Safeguards, as applicable, within the notice period. Disbursements or payments from the trust, escrow account, Government fund, or other account used to segregate and manage the funds, other than for payment of ordinary administrative costs (including taxes) and other incidental expenses of the fund (including legal, accounting, actuarial, and trustee expenses) in connection with the operation of the fund, are restricted to decommissioning expenses or transfer to another financial assurance method acceptable under § 53.1040 until final decommissioning has been completed. After decommissioning has begun and withdrawals from the decommissioning fund are made under paragraph (a) of this section, no further notification need be made to the NRC.

- (c) Licensees that are “electric utilities” under § 53.020 that use prepayment or an external sinking fund to provide financial assurance must include a provision in the terms of the trust, escrow account, Government fund, or other account used to segregate and manage funds that except for withdrawals being made under paragraph (a) of this section or for payments of ordinary administrative costs (including taxes) and other incidental expenses of the fund (including legal, accounting, actuarial, and trustee expenses) in connection with the operation of the fund, no disbursement or payment may be made from the trust, escrow account, Government fund, or other account used to segregate and manage the funds until written notice of the intention to make a disbursement or payment has been given the Director, Office of Nuclear Reactor Regulation, or Director, Office of Nuclear Material Safety and Safeguards, as applicable, at least 30 working days before the date of the intended disbursement or payment. The disbursement or payment from the trust, escrow account, Government fund or other account may be made following the 30 working day notice period if the person responsible for managing the trust, escrow account, Government fund, or other account does not receive written notice of objection from the Director, Office of Nuclear Reactor Regulation, or Director, Office of Nuclear Material Safety and Safeguards, as applicable, within the notice period. Disbursements or payments from the trust, escrow account, Government fund, or other account used to segregate and manage the funds, other than for payment of ordinary administrative costs (including taxes) and other incidental expenses of the fund (including legal, accounting, actuarial, and trustee expenses) in connection with the operation of the fund, are restricted to decommissioning expenses or transfer to another financial assurance method acceptable under § 53.1040 until final decommissioning has been completed. After decommissioning has begun and withdrawals from the decommissioning fund are made under paragraph (a) of this section, no further notification need be made to the NRC.
- (d) A licensee that is not an “electric utility” under § 53.020 and using a surety method, insurance, or other guarantee method to provide financial assurance must provide that the trust established for decommissioning costs to which the surety or insurance is payable contains in its terms the requirements in § 53.1045(b)(1) through (b)(4).

§ 53.1050 NRC oversight.

The NRC reserves the right to take the following steps in order to ensure a licensee's adequate accumulation of decommissioning funds: review, as needed, the rate of accumulation of decommissioning funds and, either independently or in cooperation with FERC and the licensee's State Public Utility Commission, take additional actions as appropriate on a case-by-case basis, including modification of a licensee's schedule for the accumulation of decommissioning funds.

§ 53.1060 Reporting and recordkeeping requirements.

- (a) Each holder of an OL under this part or holder of a COL under this part after the date that the Commission has made the finding under § 53.1452(g) must report, at least once every 2 years, by March 31, on the status of its certification of decommissioning funding for each commercial nuclear reactor or part of a commercial nuclear reactor that it owns. The information in this report must include, at a minimum, the amount of decommissioning funds estimated to be required under §§ 53.1020 and 53.1030; the amount of decommissioning funds accumulated to the end of the calendar year preceding the date of the report; a schedule of the annual amounts remaining to be collected; the assumptions used regarding rates of escalation in decommissioning costs, rates of earnings on decommissioning funds, and rates of other factors used in funding projections; any contracts upon which the licensee is relying under § 53.1040(e); any modifications occurring to a licensee's method of providing financial assurance since the last submitted report; and any material changes to trust agreements. If any of the preceding items is not

applicable, the licensee should so state in its report. Any licensee for a plant that is within 5 years of the projected end of its operation, or where conditions have changed such that it will close within 5 years (before the end of its licensed life), or that has already closed (before the end of its licensed life), or that is involved in a merger or an acquisition must submit this report annually.

- (b) Each holder of a COL under this part must, 2 years before and 1 year before the scheduled date for initial loading of fuel (or, for a fueled manufactured reactor, 2 years before and 1 year before the scheduled date for initiating the removal of the features to prevent criticality required under § 53.620(d)(1)) submit a report to the NRC containing a certification updating the DCEs and a copy of the financial instrument to be used to satisfy § 53.1040. No later than 30 days after the Commission publishes notice in the FEDERAL REGISTER under § 53.1452(a), the licensee must submit an updated decommissioning report required under § 53.1010(b)(1)(ii), including a copy of the financial instrument obtained to satisfy § 53.1040.
- (c) Each licensee must keep records of information important to the safe and effective decommissioning of the facility in an identified location until the license is terminated by the Commission. If records of relevant information are kept for other purposes, reference to these records and their locations may be used. Information the Commission considers important to decommissioning consists of—
 - (1) Records of spills or other unusual occurrences involving the spread of contamination in and around the facility, equipment, or site. These records may be limited to instances when significant contamination remains after any cleanup procedures or when there is reasonable likelihood that contaminants may have spread to inaccessible areas as in the case of possible seepage into porous materials such as concrete. These records must include any known information on identification of involved nuclides, quantities, forms, and concentrations.
 - (2) As-built drawings and modifications of structures and equipment in restricted areas where radioactive materials are used and/or stored and of locations of possible inaccessible contamination such as buried pipes that may be subject to contamination. If required drawings are referenced, each relevant document need not be indexed individually. If drawings are not available, the licensee must substitute appropriate records of available information concerning these areas and locations.
 - (3) Records of the cost estimate performed for the decommissioning funding plan or of the amount certified for decommissioning, and records of the funding method used for assuring funds if either a funding plan or certification is used.
 - (4) Records of—
 - (i) The licensed site area, as originally licensed and any revisions, which must include a site map and any acquisition or use of property outside the originally licensed site area for the purpose of receiving, possessing, or using licensed materials;
 - (ii) The licensed activities carried out on the acquired or used property; and
 - (iii) The release and final disposition of any property recorded in paragraph (c)(4)(i) of this section, the historical site assessment performed for the release, radiation surveys performed to support release of the property, submittals to the NRC made under § 53.1070, and the methods employed to ensure that the property met the radiological criteria of subpart E of 10 CFR part 20 at the time the property was released.

- (d) Each holder of an OL or COL under this part must at or about 5 years prior to the projected end of operations submit a preliminary DCE which includes an up-to-date assessment of the major factors that could affect the cost to decommission.
- (e) Prior to or within 2 years following permanent cessation of operations, the licensee must submit a PSDAR to the NRC, and a copy to the affected State(s). The PSDAR must contain a description of the planned decommissioning activities along with a schedule for their accomplishment, a discussion that provides the reasons for concluding that the environmental impacts associated with site-specific decommissioning activities will be bounded by appropriate previously issued environmental impact statements, and a site-specific DCE, including the projected cost of managing irradiated fuel.
- (f) For decommissioning activities that delay completion of decommissioning by including a period of storage or surveillance, the licensee must provide a means of adjusting cost estimates and associated funding levels over the storage or surveillance period.
- (g) After submitting its site-specific DCE required by paragraph (e) of this section, and until the licensee has completed its final radiation survey and demonstrated that residual radioactivity has been reduced to a level that permits termination of its license, the licensee must annually submit to the NRC, by March 31, a financial assurance status report. The report must include the following information, current through the end of the previous calendar year:
 - (1) The amount spent on decommissioning, both cumulative and over the previous calendar year, the remaining balance of any decommissioning funds, and the amount provided by other financial assurance methods being relied upon;
 - (2) An estimate of the costs to complete decommissioning, reflecting any difference between actual and estimated costs for work performed during the year, and the decommissioning criteria upon which the estimate is based;
 - (3) Any modifications occurring to a licensee's current method of providing financial assurance since the last submitted report; and
 - (4) Any material changes to trust agreements or financial assurance contracts.
 - (5) If the sum of the balance of any remaining decommissioning funds, plus earnings on such funds calculated at not greater than a 2 percent real rate of return, together with the amount provided by other financial assurance methods being relied upon, does not cover the estimated cost to complete the decommissioning, the financial assurance status report must include additional financial assurance to cover the estimated cost of completion.
- (h) After submitting its site-specific DCE required by paragraph (e) of this section, the licensee must annually submit to the NRC, by March 31, a report on the status of its funding for managing irradiated fuel. The report must include the following information, current through the end of the previous calendar year:
 - (1) The amount of funds accumulated to cover the cost of managing the irradiated fuel;
 - (2) The projected cost of managing irradiated fuel until title to the fuel and possession of the fuel is transferred to the Secretary of Energy; and
 - (3) If the funds accumulated do not cover the projected cost, a plan to obtain additional funds to cover the cost.

§ 53.1070 Termination of license.

For each holder of an OL or COL under this part—

(a)

- (1) When the licensee has determined to permanently cease operations the licensee must, within 30 days, submit a written certification to the NRC, consistent with the requirements of § 53.040(b)(8);
- (2) When appropriate to support decommissioning activities and the eventual permanent removal of fuel from the reactor vessel, the licensee must develop defueled technical specifications by reviewing the operational technical specifications and determining which specifications no longer apply during decommissioning and which ones should remain applicable. The licensee must make the appropriate submittals to the NRC in accordance with § 53.1510 to request changes to the technical specifications; and

(3)

- (i) Once fuel has been permanently removed from the reactor vessel, the licensee must submit a written certification to the NRC that meets the requirements of § 53.040(b)(9); and
- (ii) The licensee must establish and maintain staffing consisting of certified fuel handlers, as defined under § 53.020, and other non-licensed personnel with appropriate qualifications, and in sufficient numbers, to ensure support for facility operations and radiological control activities, as required by the facility defueled technical specifications. These personnel must be subject to the training requirements of § 53.830.

(b) Upon docketing of the certifications for permanent cessation of operations and permanent removal of fuel from the reactor vessel, or when a final legally effective order to permanently cease operations has come into effect, the license issued under this part no longer authorizes operation of the reactor or emplacement or retention of fuel into the reactor vessel.

(c) Decommissioning will be completed within 60 years of permanent cessation of operations. Completion of decommissioning beyond 60 years will be approved by the Commission only when necessary to protect public health and safety. Factors that will be considered by the Commission in evaluating an alternative that provides for completion of decommissioning beyond 60 years of permanent cessation of operations include unavailability of waste disposal capacity and other site-specific factors affecting the licensee's capability to carry out decommissioning, including presence of other nuclear facilities at the site.

(d)

- (1) Prior to or within 2 years following permanent cessation of operations, the licensee must submit a PSDAR and site-specific DCE in accordance with § 53.1060(e).
- (2) The NRC must notice receipt of the PSDAR and make the PSDAR publicly available and publish notice of its availability for public comment in the FEDERAL REGISTER. The NRC must also schedule a public meeting readily accessible to individuals in the vicinity of the licensee's facility. The NRC must publish a notice in the FEDERAL REGISTER and in a forum, such as local newspapers, that is readily accessible to individuals in the vicinity of the site, announcing the date, time, and location of the meeting, along with a brief description of the purpose of the meeting.

- (e) Licensees must not perform any major decommissioning activities, as defined in § 53.020, until 90 days after the NRC has received the licensee's PSDAR submittal and until certifications of permanent cessation of operations and permanent removal of fuel from the reactor vessel, as required under paragraph (a) of this section, have been submitted.
- (f) Licensees must not perform any decommissioning activities, as defined in § 53.020, that—
 - (1) Foreclose release of the site for possible unrestricted use;
 - (2) Result in significant environmental impacts not previously reviewed; or
 - (3) Result in there no longer being reasonable assurance that adequate funds will be available for decommissioning.
- (g) In taking actions permitted under § 53.1540 following submittal of the PSDAR, the licensee must notify the NRC in writing, and send a copy to the affected State(s), before performing any decommissioning activity inconsistent with, or making any significant schedule change from, those actions and schedules described in the PSDAR, including changes that increase the decommissioning cost by more than 20 percent from the previously provided DCE.
- (h) Licensees may use decommissioning trust funds consistent with the limitations of § 53.1045(a). Licensees must report on the status of decommissioning trust funds consistent with the requirements of § 53.1060.
- (i) Licensees must submit an application for termination of license in accordance with § 53.1070. The application for termination of license must be accompanied or preceded by a license termination plan to be submitted for NRC approval.
 - (1) The license termination plan must be a supplement to the Final Safety Analysis Report or equivalent and must be submitted at least 2 years before termination of the license date.
 - (2) The license termination plan must include—
 - (i) A site characterization;
 - (ii) Identification of remaining dismantlement activities;
 - (iii) Plans for site remediation;
 - (iv) Detailed plans for the final radiation survey;
 - (v) A description of the end use of the site, if restricted;
 - (vi) An updated site-specific estimate of remaining decommissioning costs;
 - (vii) A supplement to the environmental report, pursuant to § 51.53 of this chapter, describing any new information or significant environmental change associated with the licensee's proposed termination activities; and
 - (viii) Identification of parts, if any, of the facility or site that were released for use before approval of the license termination plan.
 - (3) Following receipt of the license termination plan, the NRC must make the license termination plan publicly available and publish notice of its availability for public comment in the FEDERAL REGISTER. The NRC must also schedule a public meeting readily accessible to individuals in the vicinity of the licensee's facility upon receipt of the license termination plan. The NRC must publish a notice in the

FEDERAL REGISTER and in a forum, such as local newspapers, that is readily accessible to individuals in the vicinity of the site, announcing the date, time, and location of the meeting, along with a brief description of the purpose of the meeting.

- (j) If the license termination plan demonstrates that the remainder of decommissioning activities will be performed in accordance with the regulations in this chapter, will not be inimical to the common defense and security or to the health and safety of the public, and will not have a significant effect on the quality of the environment and after notice to interested persons, the Commission will approve the plan, by license amendment, subject to such conditions and limitations as it deems appropriate and necessary and authorize implementation of the license termination plan.
- (k) The Commission will terminate the license if it determines that—
 - (1) The remaining dismantlement has been performed in accordance with the approved license termination plan; and
 - (2) The final radiation survey and associated documentation, including an assessment of dose contributions associated with parts released for use before approval of the license termination plan, demonstrate that the facility and site have met the criteria for decommissioning in subpart E of 10 CFR part 20.

§ 53.1075 Program requirements during decommissioning.

- (a) Licensees that have submitted the certifications required under § 53.1070 must maintain a decommissioning fire protection program to address the potential for fires that could cause the release or spread of radioactive materials.
 - (1) The objectives of the decommissioning fire protection program are to
 - (i) Reasonably prevent these fires from occurring;
 - (ii) Rapidly detect, control, and extinguish those fires that do occur and that could result in a radiological hazard; and
 - (iii) Ensure that the risk of fire-induced radiological hazards to the public, environment, and plant personnel is minimized.
 - (2) The licensee must assess the decommissioning fire protection program on a regular basis. The licensee must revise the decommissioning fire protection program documentation as appropriate throughout the various stages of facility decommissioning.
 - (3) The licensee may make changes to the decommissioning fire protection program without NRC approval if these changes do not reduce the effectiveness of fire protection for structures, systems, and components that could result in a radiological hazard, taking into account the decommissioning plant conditions and activities.
- (b) [Reserved]

§ 53.1080 Release of part of a commercial nuclear plant or site for unrestricted use.

- (a) Prior written NRC approval is required to release part of a commercial nuclear plant or site for unrestricted use at any time before receiving approval of a license termination plan. Section 53.1060 specifies recordkeeping requirements associated with partial release. Holders of an OL or COL under this part seeking NRC review and approval must—

- (1) Evaluate the effect of releasing the property to ensure that—
 - (i) The dose to individual members of the public does not exceed the limits and standards of subpart D of 10 CFR part 20;
 - (ii) There is no reduction in the effectiveness of emergency planning or physical security;
 - (iii) Effluent releases remain within license conditions;
 - (iv) The environmental monitoring program and offsite dose calculation manual are revised to account for the changes;
 - (v) The siting criteria of subpart D of this part continue to be met; and
 - (vi) All other applicable statutory and regulatory requirements continue to be met.
 - (2) Perform a historical site assessment of the part of the commercial nuclear plant or site to be released; and
 - (3) Perform surveys adequate to demonstrate compliance with the radiological criteria for unrestricted use specified in § 20.1402 of this chapter for impacted areas.
- (b) For release of non-impacted areas, the licensee may submit a written request for NRC review and approval of the release if a license amendment is not otherwise required. The request submittal must include—
- (1) The results of the evaluations performed in accordance with paragraphs (a)(1) and (a)(2) of this section;
 - (2) A description of the part of the commercial nuclear plant or site to be released;
 - (3) The schedule for release of the property;
 - (4) The results of the evaluations performed in accordance with § 53.1540; and
 - (5) A discussion that provides the reasons for concluding that the environmental impacts associated with the licensee's proposed release of the property will be bounded by appropriate previously issued environmental impact statements.
- (c) After receiving a request from the licensee for NRC approval of the release of a non-impacted area, the NRC must—
- (1) Determine whether the licensee has adequately evaluated the effect of releasing the property as required by paragraph (a)(1) of this section;
 - (2) Determine whether the licensee's classification of any release areas as non-impacted is adequately justified; and
 - (3) If determining that the licensee's submittal is adequate, inform the licensee in writing that the release is approved.
- (d) For release of impacted areas, the licensee must submit an application for amendment of its license for the release of the property. The application must include—
- (1) The information specified in paragraphs (b)(1) through (b)(3) of this section;
 - (2) The methods used for and results obtained from the radiation surveys required to demonstrate compliance with the radiological criteria for unrestricted use specified in § 20.1402; and

- (3) A supplement to the environmental report, under § 51.53 of this chapter.
- (e) After receiving a license amendment application from the licensee for the release of an impacted area, the NRC must—
 - (1) Determine whether the licensee has adequately evaluated the effect of releasing the property as required by paragraph (a)(1) of this section;
 - (2) Determine whether the licensee's classification of any release areas as non-impacted is adequately justified;
 - (3) Determine whether the licensee's radiation survey for an impacted area is adequate; and
 - (4) If determining that the licensee's submittal is adequate, approve the licensee's amendment application.
- (f) The NRC must publish notice receipt of the release approval request or license amendment application in the FEDERAL REGISTER and make the approval request or license amendment application available for public comment. Before acting on an approval request or license amendment application submitted in accordance with this section, the NRC must conduct a public meeting readily accessible to individuals in the vicinity of the licensee's facility for the purpose of obtaining public comments on the proposed release of part of the commercial nuclear plant or site. The NRC must publish a document in the FEDERAL REGISTER and in a forum, such as local newspapers, which is readily accessible to individuals in the vicinity of the site, announcing the date, time, and location of the meeting, along with a brief description of the purpose of the meeting.

Subpart H—Licenses, Certifications, and Approvals

§ 53.1100 Filing of application for licenses, certifications, or approvals; oath or affirmation.

- (a) *Serving of applications.*
 - (1) Each filing of an application for a standard design approval, standard design certification, or license under this part, and any amendments to the applications, must be submitted to the U.S. Nuclear Regulatory Commission (NRC) under § 53.040, as applicable.
 - (i) Any person, except one excluded by § 53.1118, may file an application for a manufacturing license (ML), combined license (COL), construction permit (CP), or operating license (OL) under this part with the Director, Office of Nuclear Reactor Regulation.
 - (ii) Any person who may apply for a CP or for a COL under this part, may file an application for an early site permit (ESP) with the Director, Office of Nuclear Reactor Regulation. An application for an early site permit may be filed notwithstanding the fact that an application for a CP or a COL has not been filed in connection with the site for which a permit is sought.
 - (iii) Any person may submit a proposed standard design for a commercial nuclear plant to the NRC for its review. The submittal may consist of either the final design for the entire facility or the final design for major portions thereof.
 - (iv) An application for design certification may be filed notwithstanding the fact that an application for a CP, COL, or ML for such a facility has not been filed. The application must comply with §§ 2.811 through 2.819 of this chapter.

- (2) The applicant must make a copy of the updated application available at the public hearing for the use of any other parties to the proceeding and must certify that the updated copies of the application contain the current contents of the application submitted in accordance with the requirements under this part.
- (3) At the time of filing an application, the Commission will make available at the NRC website, <https://www.nrc.gov>, a copy of the application, subsequent amendments, and other records pertinent to the matter that is the subject of the application for public inspection and copying.
- (4) The serving of copies required by this section must not occur until the application has been docketed under § 2.101(a) of this chapter. Copies must be submitted to the Commission, as specified in § 53.040, as applicable, to enable the Director, Office of Nuclear Reactor Regulation to determine whether the application is sufficiently complete to permit docketing.
- (b) **Oath or affirmation.** Each application for a standard design approval, standard design certification, or license, including, whenever appropriate, a CP or early site permit, or amendment of it, and each amendment of each application must be executed in a signed original by the applicant or duly authorized officer thereof under oath or affirmation.
- (c)-(d) [Reserved]
- (e) **Filing fees.** Each application for a standard design approval, standard design certification, or commercial nuclear plant license under this part, including, whenever appropriate, a CP, COL, operating license (OL), ML, or early site permit, other than a license exempted from 10 CFR part 170, must be accompanied by the fee prescribed in 10 CFR part 170. No fee will be required to accompany an application for renewal, amendment, or termination of a CP, OL, COL, or ML, except as provided in § 170.21 of this chapter.
- (f) **Environmental report.** An application for a CP, OL, early site permit, design certification, COL, or ML for a commercial nuclear plant must be accompanied by an environmental report required under 10 CFR part 51.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20067, Apr. 15, 2026]

§ 53.1101 Requirement for license.

Except as provided in § 53.1120, no person within the United States may transfer or receive in interstate commerce, manufacture, produce, transfer, acquire, possess, or use any utilization facility except as authorized by a license issued by the Commission.

§ 53.1103 Combining applications and licenses.

- (a) An applicant may combine several applications in one application for different kinds of licenses under the regulations in this chapter.
- (b) The Commission may combine in a single license the activities of an applicant which would otherwise be licensed separately.

§ 53.1106 Elimination of repetition.

An applicant may incorporate by reference in its application information contained in previous applications, statements, or reports filed with the Commission, provided, however, that such references are clear and specific.

§ 53.1109 Contents of applications; general information.

Each application must include, unless otherwise indicated in this subpart—

- (a) Name of applicant;
- (b) Address of applicant;
- (c) Description of business or occupation of applicant;
- (d)
 - (1) If applicant is an individual, the citizenship of applicant;
 - (2) If applicant is a partnership, the name, citizenship and address of each partner and the principal location where the partnership does business;
 - (3) If applicant is a corporation or an unincorporated association, the following information:
 - (i) The State where it is incorporated or organized and the principal location where it does business;
 - (ii) The names, addresses and citizenship of its directors and of its principal officers; and
 - (iii) Whether it is owned, controlled, or dominated by an alien, a foreign corporation, or foreign government, and if so, give details; or
 - (4) If the applicant is acting as agent or representative of another person in filing the application, identify the principal and furnish information required under this paragraph (d) with respect to such principal;
- (e) The class and type of license applied for, the use to which the facility will be put, the period of time for which the license is sought, and a list of other licenses, except operator's licenses, issued or applied for in connection with the proposed facility;
- (f) [Reserved]
- (g)
 - (1) Except as provided in paragraph (g)(2) of this section, if the application is for an OL or COL for a commercial nuclear plant, or if the application is for an early site permit for a commercial nuclear plant and contains plans for coping with emergencies under § 53.1146(b)(2)(ii), the applicant must submit the radiological emergency response plans of State, local, and participating Tribal governmental entities in the United States that are wholly or partially within the plume exposure pathway emergency planning zone (EPZ),^[1] and the plans of State governments wholly or partially within the ingestion pathway EPZ.^[2] If the application is for an early site permit that, under § 53.1146(b)(2)(i), proposes major features of the emergency plans describing the EPZs, then the descriptions of the EPZs must meet the requirements of this paragraph (g)(1). Generally, the plume exposure pathway EPZ for a commercial nuclear plant must consist of an area about 10 miles (16 km) in radius and the ingestion pathway EPZ must consist of an area about 50 miles (80 km) in radius. The exact size and configuration of the EPZs surrounding a particular commercial nuclear plant must be determined in relation to the local emergency response needs and capabilities as they are affected by such conditions as demography, topography, land characteristics, access routes, and jurisdictional boundaries. The size of the EPZs also may be determined on a case-by-case basis for

gas-cooled reactors and for reactors with an authorized power level less than 250 megawatt thermal. The plans for the ingestion pathway must focus on such actions as are appropriate to protect the food ingestion pathway.

- (2) Applicants for commercial nuclear plants consisting of either small modular reactors or non-light-water reactors complying with § 50.160 of this chapter who apply for a CP, an OL, a COL, or an early site permit under this part must submit as part of the application the analysis used to determine whether the criteria in § 53.1109(g)(2)(i)(A) and (B) are met and, if they are met, the size of the plume exposure pathway EPZ.
 - (i) The plume exposure pathway EPZ is the area within which:
 - (A) Public dose, as defined in § 20.1003 of this chapter, is projected to exceed 10 millisieverts (1 rem) total effective dose equivalent over 96 hours from the release of radioactive materials from the facility considering accident likelihood and source term, timing of the accident sequence, and meteorology; and
 - (B) Pre-determined, prompt protective measures are necessary.
 - (ii) If the application is for an OL or COL or if the application is for an early site permit and contains plans for coping with emergencies under § 53.1146(b)(2)(ii), and if the plume exposure pathway EPZ extends beyond the site boundary:
 - (A) The applicant must submit radiological emergency response plans of State, local, and participating Tribal governmental entities in the United States that are wholly or partially within the plume exposure pathway EPZ.
 - (B) The exact configuration of the plume exposure pathway EPZ surrounding the facility shall be determined in relation to the local emergency response needs and capabilities as they are affected by such conditions as demography, topography, land characteristics, access routes, and jurisdictional boundaries.
 - (iii) If the application is for an early site permit that, under § 53.1146(b)(2)(i), proposes major features of the emergency plans and describes the EPZ, and if the EPZ extends beyond the site boundary, then the exact configuration of the plume exposure pathway EPZ surrounding the facility must be determined in relation to the local emergency response needs and capabilities as they are affected by such conditions as demography, topography, land characteristics, access routes, and jurisdictional boundaries.
- (h) [Reserved]

^[2] If the State, local, and participating Tribal emergency response plans have been previously provided to the NRC for inclusion in the facility docket, the applicant need only provide the appropriate reference to meet this requirement.

^[1] EPZs are discussed in NUREG-0396, U.S. Environmental Protection Agency 520/1-78-016, "Planning Basis for the Development of State and Local Government Radiological Emergency Response Plans in Support of Light-Water Nuclear Power Plants," December 1978.

- (i) A list of the names and addresses of such regulatory agencies as may have jurisdiction over the rates and services incident to the proposed activity, and a list of trade and news publications which circulate in the area where the proposed activity will be conducted and which are considered appropriate to give reasonable notice of the application to those municipalities, private utilities, public bodies, and cooperatives, which might have a potential interest in the facility; and
- (j) If the application contains Restricted Data or classified National Security information, confirmation that all Restricted Data and classified National Security information are separated from the unclassified information.

§ 53.1112 Environmental conditions.

- (a) Each CP, early site permit, and COL under this part may include conditions to address environmental issues during construction. These conditions are to be set out in an attachment to the license, which is incorporated in and made a part of the license. These conditions will be derived from information contained in the environmental report submitted pursuant to § 51.50 of this chapter, as analyzed and evaluated in the NRC record of decision and will identify the obligations of the licensee in the environmental area, including, as appropriate, requirements for reporting and keeping records of environmental data, and any conditions and monitoring requirement for the protection of the nonaquatic environment.
- (b) Each license authorizing operation of a commercial nuclear plant under this part, and each license for a commercial nuclear plant for which the certification of permanent cessation of operations required under § 53.1070 has been submitted may include conditions to address environmental issues during operation and decommissioning. These conditions are to be set out in an attachment to the license, which is incorporated in and made a part of the license. These conditions will be derived from information contained in the environmental report or the supplement to the environmental report submitted under §§ 51.50 and 51.53 of this chapter as analyzed and evaluated in the NRC record of decision, and will identify the obligations of the licensee in the environmental area, including, as appropriate, requirements for reporting and keeping records of environmental data and any conditions and monitoring requirement for the protection of the nonaquatic environment.

§ 53.1115 Agreement limiting access to classified information.

As part of its application and in any event before the receipt of Restricted Data or classified National Security Information or the issuance of a license or standard design approval under this part, or before the Commission has adopted a final standard design certification rule under this part, the applicant must agree in writing that it will not permit any individual to have access to or any facility to possess Restricted Data or classified National Security Information until the individual and/or facility has been approved for access under the provisions of 10 CFR parts 25 and/or 95. The agreement of the applicant becomes part of the license or standard design approval.

§ 53.1118 Ineligibility of certain applicants.

Any person who is a citizen, national, or agent of a foreign country, or any corporation, or other entity which the Commission knows or has reason to believe is owned, controlled, or dominated by an alien, a foreign corporation, or a foreign government, will be ineligible to apply for and obtain a license unless—

- (a) The Commission determines that issuance of the applicable license to the entity is not inimical to the common defense and security or the health and safety of the public; and

- (b) The entity is an alien, corporation, or other entity that is owned, controlled, or dominated by the government of, a corporation that is incorporated in, or an alien who is a citizen or national of Australia, Austria, Belgium, Canada, Chile, Colombia, Costa Rica, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, India, Ireland, Israel, Italy, Japan, Korea, Latvia, Lithuania, Luxembourg, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, or the United Kingdom.

§ 53.1120 Exceptions and exemptions from licensing requirements.

Nothing in this part must be deemed to require a license for—

- (a) The manufacture, production, or acquisition by the Department of Defense of any utilization facility authorized pursuant to section 91 of the Act or the use of such facility by the Department of Defense or by a person under contract with and for the account of the Department of Defense;
- (b) Except to the extent that the Department of Energy facilities of the types subject to licensing pursuant to section 202 of the Energy Reorganization Act of 1974, as amended, are involved—
 - (1)
 - (i) The processing, fabrication or refining of special nuclear material (SNM) or the separation of SNM, or the separation of SNM from other substances by a prime contractor of the Department of Energy under a prime contract for—
 - (A) The performance of work for the Department of Energy at a United States Government-owned or controlled site;
 - (B) Research in, or development, manufacture, storage, testing or transportation of, atomic weapons or components thereof; or
 - (C) The use or operation of a utilization facility in a United States owned vehicle or vessel; or
 - (ii) The processing, fabrication or refining of SNM of the separation of SNM, or the separation of SNM from other substances by a prime contractor or subcontractor of the Commission or the Department of Energy under a prime contract or subcontract when the Commission determines that the exemption of the prime contractor or subcontractor is authorized by law; and that, under the terms of the contract or subcontract, there is adequate assurance that the work thereunder can be accomplished without undue risk to the public health and safety; or
 - (2)
 - (i) The construction or operation of a utilization facility for the Department of Energy at a United States Government-owned or controlled site, including the transportation of the utilization facility to or from such site and the performance of contract services during temporary interruptions of such transportation; or the construction or operation of a utilization facility for the Department of Energy in the performance of research in, or development, manufacture, storage, testing, or transportation of, atomic weapons or components thereof; or the use or operation of a utilization facility for the Department of Energy in a United States Government-owned vehicle or vessel; provided that such activities are conducted by a prime contractor of the Department of Energy under a prime contract with the Department of Energy; or

- (ii) The construction or operation of a utilization facility by a prime contractor or subcontractor of the Commission or the Department of Energy under his or her prime contract or subcontract when the Commission determines that the exemption of the prime contractor or subcontractor is authorized by law; and that, under the terms of the contract or subcontract, there is adequate assurance that the work thereunder can be accomplished without undue risk to the public health and safety; or
- (c) The transportation or possession of any utilization facility by a common or contract carrier or warehouse employee in the regular course of carriage for another or storage incident thereto.

§ 53.1121 Public inspection of applications.

Applications and documents submitted to the Commission in connection with applications may be made available for public inspection under the provisions of part 2 of this chapter.

§ 53.1124 Relationship between sections.

- (a) **Limited work authorization.** An application for a limited work authorization (LWA) under this part may be submitted as part of an application for an early site permit, CP, or COL under this part as required in § 53.1130(a)(2).
- (b) **Early site permit.**
 - (1) A holder of an early site permit may request an LWA.
 - (2) An application for a CP or COL under this part may, but need not, reference an early site permit.
- (c) **Standard design approval.** An application for a standard design approval under this part may, but need not, reference an OL or custom COL under this part that is essentially the same as the information supporting the standard design for which approval is being requested.
- (d) **Standard design certification.** An application for a standard design certification under this part may, but need not, reference an OL or custom COL under this part that is essentially the same as the standard design for which certification is being requested.
- (e) **Manufacturing license.**
 - (1) A manufactured reactor or portions thereof as defined in an ML issued under this part may be either transported to and installed at a site for which a COL or CP under this part has been issued or exported in accordance with part 110.
 - (2) An ML applicant under this part may reference a standard design certification or a standard design approval under this part in its application.
- (f) **Construction permit.** An application for a CP may, but need not, reference a standard design certification, standard design approval, or ML issued under this part, respectively, and may also reference an early site permit issued under this part. In the absence of a demonstration that an entity other than the one originally sponsoring a standard design certification is qualified to supply a design, the Commission will entertain an application for a CP that references a standard design certification issued under this part only if the entity that sponsored the certification supplies the design for the applicant's use.
- (g) **Operating license.**

- (1) An application for an OL under this part may, but need not, reference an early site permit, standard design certification, or standard design approval issued under this part. In the absence of a demonstration that an entity other than the one originally sponsoring a standard design certification is qualified to supply a design, the Commission will entertain an application for an OL that references a standard design certification issued under this part only if the entity that sponsored the certification supplies the design for the applicant's use.
- (2) The holder of a CP must, at the time of submission of the Final Safety Analysis Report (FSAR), file an application for an OL.
- (h) **Combined licenses.** An application for a COL under this part may, but need not, reference an early site permit, standard design certification, standard design approval, or ML issued under this part. In the absence of a demonstration that an entity other than the one originally sponsoring and obtaining a standard design certification is qualified to supply a design, the Commission will entertain an application for a COL that references a standard design certification issued under this part only if the entity that sponsored the certification supplies the design for the applicant's use.

§ 53.1130 Limited work authorizations.

- (a) **Request for limited work authorization.**
 - (1) Any person to whom the Commission may otherwise issue either a license or permit related to a commercial nuclear plant may request an LWA allowing that person to perform the driving of piles, subsurface preparation, placement of backfill, concrete, or permanent retaining walls within an excavation, and installation of the foundation, including placement of concrete, any of which are for a structure, system, or component (SSC) of the facility for which either a CP or COL is otherwise required under § 53.610.
 - (2) An application for an LWA may be submitted as part of a complete application for a CP or COL in accordance with § 2.101(a)(1) through (a)(5) of this chapter, or as a partial application in accordance with § 2.101(a)(9) of this chapter. An application for an LWA by the holder of an early site permit must be submitted as a complete application in accordance with § 2.101(a)(1) through (a)(4) of this chapter.
 - (3) The application must include—
 - (i) A Safety Analysis Report required by § 53.1146, § 53.1309 or § 53.1416, as applicable, a description of the activities requested to be performed, and the design and construction information otherwise required by the Commission's rules and regulations to be submitted for a CP or COL under this part but limited to those portions of the facility that are within the scope of the LWA. The Safety Analysis Report must demonstrate that activities conducted under the LWA will be conducted in compliance with the technically relevant Commission requirements in 10 CFR chapter I applicable to the design of those portions of the facility within the scope of the LWA;
 - (ii) An environmental report in accordance with § 51.49 of this chapter; and
 - (iii) A plan for redress of activities performed under the LWA, should limited work activities be terminated by the holder, or the LWA be revoked by the NRC or upon effectiveness of the Commission's final decision denying the associated CP or COL application, as applicable.
- (b) **Issuance of limited work authorization.**

- (1) The Director, Office of Nuclear Reactor Regulation may issue an LWA only after—
 - (i) The NRC staff issues the final environmental impact statement for the LWA under part 51 of this chapter;
 - (ii) The Director determines that the applicable standards and requirements of the Act, and the Commission's regulations applicable to the activities to be conducted under the LWA, have been met, the applicant is technically qualified to engage in the activities authorized, and that issuance of the LWA will provide reasonable assurance of adequate protection to public health and safety and will not be inimical to the common defense and security; and
 - (iii) If a contested hearing is held, the presiding officer finds that there are no unresolved safety issues relating to the activities to be conducted under the LWA that would constitute good cause for withholding the authorization.
 - (2) Each LWA will specify the activities that the holder is authorized to perform.
- (c) **Effect of limited work authorization.** Any activities undertaken under an LWA are entirely at the risk of the applicant and, except as to the matters determined under paragraph (b)(1) of this section, the issuance of the LWA has no bearing on the issuance of a CP or COL with respect to the requirements of the Act and rules, regulations, or orders issued under the Act. The environmental impact statement for a CP or COL application for which an LWA was previously issued will not address, and the presiding officer in a contested hearing will not consider, the sunk costs of the holder of the LWA in determining the proposed action (i.e., issuance of the CP or COL).
- (d) **Implementation of redress plan.** If construction is terminated by the holder, the underlying application is withdrawn by the applicant or denied by the NRC, or the LWA is revoked by the NRC, then the holder must begin implementation of the redress plan in a reasonable time. The holder must also complete the redress of the site no later than 18 months after termination of construction, revocation of the LWA, or upon effectiveness of the Commission's final decision denying the associated CP application or the associated COL application, as applicable.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20068, Apr. 15, 2026]

§ 53.1140 Early site permits.

Sections 53.1140 through 53.1188 set out the requirements and procedures applicable to Commission issuance of an early site permit under this part for approval of a site for a commercial nuclear plant separate from the filing of an application for a CP or COL for the facility.

§ 53.1144 Contents of applications for early site permits; general information.

The application must contain all of the information required by § 53.1109(a) through (d) and (j).

§ 53.1146 Contents of applications for early site permits; technical information.

- (a) The application must contain—
 - (1) A Site Safety Analysis Report that must include the following:
 - (i) The specific number, type, and thermal power level of the facilities, or range of possible facilities, for which the site may be used;

- (ii) The anticipated maximum levels of radiological and thermal effluents each facility will produce;
- (iii) The type of cooling systems, including intakes and outflows, where appropriate, that may be associated with each facility;
- (iv) The boundaries of the site;
- (v) The proposed general location of each facility on the site;
- (vi) The external hazards and site characteristics required by this part;
- (vii) The location and description of any nearby industrial, military, or transportation facilities and routes;
- (viii) The existing and projected future population profile of the area surrounding the site;
- (ix) A description and assessment of the site on which a facility is to be located. The assessment must address the requirements of subpart D of this part;
- (x) Information demonstrating that site characteristics are such that adequate security plans and measures can be developed; and
- (xi) A description of the quality assurance program (QAP) required by appendix B to part 50 of this chapter applied to site-related activities for the future design, fabrication, construction, and testing of the SSCs of a facility or facilities that may be constructed on the site.

- (2) A complete environmental report as required by § 51.50(b) of this chapter.

(b)

- (1) The Site Safety Analysis Report must identify physical characteristics of the proposed site, such as egress limitations from the area surrounding the site, that could pose a significant impediment to the development of emergency plans. If physical characteristics are identified that could pose a significant impediment to the development of emergency plans, the application must identify measures that would, when implemented, mitigate or eliminate the significant impediment.
- (2) The Site Safety Analysis Report may also—
 - (i) Propose major features of the emergency plans, under either § 50.160 or the requirements in appendix E to part 50 and § 50.47(b) of this chapter, as applicable, such as the exact size and configuration of the EPZs, for review and approval by the NRC, in consultation with the Federal Emergency Management Agency (FEMA), as applicable, in the absence of complete and integrated emergency plans; or
 - (ii) Propose complete and integrated emergency plans for review and approval by the NRC, in consultation with FEMA, as applicable, in accordance with either § 50.160 or the requirements in appendix E to part 50 and § 50.47(b) of this chapter. To the extent approval of emergency plans is sought, the application must contain the information required by § 53.1109(g).
- (3) Emergency plans submitted under paragraph (b)(2)(ii) of this section must include the proposed inspections, tests, and analyses that the holder of a COL referencing the early site permit must perform, and the acceptance criteria that are necessary and sufficient to provide reasonable assurance that, if the inspections, tests, and analyses are performed and the acceptance criteria met, the facility has been constructed and will be operated in conformity with the emergency plans,

the provisions of the Act, and the Commission's rules and regulations. Major features of an emergency plan submitted under paragraph (b)(2)(i) of this section may include proposed inspections, tests, analyses, and acceptance criteria (ITAAC).

- (4) Under paragraphs (b)(1) and (b)(2)(i) of this section, the Site Safety Analysis Report must include, where appropriate, a description of contacts and arrangements made with Federal, State, participating Tribal, and local governmental agencies with emergency planning responsibilities. The Site Safety Analysis Report must contain any certifications that have been obtained. If these certifications, where appropriate, cannot be obtained, the Site Safety Analysis Report must contain information, including a utility plan, sufficient to show that the proposed plans provide reasonable assurance that adequate protective measures can and will be taken in the event of a radiological emergency at the site. Under the option set forth in paragraph (b)(2)(ii) of this section, the applicant must make good faith efforts, where appropriate, to obtain from the same governmental agencies certifications that—
 - (i) The proposed emergency plans are practicable;
 - (ii) These agencies are committed to participating in any further development of the plans, including any required field demonstrations; and
 - (iii) That these agencies are committed to executing their responsibilities under the plans in the event of an emergency.
- (c) An applicant may request that an LWA under § 53.1130 be issued in conjunction with the early site permit. The application must include the information otherwise required by § 53.1130.
- (d) Each applicant for an early site permit under this part must protect safeguards information against unauthorized disclosure in accordance with the requirements in §§ 73.21 and 73.22 of this chapter, as applicable.

§ 53.1149 Review of applications.

- (a) **Standards for review of applications.** Applications filed under this part will be reviewed according to the applicable standards set out in this part. In addition, the Commission must prepare an environmental impact statement during review of the application, under the applicable provisions of 10 CFR part 51. The Commission must determine, after consultation with FEMA, as applicable, whether the information required of the applicant by § 53.1146(b)(1) shows that there is no significant impediment to the development of emergency plans that cannot be mitigated or eliminated by measures proposed by the applicant, whether any major features of emergency plans submitted by the applicant under § 53.1146(b)(2)(i) are acceptable under either § 50.160 or appendix E to part 50 and § 50.47(b) of this chapter, and whether any emergency plans submitted by the applicant under § 53.1146(b)(2)(ii) provide reasonable assurance that adequate protective measures can and will be taken in the event of a radiological emergency.
- (b) **Administrative review of applications; hearings.** An early site permit application is subject to all applicable procedural requirements in 10 CFR part 2, including the requirements for docketing in § 2.101(a)(1) through (4) of this chapter, and the requirements for issuance of a notice of hearing in § 2.104(a) and (d) of this chapter, provided that the designated sections may not be construed to require that the environmental report, or draft or final environmental impact statement includes an assessment of the benefits of construction and operation of the reactor or reactors, or an analysis of alternative energy sources. The presiding officer in a contested early site permit hearing must not admit contentions proffered by any party concerning an assessment of the benefits of construction and operation of the

reactor or reactors, or an analysis of alternative energy sources if those issues were not addressed by the applicant in the early site permit application. All contested hearings conducted on applications for early site permits filed under this part are governed by the procedures contained in subparts C, G, L, and N of 10 CFR part 2, as applicable.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20068, Apr. 15, 2026]

§ 53.1155 Referral to the Advisory Committee on Reactor Safeguards.

The Commission must refer a copy of the application for an early site permit to the Advisory Committee on Reactor Safeguards (ACRS). The ACRS must report on those portions of the application which concern safety.

§ 53.1158 Issuance of early site permit.

- (a) The Commission may issue an early site permit, in the form the Commission deems appropriate, if the Commission finds that—
 - (1) An application for an early site permit demonstrates compliance with the applicable standards and requirements of the Act and the Commission's regulations;
 - (2) Notifications, if any, to other agencies or bodies have been duly made;
 - (3) There is reasonable assurance that the site is in conformity with the provisions of the Act and the Commission's regulations;
 - (4) The applicant is technically qualified to engage in any activities authorized;
 - (5) The proposed ITAAC, including any on emergency planning, are necessary and sufficient, within the scope of the early site permit, to provide reasonable assurance that the facility has been constructed and will be operated in conformity with the license, the provisions of the Act, and the Commission's regulations;
 - (6) Issuance of the permit will not be inimical to the common defense and security or to the health and safety of the public;
 - (7) Any significant adverse environmental impact resulting from activities requested under § 53.1146(c) can be redressed; and
 - (8) The findings required by 10 CFR part 51 have been made.
- (b) The early site permit must specify the site characteristics, design parameters, and terms and conditions of the early site permit the Commission deems appropriate. Before issuance of either a CP or COL referencing an early site permit, the Commission must find that any relevant terms and conditions of the early site permit have been met. Any terms or conditions of the early site permit that could not be met by the time of issuance of the CP or COL, must be set forth as terms or conditions of the CP or COL.
- (c) The early site permit must specify those § 53.1130(b) activities requested under § 53.1146(c) that the permit holder is authorized to perform.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20068, Apr. 15, 2026]

§ 53.1161 Extent of activities permitted.

If the activities authorized by § 53.1158(c) are performed and the site is not referenced in an application for a CP or a COL issued under this part while the permit remains valid, then the early site permit remains in effect solely for the purpose of site redress, and the holder of the permit must redress the site under the terms of the site redress plan required by § 53.1146(c). If, before redress is complete, a use not envisaged in the redress plan is found for the site or parts thereof, the holder of the permit must carry out the redress plan to the greatest extent possible consistent with the alternate use.

§ 53.1164 Duration of permit.

- (a) Except as provided in paragraph (b) of this section, an early site permit issued under this subpart may be valid for not less than 10, nor more than 20 years from the date of issuance.
- (b) An early site permit continues to be valid beyond the date of expiration in any proceeding on a CP application or a COL application that references the early site permit and is docketed before the date of expiration of the early site permit, or, if a timely application for renewal of the permit has been docketed, before the Commission has determined whether to renew the permit.
- (c) An applicant for a CP or COL may, at its own risk, reference in its application a site for which an early site permit application has been docketed but not granted.
- (d) Upon issuance of a CP or COL, a referenced early site permit is subsumed, to the extent referenced, into the CP or COL.

§ 53.1167 Limited work authorization after issuance of early site permit.

A holder of an early site permit may request an LWA under § 53.1130.

§ 53.1170 Transfer of early site permit.

An application to transfer an early site permit will be processed under § 53.1570.

§ 53.1173 Application for renewal.

- (a) Not less than 12, nor more than 36 months before the expiration date stated in the early site permit, or any later renewal period, the permit holder may apply for a renewal of the permit. An application for renewal must contain all information necessary to bring up to date the information and data contained in the previous application.
- (b) Any person whose interests may be affected by renewal of the permit may request a hearing on the application for renewal. The request for a hearing must comply with § 2.309 of this chapter. If a hearing is granted, notice of the hearing will be published under § 2.309 of this chapter.
- (c) An early site permit, either original or renewed, for which a timely application for renewal has been filed, remains in effect until the Commission has determined whether to renew the permit. If the permit is not renewed, it continues to be valid in certain proceedings in accordance with the provisions of § 53.1164(b).

§ 53.1176 Criteria for renewal.

- (a) The Commission must grant the renewal if it determines that—

- (1) The site complies with the Act, the Commission's regulations, and orders applicable and in effect at the time the site permit was originally issued; and
- (2) Any new requirements the Commission may wish to impose—
 - (i) Are necessary for adequate protection to public health and safety or common defense and security;
 - (ii) Are necessary for compliance with the Commission's regulations, and orders applicable and in effect at the time the site permit was originally issued; or
 - (iii) Would provide a substantial increase in overall protection of the public health and safety or the common defense and security to be derived from the new requirements, and the direct and indirect costs of implementation of those requirements are justified in view of this increased protection.
- (b) A denial of renewal under the provisions of § 53.1176(a) does not bar the permit holder or another applicant from filing a new application for the site which proposes changes to the site or the way that it is used to correct the deficiencies cited in the denial of the renewal.

§ 53.1179 Duration of renewal.

Each renewal of an early site permit may be for not less than 10, nor more than 20 years, plus any remaining years on the early site permit then in effect before renewal.

§ 53.1182 Use of site for other purposes.

A site for which an early site permit has been issued under this part may be used for purposes other than those described in the permit, including the location of other types of energy facilities. The permit holder must inform the Director, Office of Nuclear Reactor Regulation (Director), of any significant uses for the site which have not been approved in the early site permit. The information about the activities must be given to the Director at least 30 days in advance of any actual construction or site modification for the activities. The information provided could be the basis for imposing new requirements on the permit, under the provisions of § 53.1188. If the permit holder informs the Director that the holder no longer intends to use the site for a commercial nuclear plant, the Director may terminate the permit.

§ 53.1188 Finality of early site permit determinations.

- (a) *Commission finality.*
 - (1) While an early site permit is in effect under § 53.1164 or § 53.1179, the Commission may not change or impose new site characteristics, design parameters, or terms and conditions, including emergency planning requirements, on the early site permit unless the Commission—
 - (i) Determines that a modification is necessary to bring the permit or the site into compliance with the Commission's regulations and orders applicable and in effect at the time the permit was issued;
 - (ii) Determines the modification is necessary to assure adequate protection of the public health and safety or the common defense and security;
 - (iii) Determines that a modification is necessary based on an update under paragraph (b) of this section; or

(iv) Issues a variance requested under paragraph (d) of this section.

(2) In making the findings required for issuance of a CP, COL, or OL, or the findings required by § 53.1452(g), or in any enforcement hearing other than one initiated by the Commission under paragraph (a)(1) of this section, if the application for the CP, COL, or OL references an early site permit, the Commission must treat as resolved those matters resolved in the proceeding on the application for issuance or renewal of the early site permit, except as provided for in paragraphs (b), (c), and (d) of this section.

(i) If the Commission grants a CP application that references an early site permit and an application for an OL references the CP, the Commission must treat as resolved those matters resolved in the proceeding for the issuance or renewal of the early site permit, except as provided for in paragraphs (b), (c), and (d) of this section.

(ii) If the early site permit approved an emergency plan (or major features thereof) that is in use by a licensee of a commercial nuclear plant, the Commission must treat as resolved changes to the early site permit emergency plan (or major features thereof) that are identical to changes made to the licensee's emergency plans under § 53.1565 occurring after issuance of the early site permit.

(iii) If the early site permit approved an emergency plan (or major features thereof) that is not in use by a licensee of a commercial nuclear plant, the Commission must treat as resolved changes that are equivalent to those that could be made under § 53.1565 without prior NRC approval had the emergency plan been in use by a licensee.

(b) **Updating of early site permit-emergency preparedness.** An applicant for a CP, OL, or COL who has filed an application referencing an early site permit issued under this subpart must update the emergency preparedness information that was provided under § 53.1146(b) and discuss whether the updated information materially changes the bases for compliance with applicable NRC requirements.

(c) **Hearings and petitions.**

(1) In any proceeding for the issuance of a CP, OL, or COL referencing an early site permit, contentions on the following matters may be litigated in the same manner as other issues material to the proceeding:

(i) The nuclear reactor proposed to be built does not fit within one or more of the site characteristics or design parameters included in the early site permit;

(ii) One or more of the terms and conditions of the early site permit have not been met;

(iii) A variance requested under paragraph (d) of this section is unwarranted or should be modified;

(iv) New or additional information is provided in the application that substantially alters the bases for a previous NRC conclusion or constitutes a sufficient basis for the Commission to modify or impose new terms and conditions related to emergency preparedness; or

(v) Any significant environmental issue that was not resolved in the early site permit proceeding, or any issue involving the impacts of construction and operation of the facility that was resolved in the early site permit proceeding for which significant new information has been identified.

(2) Any person may file a petition requesting that the site characteristics, design parameters, or terms and conditions of the early site permit be modified, or that the permit be suspended or revoked. The petition will be considered under § 2.206 of this chapter. Before construction commences, the

Commission must consider the petition and determine whether any immediate action is required. If the petition is granted, an appropriate order will be issued. Construction under the CP or COL will not be affected by the granting of the petition unless the order is made immediately effective. Any change required by the Commission in response to the petition must demonstrate compliance with the requirements of paragraph (a)(1) of this section.

- (d) **Variances.** An applicant for a CP, OL, or COL referencing an early site permit may include in its application a request for a variance from one or more site characteristics, design parameters, or terms and conditions of the early site permit, or from the Site Safety Analysis Report. In determining whether to grant the variance, the Commission must apply the same technically relevant criteria applicable to the application for the original or renewed early site permit. Once a CP or COL referencing an early site permit is issued, variances from the early site permit will not be granted for that CP or COL.
- (e) **Early site permit amendment.** The holder of an early site permit may not make changes to the early site permit or the Site Safety Analysis Report without prior Commission approval. The request for a change to the early site permit must be in the form of an application for a license amendment and must demonstrate compliance with the requirements of §§ 53.1510 and 53.1520.

§ 53.1200 Standard design approvals.

Sections 53.1200 through 53.1221 set out procedures for the filing, NRC staff review, and referral to the ACRS of standard designs, or major portions thereof, for a commercial nuclear plant under this part.

§ 53.1206 Contents of applications for standard design approvals; general information.

The application must contain all of the information required by § 53.1109(a) through (c) and (j).

§ 53.1209 Contents of applications for standard design approvals; technical information.

- (a) **Major portion of a standard design.** If the applicant seeks review of a major portion of a standard design, the application need only contain the information required by this section to the extent the requirements are applicable to the major portion of the standard design for which NRC staff approval is sought. If an applicant seeks approval of a major portion of the design, the scope of the application for which approval is sought must include all functional design criteria necessary to demonstrate compliance with the safety criteria in §§ 53.210, 53.220 and 53.450(e), as applicable, for the major portion of the standard design for which NRC staff approval is sought. Such applicants must identify conditions related to interfaces with systems outside the scope of the major portion of the standard design for which NRC staff approval is sought, and functional or physical boundary conditions between the major portion of the standard design for which NRC staff approval is sought and the remainder of the standard design. These conditions must be demonstrated when the standard design approval is incorporated into a subsequent CP, design certification, ML, or COL application.
- (b) **Final Safety Analysis Report.** The application must contain an FSAR that describes the facility and the limits on its operation, presents a safety analysis of the SSCs and of the facility, or major portions thereof, for which the applicant seeks design approval, and must include the following information:
 - (1) **Site parameters.** The site parameters postulated for the design under this part, including the design-basis external hazard levels for the relevant external hazards, and an analysis and evaluation of the design in terms of those site parameters.

- (2) **Design information.** Except as specified in this paragraph (b), an application for a standard design approval for a commercial nuclear plant must include the design information equivalent to that required for a standard design certification under § 53.1239(a)(2) through (27) for those portions of a commercial nuclear plant included in the standard design approval.

§ 53.1210 Contents of applications for standard design approvals; other application content.

- (a) In addition to the FSAR, the application must also include the following:
- (1) **Availability controls (if not included in the FSAR).** A description of the controls on plant operations, including availability controls, to provide reasonable confidence that the configurations and special treatments for safety-related (SR) SSCs and non-safety-related but safety-significant (NSRSS) SSCs provide the capabilities and reliabilities required to demonstrate compliance with the safety criteria of § 53.220.
- (2) **Safeguards information.** A description of the program to protect Safeguards Information against unauthorized disclosure in accordance with the requirements in §§ 73.21 and 73.22 of this chapter, as applicable.
- (b) If there are SSCs of the plant which required research and development to confirm the adequacy of their design, provide a report in the application which documents the resolution of any safety questions associated with such SSCs.
- (c) A description of how the performance of each design feature has been demonstrated capable of fulfilling functional design criteria considering interdependent effects through either analysis, appropriate test programs, prototype testing, operating experience, or a combination thereof, in accordance with § 53.440(a).

§ 53.1212 Standards for review of applications.

Applications filed under this part will be reviewed under the standards set out in 10 CFR parts 20, 53, and 73.

§ 53.1215 Referral to the Advisory Committee on Reactor Safeguards.

The Commission must refer a copy of the application to the ACRS. The ACRS must report on those portions of the application which concern safety.

§ 53.1218 Staff approval of design.

- (a) Upon completion of its review of a submittal under §§ 53.1200 through 53.1221 and receipt of a report by the ACRS under § 53.1215, the NRC staff must publish a determination in the FEDERAL REGISTER as to whether or not the design is acceptable, subject to appropriate terms and conditions, and make an analysis of the design in the form of a report available at the NRC website, <https://www.nrc.gov>.
- (b) A standard design approval issued under this section is valid for 15 years from the date of issuance and may not be renewed. A design approval continues to be valid beyond the date of expiration in any proceeding on an application for a CP, OL, COL, or ML under this part that references the design approval and is docketed before the date of expiration of the design approval.

§ 53.1221 Finality of standard design approvals; information requests.

- (a) An approved design must be used by and relied upon by the NRC staff and the ACRS in their reviews of any standard design certification or individual facility license application under this part that incorporates by reference a standard design approved under this part unless there exists significant new information that substantially affects the earlier determination or other good cause.
- (b) The determination and report by the NRC staff do not constitute a commitment to issue a permit or license, or in any way affect the authority of the Commission, Atomic Safety and Licensing Board Panel, or presiding officers in any proceeding under part 2 of this chapter.
- (c) Except for information requests seeking to verify compliance with the current licensing basis of the standard design approval, information requests to the holder of a standard design approval must be evaluated before issuance to ensure that the burden to be imposed on respondents is justified in view of the potential safety significance of the issue to be addressed in the requested information. Each evaluation performed by the NRC staff must be in accordance with § 53.1580 and must be approved by the Executive Director for Operations or authorized designee before issuance of the request.
- (d) The Commission will require, before granting a CP, COL, OL, or ML that references a standard design approval, that information normally contained in engineering documents, such as analyses, drawings, procurement specifications, or construction and installation specifications, be completed and available for audit if the more detailed information is necessary for the Commission to verify the information in the application and make its safety determination, including the determination that the application is consistent with the design approval information. This information may be acquired by appropriate arrangements with the design approval applicant.

§ 53.1230 Standard design certifications.

Sections 53.1230 through 53.1263 set forth the requirements and procedures applicable to the Commission's issuance of rules granting standard design certifications for commercial nuclear plants under this part separate from the filing of an application for a CP or COL for such a facility.

§ 53.1236 Contents of applications for standard design certifications; general information.

The application must contain all of the information required by § 53.1109(a) through (c) and (j).

§ 53.1239 Contents of applications for standard design certifications; technical information.

The application must contain a level of design information sufficient to enable the Commission to judge the applicant's proposed means of assuring that construction conforms to the design and to reach a final conclusion on all safety questions associated with the design before the certification is granted. The information submitted for a design certification must include performance requirements and design information sufficiently detailed to permit the preparation of acceptance and inspection requirements by the NRC. The Commission will require, before design certification, that information normally contained in engineering documents, such as analyses, drawings, procurement specifications, or construction and installation specifications, be completed and available for audit if the more detailed information is necessary for the Commission to verify the information in the application and make its safety determination.

- (a) **Final Safety Analysis Report.** The application must contain an FSAR that describes the facility and the limits on its operation, and presents a safety analysis of the SSCs, and must include the following information:

- (1) **Site parameters.** The site parameters postulated for the design under this part, including the design-basis external hazard levels for the relevant external hazards, and an analysis and evaluation of the design in terms of those site parameters.
- (2) **Plant description and safety functions –**
 - (i) **General plant description.** A general description of the commercial nuclear plant including reactor type, the intended use of the reactor, nuclear design (e.g., neutron spectrum, reactor control, multi-unit reactor control), overall layout of the plant including significant plant features and SSCs, maximum power level and the nature and inventory of radioactive materials.
 - (ii) **Safety functions.** A description of the primary and additional safety functions required under § 53.230 and a summary of how each safety function is satisfied.
- (3) **Design features and functional design criteria—licensing-basis events.**
 - (i) A description of the design features required by § 53.400 and the functional design criteria required by §§ 53.410 and 53.420 that, when combined with corresponding human actions and programmatic controls, demonstrate that the plant will demonstrate compliance with the safety criteria defined in § 53.210 and established in accordance with § 53.220 during licensing-basis events (LBEs).
 - (ii) A description of how design features demonstrate compliance with the requirements of § 53.440(a) through (j) and (k) through (m).
- (4) **Design features supporting normal operations.** A description of the design features required by § 53.425 to support the holder of an OL or COL complying with § 53.260 during normal operations.
- (5) [Reserved]
- (6) **Earthquake engineering.** The information necessary to demonstrate that the commercial nuclear plant complies with the earthquake engineering criteria in § 53.480.
- (7) **Programmatic controls and interfaces.**
 - (i) A description of the corresponding programmatic controls and interfaces necessary to achieve and maintain the reliability and capability of SSCs relied upon to demonstrate compliance with the functional design criteria required by §§ 53.410 and 53.420 and the safety criteria in §§ 53.210 and 53.220 and necessary to maintain consistency with analyses required by § 53.450.
 - (ii) For an application for a multi-unit commercial nuclear plant, the programmatic controls and interfaces must also be described for different modular configurations, as required by § 53.440(i), including any restrictions that will be necessary during the construction and startup of any given unit to ensure the safe operation of the overall commercial nuclear plant to be licensed under this part.
- (8) **Programmatic controls for normal operations.** A description of how programmatic controls, including monitoring programs, would provide assurance that design features and procedures will enable the holder of an OL or COL to comply with § 53.260.
- (9) **Design features supporting the protection of plant workers.** A description of the design features required by § 53.430 to support the holder of an OL or COL complying with § 53.270.

- (10) **Programmatic controls for protection of plant workers.** A description of how programmatic controls, including monitoring programs, would provide assurance that design features and procedures will enable the holder of an OL or COL to comply with § 53.270.
- (11) **Codes and standards.** A description of generally accepted consensus codes and standards used to design the design features, as required by § 53.440(b).
- (12) **Materials.** A description of the materials used for SR and NSRSS SSCs and a description of the qualification of these materials for their service conditions over the plant lifetime, as required by § 53.440(c).
- (13) **Integrity assessment program.** A description of a design integrity assessment program that addresses the elements described in § 53.440(d).
- (14) [Reserved]
- (15) **Criticality.** Information demonstrating how the applicant will comply with requirements for criticality accidents in § 53.440(m).
- (16) **Multi-unit plants.** For an application for standard design certification of a multi-unit commercial nuclear plant, the possible operating configurations of the reactor units, including common systems, interface requirements, and system interactions, as required by § 53.440(i).
- (17) **SSC classification.**
 - (i) The classification of SSCs according to their safety significance under § 53.460(a).
 - (ii) For SR and NSRSS SSCs, the conditions under which they must perform the safety functions required by § 53.230, including environmental conditions.
- (18) **Probabilistic risk assessment or other systematic risk evaluations (SREs).** A description of the probabilistic risk assessment (PRA), other SREs, or a combination thereof required by § 53.450(a) and associated results.
- (19) **Analyses.** A description of the analyses performed under § 53.450(b) through (g) that includes the following information:
 - (i) A description of the analysis of LBEs and its results, as described in § 53.240. This analysis description must—
 - (A) Address the elements in § 53.450(e) and (f); and
 - (B) Under § 53.460(c)—
 - (1) Describe any human actions that are necessary to prevent or mitigate LBEs;
 - (2) Describe how those human actions are capable of being reliably performed under the postulated environmental conditions present; and
 - (3) Describe how those human actions would be addressed by programs established under subpart F of this part.
 - (ii)
 - (A) A description of how SSCs relied on to meet the safety criteria defined in § 53.210 are protected against or designed to withstand the effects of external hazards under § 53.510.

- (B) The information necessary to demonstrate that the commercial nuclear plant complies with the earthquake engineering criteria in § 53.480.
- (iii) A description of the defense-in-depth measures required by § 53.250.
- (iv) A description of all plant operating states where there is the potential for the uncontrolled release of radioactive material to the environment, as required by § 53.450(b)(4).
- (v) A description of the events that challenge plant control and safety systems whose failure could lead to an undesirable end state and/or radioactive material release, as required by § 53.450(b)(5).
- (vi) A description of the analytical codes used in modeling plant behavior in analyses of LBEs and how these codes are qualified for the range of conditions for which they were used, as required by § 53.450(d).
- (vii) A description of the results of other analyses required by § 53.450(g).
- (20) **Special treatments.** A description of special treatments established as required by § 53.460.
- (21) [Reserved]
- (22) **Quality assurance.** A description of the QAP applied to the design of the SSCs of the commercial nuclear plant, as required by § 53.460(b). The description of the QAP for a commercial nuclear plant must include a discussion of how the applicable requirements of appendix B to part 50 of this chapter were satisfied.
- (23) **Design features and controls to address the minimization of contamination.** The information required by § 20.1406 of this chapter.
- (24) **Interface requirements.**
 - (i) A description analysis, and evaluation of the interfaces between the standard design and the balance of the commercial nuclear plant that may impact the ability of the plant to demonstrate compliance with the functional design criteria or the safety criteria of subparts B and C of this part.
 - (ii) Confirmation that interface requirements are verifiable through inspections, testing, or analysis. These requirements must be sufficiently detailed to allow for completion of the final safety analysis by license applicants that reference the certified design under this subpart. The method to be used for verification of interface requirements must be included as part of the proposed ITAAC required by § 53.1241(a)(3).
 - (iii) A representative conceptual design for those portions of the plant for which the application does not seek certification to aid the NRC in its review of the FSAR and to permit assessment of the adequacy of the interface requirements under paragraph (a)(24)(i) of this section.
- (25) **Technical qualifications.** A description of the technical qualifications of the applicant to engage in the proposed activities in accordance with the regulations in this chapter.
- (26) **Technical specifications.** Proposed technical specifications prepared under § 53.710(a) for those areas addressed by the design certification.
- (27) **Role of personnel.** Information to address the following for the role of personnel in ensuring safe operations:

- (i) A description of how the human factors engineering design requirements of § 53.440(n)(1) are addressed;
- (ii) A description of how the human system interface design requirements of § 53.440(n)(2) are addressed;
- (iii) A concept of operations that is of sufficient scope and detail to address the requirements of § 53.440(n)(3);
- (iv) A functional requirements analysis and function allocation that is of sufficient scope and detail to address the requirements of § 53.440(n)(4).

(b) [Reserved]

§ 53.1241 Contents of applications for standard design certifications; other application content.

- (a) In addition to the FSAR, the application must also include the following:
 - (1) **Environmental report.** An environmental report as required by § 51.55 of this chapter.
 - (2) **Availability controls (if not included in the FSAR).** A description of the controls on plant operations, including availability controls, to provide reasonable confidence that the configurations and special treatments for SR and NSRSS SSCs provide the capabilities and reliabilities required to demonstrate compliance with the safety criteria of § 53.220.
 - (3) **Inspections, tests, analyses, and acceptance criteria.** The proposed ITAAC that are necessary and sufficient to provide reasonable assurance that, if the inspections, tests, and analyses are performed and the acceptance criteria met, a facility that incorporates the design certification has been constructed and will be operated in conformity with the design certification, the provisions of the Act, and the Commission's rules and regulations.
 - (4) **Safeguards information.** A description of the program to protect Safeguards Information against unauthorized disclosure in accordance with the requirements in §§ 73.21 and 73.22 of this chapter, as applicable.
- (b) If there are SSCs of the plant which required research and development to confirm the adequacy of their design, provide a report in the application which documents the resolution of any safety questions associated with such SSCs.
- (c) A description of how the performance of each design feature has been demonstrated capable of fulfilling functional design criteria considering interdependent effects through either analysis, appropriate test programs, prototype testing, operating experience, or a combination thereof, in accordance with § 53.440(a).

§ 53.1242 Review of applications.

- (a) **Standards for review of applications.** Applications filed under this part will be reviewed for compliance with the standards set out in 10 CFR parts 20, 51, 53, and 73.
- (b) **Administrative review of applications; hearings.**
 - (1) A standard design certification is a rule that will be issued under the provisions of subpart H of 10 CFR part 2, as supplemented by the provisions of this section. The Commission must initiate the rulemaking after an application has been filed under § 53.1100(a)(1)(iii) and must specify the procedures to be used for the rulemaking. The notice of proposed rulemaking published in the

FEDERAL REGISTER must provide an opportunity for the submission of comments on the proposed design certification rule. If, at the time a proposed design certification rule is published in the FEDERAL REGISTER under this paragraph (b)(1), the Commission decides that a legislative hearing should be held, the information required by § 2.1502(c) of this chapter must be included in the FEDERAL REGISTER document for the proposed design certification.

- (2) Following the submission of comments on the proposed design certification rule, the Commission may, at its discretion, hold a legislative hearing under the procedures in subpart O of part 2 of this chapter. The Commission must publish a document in the FEDERAL REGISTER of its decision to hold a legislative hearing. The document must contain the information specified in § 2.1502(c) of this chapter and specify whether the Commission or a presiding officer will conduct the legislative hearing.
 - (3) Notwithstanding anything in § 2.390 of this chapter to the contrary, proprietary information will be protected in the same manner and to the same extent as proprietary information submitted in connection with applications for licenses, provided that the design certification will be published in chapter I of this title.
- (c) *Reference to an issued operating license or combined license.* In those cases where a design certification application is preceded by the issuance of an OL or custom COL for a commercial nuclear plant that is essentially the same as the standard design for which certification is being requested, the NRC review will follow the processes for referencing a standard design approval in § 53.1221, to the extent practicable.

§ 53.1245 Referral to the Advisory Committee on Reactor Safeguards.

The Commission must refer a copy of the application to the ACRS. The ACRS must report on those portions of the application which concern safety.

§ 53.1248 Issuance of standard design certification.

- (a) After conducting a rulemaking proceeding under § 53.1242 on an application for a standard design certification and receiving the report to be submitted by the ACRS under § 53.1245, the Commission may issue a standard design certification in the form of a rule for the design that is the subject of the application, if the Commission determines that—
 - (1) The application demonstrates compliance with the applicable standards and requirements of the Act and the Commission's regulations;
 - (2) Notifications, if any, to other agencies or bodies have been duly made;
 - (3) There is reasonable assurance that the standard design conforms with the provisions of the Act and the Commission's regulations;
 - (4) The applicant is technically qualified;
 - (5) The proposed ITAAC are necessary and sufficient, within the scope of the standard design, to provide reasonable assurance that, if the inspections, tests, and analyses are performed and the acceptance criteria met, the facility has been constructed and will be operated in accordance with the design certification, the provisions of the Act, and the Commission's regulations;
 - (6) Issuance of the standard design certification will not be inimical to the common defense and security or to the health and safety of the public;
 - (7) The findings required by part 51 of this chapter have been made; and

- (8) The applicant has implemented the QAP described or referenced in the Safety Analysis Report.
- (b) The design certification rule must specify the site parameters, design characteristics, and any additional requirements and restrictions of the design certification rule.
- (c) After the Commission has adopted a final design certification rule, the applicant must not permit any individual to have access to or any facility to possess restricted data or classified National Security Information until the individual and/or facility has been approved for access under the provisions of 10 CFR parts 25 and/or 95, as applicable.

§ 53.1251 Duration of certification.

- (a) Except as provided in paragraph (b) of this section, a standard design certification issued under this subpart is valid for 40 years from the effective date of the rule.
- (b) A standard design certification continues to be valid beyond the date of expiration in any proceeding on an application for a COL or an OL under this part that references the standard design certification and is docketed either before the date of expiration of the certification, or, if a timely application for renewal of the certification has been filed, before the Commission has determined whether to renew the certification. A design certification also continues to be valid beyond the date of expiration in any hearing held under § 53.1452 before operation begins under a COL that references the design certification.
- (c) An applicant for a CP, OL, COL, or ML under this part may, at its own risk, reference in its application a design for which a design certification application has been docketed but not granted.

§ 53.1254 Application for renewal.

- (a) Not less than 12 nor more than 36 months before the expiration of the initial 40-year period, or any later renewal period, any person may apply for renewal of the certification. An application for renewal must contain all information necessary to bring up to date the information and data contained in the previous application. The Commission will require, before renewal of certification, that information normally contained in engineering documents, such as analyses, drawings, procurement specifications, or construction and installation specifications, be completed and available for audit if the more detailed information is necessary for the Commission to verify the information in the application and make its safety determination. Notice and comment procedures must be used for a rulemaking proceeding on the application for renewal. The Commission, in its discretion, may require the use of additional procedures in individual renewal proceedings.
- (b) A design certification, either original or renewed, for which a timely application for renewal has been filed remains in effect until the Commission has determined whether to renew the certification. If the certification is not renewed, it continues to be valid in certain proceedings under § 53.1251.

§ 53.1257 Criteria for renewal.

- (a) The Commission must issue a rule granting the renewal if the design, either as originally certified or as modified during the rulemaking on the renewal, complies with the Act and the Commission's regulations applicable and in effect at the time the certification was issued.
- (b) The Commission may impose other requirements if it determines that—
 - (1) They are necessary for adequate protection to public health and safety or common defense and security;

- (2) They are necessary for compliance with the Commission's regulations and orders applicable and in effect at the time the design certification was issued; or
 - (3) There is a substantial increase in overall protection of the public health and safety or the common defense and security to be derived from the new requirements, and the direct and indirect costs of implementing those requirements are justified in view of this increased protection.
- (c) In addition, the applicant for renewal may request an amendment to the design certification. The Commission must grant the amendment request if it determines that the amendment will comply with the Act and the Commission's regulations in effect at the time of renewal. If the amendment request entails such an extensive change to the design certification that an essentially new standard design is being proposed, an application for a design certification must be filed in accordance with this subpart.
- (d) Denial of renewal does not bar the applicant, or another applicant, from filing a new application for certification of the design, which proposes design changes that correct the deficiencies cited in the denial of the renewal.

§ 53.1260 Duration of renewal.

Each renewal of certification for a standard design will be for not less than 10, nor more than 40 years.

§ 53.1263 Finality of standard design certifications.

- (a)
- (1) While a standard design certification rule is in effect under § 53.1251 or § 53.1260, the Commission may not modify, rescind, or impose new requirements on the certification information, whether on its own motion, or in response to a petition from any person, unless the Commission determines in a rulemaking that the change—
 - (i) Is necessary either to bring the certification information or the referencing plants into compliance with the Commission's regulations applicable and in effect at the time the certification was issued;
 - (ii) Is necessary to provide adequate protection of the public health and safety or the common defense and security;
 - (iii) Reduces unnecessary regulatory burden and maintains protection to public health and safety and the common defense and security;
 - (iv) Provides the detailed design information to be verified under those ITAAC that are directed at certification information (*i.e.*, design acceptance criteria);
 - (v) Is necessary to correct material errors in the certification information;
 - (vi) Substantially increases overall safety, reliability, or security of facility design, construction, or operation, and the direct and indirect costs of implementation of the rule change are justified in view of this increased safety, reliability, or security; or
 - (vii) Contributes to increased standardization of the certification information.
 - (2)

- (i) In a rulemaking under § 53.1263(a)(1), except for § 53.1263(a)(1)(ii), the Commission will give consideration to whether the benefits justify the costs for plants that are already licensed or for which an application for a permit or license is under consideration.
 - (ii) The rulemaking procedures for changes under § 53.1263(a)(1) must provide for notice and opportunity for public comment.
- (3) Any modification the NRC imposes on a design certification rule under paragraph (a)(1) of this section will be applied to all plants referencing the certified design, except those to which the modification has been rendered technically irrelevant by action taken under paragraphs (a)(4) or (b) of this section.
- (4) The Commission may not impose new requirements by plant-specific order on any part of the design of a specific plant referencing the design certification rule if that part was approved in the design certification while a design certification rule is in effect under § 53.1248, unless—
 - (i) A modification is necessary to secure compliance with the Commission's regulations applicable and in effect at the time the certification was issued, or to assure adequate protection of the public health and safety or the common defense and security; and
 - (ii) Special circumstances as defined in § 53.080 are present. In addition to the factors listed in § 53.080, the Commission must consider whether the special circumstances which § 53.080 requires to be present outweigh any decrease in safety that may result from the reduction in standardization caused by the plant-specific order.
- (5) Except as provided in § 2.335 of this chapter, in making the findings required for issuance of a COL, CP, OL, or ML, or for any hearing under § 53.1452, the Commission must treat as resolved those matters resolved in connection with the issuance or renewal of a design certification rule.
- (b) An applicant who references a design certification rule may request an exemption from one or more elements of the certification information. The Commission may grant such a request only if it determines that the exemption will comply with the requirements of § 53.080. In addition to the factors listed in § 53.080, the Commission must consider whether the special circumstances that § 53.080 requires to be present outweigh any decrease in safety that may result from the reduction in standardization caused by the exemption. The granting of an exemption on request of an applicant is subject to litigation in the same manner as other issues in the OL or COL hearing.
- (c) The Commission will require, before granting a CP, COL, OL, or ML that references a design certification rule, that information normally contained in engineering documents, such as analyses, drawings, procurement specifications, or construction and installation specifications, be completed and available for audit if the more detailed information is necessary for the Commission to verify the information in the application and make its safety determination, including the determination that the application is consistent with the certification information. This information may be acquired by appropriate arrangements with the design certification applicant.

§ 53.1270 Manufacturing licenses.

Sections 53.1270 through 53.1295 set out the requirements and procedures applicable to Commission issuance of a license under this part authorizing manufacture of manufactured reactors to be installed at sites not identified in the ML application.

§ 53.1276 Contents of applications for manufacturing licenses; general information.

Each application for an ML must include the information contained in § 53.1109(a) through (e), and (j).

§ 53.1279 Contents of applications for manufacturing licenses; technical information.

- (a) **Final Safety Analysis Report-siting and design.** The application must include an FSAR containing the information set forth below, with a level of design information sufficient to enable the Commission to judge the applicant's proposed means of ensuring that the manufacturing conforms to the design and to reach a final conclusion on all safety questions associated with the design, permit the preparation of construction and installation specifications by an applicant who seeks to use the manufactured reactor, and permit the preparation of acceptance and inspection requirements by the NRC. The application must include the following information:
- (1) **Site parameters.** The site parameters postulated for the design under this part, including the design-basis external hazard levels for the relevant external hazards, and an analysis and evaluation of the design in terms of those site parameters.
 - (2) **Design information.** The design information equivalent to that required for a standard design certification as defined in § 53.1239(a)(2) through (27) for those portions of a commercial nuclear plant included in the manufactured reactor.
 - (3) **Quality assurance program.** A description of the QAP applied to the design, and to be applied to the fabrication and testing of the SSCs of the manufactured reactor under § 53.620(a)(6), including a discussion of how the applicable requirements of appendix B to part 50 of this chapter will be satisfied;
 - (4) **Conceptual designs.** Representative conceptual designs for one or more commercial nuclear plants using the manufactured reactor;
 - (5) **Operating configurations.** If multiple manufactured reactors may be installed at a commercial nuclear plant, a description of the possible operating configurations, including common systems, interface requirements, and system interactions. The final safety analysis must also account for differences among the possible configurations, including any restrictions that will be necessary during the construction and startup of a given manufactured reactor to ensure the safe operation of any commercial nuclear reactor already operating;
 - (6) **Interface requirements.**
 - (i) The interface requirements between the manufactured reactor and the remaining portions of the commercial nuclear plant or connections to other facilities outside of the commercial nuclear plant.
 - (ii) Confirmation that interface requirements are verifiable through inspections, testing, or analysis. These requirements must be sufficiently detailed to allow for completion of the final safety analysis by license applicants that reference the manufactured reactor manufactured under this subpart. Applicants for a COL under this part will need to verify the interface requirements at the installation site. The method to be used for verification of interface requirements must be included as part of the proposed ITAAC required by § 53.1282(a).
 - (iii) Information to support development of radiation monitoring programs required under subpart F of this part by an applicant for a COL, including potential pathways for radionuclides produced within the manufactured reactor to enter interfacing systems.

- (b) ***Final Safety Analysis Report—manufacturing information.*** The FSAR must include the following information related to the manufacturing processes, organization, controls, and inspections:
- (1) A description, including references to generally accepted consensus codes and standards, of the processes that will be used to procure, fabricate, and assemble components that make up the manufactured reactor. The description should clearly define which activities are proposed to be within the scope of the ML and those, such as the making of a component to be procured from a separate company for installation in the manufactured reactor, that are not considered to be within the scope of the ML;
 - (2) A description of the organizational and management structure singularly responsible for direction of design and manufacture of the manufactured reactor. The information should include a description of the management plans, technical qualifications, and controls in place to demonstrate compliance with the requirements of § 53.620;
 - (3) A description of the inspections and tests to be performed as part of the manufacturing process, including the inspection of procured components, inspection and testing of fabrication processes such as the molding, welding, or coating of components, and inspections and testing of the assembled manufactured reactor or portions of the manufactured reactor;
 - (4) A description of the fitness-for-duty program required by part 26 of this chapter and its implementation.
- (c) ***Final Safety Analysis Report—deployment of the completed manufactured reactor.*** The application must include a description of the following information related to the deployment of a manufactured reactor:
- (1) Procedures governing the preparation of the manufactured reactor or portions of the manufactured reactor for shipping to the site where it is to be operated; the conduct of shipping; and verifying the condition of the shipped items upon receipt at the site;
 - (2) Details of the interaction of the design, manufacture, and installation of a manufactured reactor within the applicant's organization and the manner by which the applicant will ensure close integration between the designer, contractors, and any facility in which the manufactured reactor is to be installed;
 - (3) Measures to be used for the control of interfaces, including the consideration of key site parameters, between the holder of the ML and the holder of the COL or CP for the commercial nuclear plant at which the manufactured reactor is to be installed.
- (d) ***Final Safety Analysis Report—special considerations for factory fueling.*** In addition to the above paragraphs (a) through (c) of this section, an application for an ML for a manufactured reactor that will be fueled at the factory under a 10 CFR part 70 license must include the following information related to loading fuel and the required features to prevent criticality and to otherwise provide assurance that the fueled manufactured reactor can be successfully transported, installed, and operated at a site for which the Commission has issued a COL or a CP and OL that authorizes construction and operation of a commercial nuclear plant using the manufactured reactor:
- (1) A description of the procedures used during the fueling of the manufactured reactor that ensure that the configuration of fuel within the fueled manufactured reactor is consistent with the design and analyses supporting operation of the manufactured reactor under the COL or OL at the place of operation. The description may reference the applicable 10 CFR part 70 application and other sections of the Safety Analysis Report supporting the ML license application.

- (i) The application must describe the measures taken for in-factory inspections and non-nuclear testing performed to ensure that the configuration of fuel within the fueled manufactured reactor is consistent with the design and analyses supporting operation of the manufactured reactor under the COL or OL at the place of operation.
 - (ii) The application must describe the design features included in the manufactured reactor to prevent criticality, the associated functional design criteria applied to those design features, and the physical and programmatic controls implemented during manufacturing, storage, and transport that are credited to assure the features function as designed when subject to potential hazards and human errors. The descriptions must include how those measures will be controlled during installation under the ML and removal under the COL or OL at the place of operation.
- (2) A description of the procedures governing the transfer of responsibilities for the fueled manufactured reactor from the holder of the ML to the holder of the COL or CP and OL for the installation site.
- (3) If available at the time of filing the ML application or, if not available at the time of filing the ML application, submitted as an amendment to the ML or ML application at the time of filing the Part 70 application, a description of the programs needed to demonstrate compliance with the requirements of § 53.620(d) and 10 CFR parts 70, 71, and 73 for the receipt, storage, and loading of SNM into a manufactured reactor and the transport of the fueled manufactured reactor to a site for which the Commission has issued a COL or CP and OL that authorizes construction and operation of a commercial nuclear plant using the manufactured reactor, including the following.
- (i) A physical security program in accordance with § 53.620(d)(2)(i).
 - (ii) A cybersecurity program in accordance with § 53.620(d)(2)(i).

§ 53.1282 Contents of applications for manufacturing licenses; other application content.

(a) *Inspections, tests, analyses, and acceptance criteria.*

- (1) The application must contain proposed inspections, tests, and analyses that the COL or CP holder must perform, and the acceptance criteria that are necessary and sufficient to provide reasonable assurance that, if the inspections, tests, and analyses are performed and the acceptance criteria met:
 - (i) The reactor has been manufactured in conformity with the ML, the provisions of the Act, and the Commission's rules and regulations; and
 - (ii) The manufactured reactor will be operated in conformity with the approved design and any license authorizing operation of the manufactured reactor.
- (2) If the application references a standard design certification, the ITAAC contained in the certified design must apply to those portions of the facility design that are covered by the design certification.
- (3) If the application references a standard design certification, the application may include a notification that a required inspection, test, or analysis in the design certification ITAAC has been successfully completed and that the corresponding acceptance criterion has been met. The FEDERAL REGISTER notice required by § 53.1285 must indicate that the application includes this notification.

(b) *Environmental report.*

- (1) The application must contain an environmental report as required by § 51.54 of this chapter.
- (2) If the ML application references a standard design certification, the environmental report need not contain a discussion of severe accident mitigation design alternatives for the manufactured reactor as used in a commercial nuclear plant.
- (c) **Safeguards information.** The application must contain a description of the program to protect safeguards information against unauthorized disclosure in accordance with the requirements in §§ 73.21 and 73.22 of this chapter, as applicable.
- (d) **Performance demonstration.** A description of how the performance of each design feature has been demonstrated capable of fulfilling functional design criteria considering interdependent effects through either analysis, appropriate test programs, prototype testing, operating experience, or a combination thereof, in accordance with § 53.440(a).

§ 53.1285 Review of applications.

- (a) **Standards for review of applications.** Applications for MLs under this part will be reviewed according to the applicable standards set out in this subpart as well as applicable standards in 10 CFR parts 20, 25, 26, 51, 53, 70, 71, 73, and 75.
- (b) **Administrative review of applications, hearings.** A proceeding on an ML is subject to all applicable procedural requirements contained in 10 CFR part 2, including the requirements for docketing in § 2.101(a)(1) through (4) of this chapter, and the requirements for issuance of a notice of proposed action in § 2.105 of this chapter, *provided, however*, that the designated sections may not be construed to require that the environmental report or draft or final environmental impact statement include an assessment of the benefits of constructing and/or operating the manufactured reactor or an evaluation of alternative energy sources. All hearings on MLs are governed by the hearing procedures contained in 10 CFR part 2, subparts C, E, G, L, and N.

§ 53.1286 Referral to the Advisory Committee on Reactor Safeguards.

The Commission must refer a copy of the application to the ACRS. The ACRS must report on those portions of the application which concern safety.

§ 53.1287 Issuance of manufacturing licenses.

- (a) After completing any hearing under § 53.1285(b), and receiving the report submitted by the ACRS, the Commission may issue an ML if the Commission finds that—
 - (1) Applicable standards and requirements of the Act and the Commission's regulations have been met;
 - (2) There is reasonable assurance that the manufactured reactor will be manufactured, and can be transported, incorporated into a commercial nuclear plant, and operated in conformity with the ML, the provision of the Act, and the Commission's regulations;
 - (3) The proposed manufactured reactor can be incorporated into a commercial nuclear plant and operated at sites having characteristics that fall within the site parameters postulated for the design of the manufactured reactors without undue risk to the health and safety of the public;
 - (4) The applicant is technically qualified to design and manufacture the proposed manufactured reactor;

- (5) The proposed ITAAC are necessary and sufficient, within the scope of the ML, to provide reasonable assurance that the manufactured reactor has been manufactured and will be operated in conformity with the license, the provisions of the Act, and the Commission's regulations;
 - (6) The issuance of a license to the applicant will not be inimical to the common defense and security or to the health and safety of the public; and
 - (7) The findings required by 10 CFR part 51 have been made.
- (b) Each ML issued under this subpart must specify—
- (1) Terms and conditions as the Commission deems necessary and appropriate;
 - (2) Technical specifications for operation of the manufactured reactor, as the Commission deems necessary and appropriate;
 - (3) Significant site parameters and significant design characteristics for the manufactured reactor;
 - (4) The interface requirements to be met by the site-specific elements of the facility, such as the energy conversions systems and ultimate heat sink, not within the scope of the manufactured reactor; and
 - (5) The entity with design authority for the manufactured reactor covered by the license.

§ 53.1288 Finality of manufacturing licenses.

- (a)
- (1) During the term of an ML issued under this part, the Commission may not modify, rescind, or impose new requirements on the design of the manufactured reactor, or the requirements for the manufacture of the manufactured reactor, unless the Commission determines that a modification is necessary to bring the design of the reactor or its manufacture into compliance with the Commission's requirements applicable and in effect at the time the ML was issued, or to provide reasonable assurance of adequate protection to public health and safety or common defense and security.
 - (2) Any modification to the design of a manufactured reactor that is imposed by the Commission under paragraph (a)(1) of this section will be applied to all manufactured reactors manufactured under the license, including those that have already been transported and sited, except those manufactured reactors to which the modification has been rendered technically irrelevant or otherwise unnecessary by action taken under § 53.1530, § 53.1550, or paragraph (b) of this section.
 - (3) In making the findings required under this part for issuance of a COL, CP, or OL, in any hearing under § 53.1452, or in any enforcement hearing other than one initiated by the Commission under paragraph (a)(1) of this section, for which a manufactured reactor manufactured under this subpart is referenced or used, the Commission must treat as resolved those matters resolved in the proceeding on the application for issuance or renewal of the ML, including the adequacy of design of the manufactured reactor, the costs and benefits of severe accident mitigation design alternatives, and the bases for not incorporating severe accident mitigation design alternatives into the design of the manufactured reactor to be manufactured.
- (b) An applicant who references or uses a manufactured reactor manufactured under an ML under this part may include in the application a request for a departure from the design characteristics, site parameters, terms and conditions, or approved design of the manufactured reactor. The Commission may grant a

request only if it determines that the departure will comply with the requirements of § 53.080. The granting of a departure on request of an applicant is subject to litigation in the same manner as other issues in the COL or CP hearing.

§ 53.1291 Duration of manufacturing licenses.

An ML issued under this part is valid for not less than 5, nor more than 40 years from the date of issuance. Upon expiration of the ML, the manufacture of any uncompleted manufactured reactors must cease unless a timely application for renewal has been docketed with the NRC.

§ 53.1293 Transfer of manufacturing licenses.

An ML may be transferred under § 53.1570.

§ 53.1295 Renewal of manufacturing licenses.

(a)

- (1) Not less than 12 months, nor more than 5 years before the expiration of the ML, or any later renewal period, the holder of the ML issued under this part may apply for a renewal of the license. An application for renewal must contain all information necessary to bring up to date the information and data contained in the previous application.
- (2) The filing of an application for a renewed license must be in accordance with subpart A of 10 CFR part 2 and § 53.1100.
- (3) An ML issued under this part, either original or renewed, for which a timely application for renewal has been filed, remains in effect until the Commission has made a final determination on the renewal application.
- (4) Any person whose interest may be affected by renewal of the license may request a hearing on the application for renewal. The request for a hearing must comply with § 2.309 of this chapter. If a hearing is granted, notice of the hearing will be published in accordance with § 2.104 of this chapter.

(b) The Commission may grant the renewal if the Commission determines—

- (1) The ML complies with the Act and the Commission's regulations and orders applicable and in effect at the time the ML was originally issued; and
- (2) Any new requirements the Commission may wish to impose are—
 - (i) Necessary for adequate protection to public health and safety or common defense and security;
 - (ii) Necessary for compliance with the Commission's regulations and orders applicable and in effect at the time the ML was originally issued; or
 - (iii) A substantial increase in overall protection of the public health and safety or the common defense and security to be derived from the new requirements, and the direct and indirect costs of implementation of those requirements are justified in view of this increased protection.

(c) A renewed ML may be issued for a term of not less than 5, nor more than 40 years, plus any remaining years on the ML then in effect before renewal. The renewed license must be subject to the requirements of § 53.1288.

§ 53.1300 Construction permits.

Sections 53.1300 through 53.1348 set out the requirements and procedures applicable to Commission issuance of a CP for commercial nuclear plants. A CP for the construction of a commercial nuclear plant under this part will be issued before the issuance of an OL if the application is otherwise acceptable and will be converted upon completion of the facility and Commission action, into an OL as provided under §§ 53.1360 through 53.1405.

§ 53.1306 Contents of applications for construction permits; general information.

An application for a CP must include the information required by § 53.1109 and the following information:

- (a) Information sufficient to demonstrate to the Commission the financial qualification of the applicant to carry out, under the regulations in this chapter, the activities for which the permit is sought. As applicable, the applicant should provide information that demonstrates that the applicant appears to be financially qualified to cover estimated construction costs and related fuel cycle costs, including estimates of the total construction costs and related fuel cycle costs of the facility, a financial capacity plan, and any source(s) of funds available at the time of application to cover these costs. If available funding at the time of application is 50 percent or less, the applicant should include proposed license conditions to facilitate verification that funding is available prior to the start of construction.
- (b) If the applicant proposes to construct or alter a facility, the application must state the earliest and latest dates for completion of the construction or alteration.

§ 53.1309 Contents of applications for construction permits; technical information.

The application must contain a Preliminary Safety Analysis Report (PSAR) that describes the facility and the limits on its operation and presents a preliminary safety analysis of the SSCs of the facility as a whole. The PSAR must include the following information, at a level of detail sufficient to enable the Commission to reach a conclusion on safety matters that must be resolved by the Commission before issuance of a CP:

- (a)
 - (1) **Site information.** An application for a CP for a commercial nuclear reactor must include the site information equivalent to that required for an early site permit in § 53.1146(a)(1)(iv) through (x).
 - (2) **Design information.** Except as specified in this paragraph (a)(2), an application for a CP for a commercial nuclear plant must include the design information equivalent to that required for a standard design certification as defined in § 53.1239(a)(2) through (a)(21), (a)(23), and § 53.1239(a)(26) through (27).
 - (i) **Quality assurance program.** A description of the QAP to be applied to the design, fabrication, construction, and testing of the SSCs of the facility under § 53.610(a)(6), including a discussion of how the requirements of appendix B to part 50 of this chapter will be satisfied.
 - (ii) **Preliminary design information.** The information provided in the application may include some aspects of the design that are not fully developed, and the information is therefore preliminary. The completed design, including any changes during construction, must be described in the FSAR required in § 53.1369 that supports an application for an OL.
 - (iii) **Planned research or testing.** Descriptions of how design features and related functional design criteria will fulfill the safety criteria in subpart B and how that has been or will be demonstrated through either analysis, appropriate test programs, experience, or a combination thereof. Where

any design feature has not been fully developed or demonstrated to fulfill the functional design criteria at the time of an application for a CP, the applicant must provide a plan for future analysis, research and development, test programs, gathering of experience, or a combination thereof to provide reasonable confidence that the required demonstration will be available for an application for an OL

- (iv) **Programmatic controls.** Descriptions of the programmatic controls may include those to be provided in the FSAR or other licensing-basis documents because they are necessary to achieve and maintain the reliability and capability of SSCs relied upon to demonstrate compliance with the established safety criteria and functional design criteria required in subpart B, and to maintain consistency with analyses required by § 53.450.
- (3) **Technical qualifications.** A description of the technical qualifications of the applicant to engage in the proposed activities under the regulations in this chapter.
- (4) **Emergency preparedness.** A description of the applicant's preliminary plans for coping with emergencies based on:
 - (i) Except as provided in paragraph (a)(4)(ii) of this section, the requirements in appendix E to part 50.
 - (ii) For a commercial nuclear plant consisting of either small modular reactors or non-light-water reactors, the requirements in either § 50.160 or appendix E to part 50.
- (5) **Physical security.** A report that provides a preliminary description of how the site characteristics support the development of adequate security plans and measures consistent with the requirements in § 53.540.
- (6) **Fitness-for-duty program.** A description of the fitness-for-duty (FFD) program required by 10 CFR part 26 and its implementation.
- (b) A description of the program to protect Safeguards Information against unauthorized disclosure in accordance with the requirements in §§ 73.21 and 73.22 of this chapter, as applicable.

§ 53.1312 Contents of applications for construction permits; other application content.

- (a) In addition to the PSAR, the application must include the following:
 - (1) An environmental report either under § 51.50(a) of this chapter if an LWA under § 53.1130 is not requested in conjunction with the CP application, or under §§ 51.49 and 51.50(a) of this chapter if an LWA is requested in conjunction with the CP application; or
 - (2) If the applicant wishes to request that an LWA under § 53.1130 be issued before issuance of the CP, the information otherwise required by § 53.1130, in accordance with either § 2.101(a)(1) through (a)(5), or § 2.101(a)(9) of this chapter.
- (b) If the CP application references an early site permit, standard design approval, standard design certification, or ML issued under this part, then the following requirements apply:
 - (1) The PSAR need not contain information or analyses submitted to the Commission in connection with the referenced NRC approval, license, or certification, provided, however, that the PSAR incorporates the material by reference and confirms that the site and design of the facility falls within parameter values postulated in the referenced NRC approval, license, or certification.

- (2) The PSAR must provide a means to demonstrate that all terms and conditions that have been included in the referenced NRC approval, license, or certification will be satisfied by the date of issuance of the OL, as appropriate. If the PSAR does not demonstrate that each site characteristic falls within the corresponding postulated site parameter and each design characteristic of the facility falls within the corresponding postulated design parameter, the application must justify a departure, variance, or exemption from the referenced NRC approval, license, or certification in regard to that particular site or design characteristic in compliance with the requirements of this part.
- (3) If a referenced early site permit approves complete and integrated emergency plans, or major features of emergency plans, then the PSAR must include any new or additional information that updates and corrects the information that was provided under § 53.1146(b)(2) and discuss whether the new or additional information materially changes the bases for compliance with the applicable requirements.

§ 53.1315 Review of applications.

- (a) **Standards for review of applications.** Applications filed under this part will be reviewed according to the standards set out in 10 CFR parts 20, 51, 53, 73, and 140.
- (b) **Administrative review of applications; hearings.** A proceeding on a CP application is subject to all applicable procedural requirements contained in 10 CFR part 2, including the requirements for docketing (§ 2.101 of this chapter) and issuance of a notice of hearing (§ 2.104 of this chapter). All contested hearings on CP applications are governed by the procedures contained in 10 CFR part 2.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20068, Apr. 15, 2026]

§ 53.1318 Finality of referenced NRC approvals, permits, and certifications.

If the application for a CP under this part references an early site permit, standard design approval, standard design certification, or ML, the scope and nature of matters resolved for the application are governed by the relevant provisions addressing finality, including §§ 53.1188, 53.1221, 53.1263, and 53.1288.

§ 53.1324 Referral to the Advisory Committee on Reactor Safeguards.

The Commission must refer a copy of the application to the ACRS. The ACRS must report on those portions of the application that concern safety and must apply the standards referenced in § 53.1315, in accordance with the finality provisions in § 53.1318.

§ 53.1327 Authorization to conduct limited work authorization activities.

- (a) If the application does not reference an early site permit which authorizes the holder to perform the activities under § 53.1130, the applicant may not perform those activities without obtaining the separate authorization required by § 53.1130. Authorization may be granted only after the presiding officer in a contested hearing on the application has made the findings and determination required by § 53.1130(b)(1)(iii), and the Director, Office of Nuclear Reactor Regulation makes the determination required by § 53.1130(b)(1)(ii).
- (b) If, after an applicant has performed the activities permitted by paragraph (a) of this section, the application for the CP is withdrawn or denied, then the applicant must implement an approved site redress plan.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20068, Apr. 15, 2026]

§ 53.1330 Exemptions, departures, and variances.

- (a) Applicants for a CP under this part, or any amendment to a CP, may include in the application a request for an exemption from one or more of the Commission's regulations. The Commission may grant a request if it determines that the exemption complies with § 53.080.
- (b) An applicant for a CP who has filed an application referencing an NRC approval, license, or certification issued under this part may include in the application a request for exemptions, departures, or variances related to the subject referenced NRC approval, license, or certification. In determining whether to grant the departure, variance, or exemption, the Commission must apply the same technically relevant criteria as were applicable to the application for the original or renewed approval, license, or certification.

§ 53.1333 Issuance of construction permits.

- (a) The Commission may issue a CP only if the Commission finds that—
 - (1) The applicant has described the proposed design of the facility and has identified the major features or components incorporated therein for the protection of the health and safety of the public;
 - (2) Such further technical or design information as may be required to complete the safety analysis, and which can reasonably be left for later consideration, will be supplied in the FSAR;
 - (3) Safety features or components, if any, that require research and development have been described by the applicant and the applicant has identified, and there will be conducted, a research and development program reasonably designed to resolve any safety questions associated with such features or components; and
 - (4) On the basis of the foregoing, there is reasonable assurance of the following—
 - (i) Such safety questions will be satisfactorily resolved at or before the latest date stated in the application for completion of construction of the proposed facility; and
 - (ii) Taking into consideration the site criteria contained in subpart D to this part, the proposed facility can be constructed and operated at the proposed location without undue risk to the health and safety of the public.
- (b) A CP must contain the terms and conditions for the permit, as the Commission deems necessary and appropriate. The Commission may, in its discretion, incorporate in any CP provisions requiring the applicant to furnish periodic reports of the progress and results of research and development programs designed to resolve safety questions.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20068, Apr. 15, 2026]

§ 53.1336 Finality of construction permits.

Notwithstanding any provision in § 53.1590, a CP constitutes an authorization to proceed with construction but does not constitute Commission approval of the safety of any design feature or specification unless the applicant specifically requests such approval and such approval is incorporated in the permit. The applicant, at its option, may request such approvals in the CP or by amendment to the CP. If approved by the NRC and included in the permit, the NRC will consider modifications to the approved design features or specifications in accordance with § 53.1590.

§ 53.1342 Duration of construction permits.

- (a) A CP will state the earliest and latest dates for completion of construction or alteration of the facility, not to exceed 40 years from date of issuance.
- (b) If the proposed construction or alteration of the facility is not completed by the latest completion date, the CP shall expire, and all rights are forfeited. However, upon good cause shown, the Commission will extend the completion date for a reasonable period of time. The Commission will recognize, among other things, developmental problems attributed to the experimental nature of the facility or fire, flood explosion, strike, sabotage, domestic violence, enemy action an act of the elements, and other acts beyond the control of the permit holder, as a basis for extending the completion date.

§ 53.1345 Transfer of construction permits.

A CP may be transferred under § 53.1570.

§ 53.1348 Termination of construction permits.

When a permit holder has determined to permanently cease construction, the holder must, within 30 days, submit a written certification to the NRC.

§ 53.1360 Operating licenses.

Sections 53.1360 through 53.1405 set out the requirements and procedures applicable to Commission issuance of an OL for a nuclear power facility.

§ 53.1366 Contents of applications for operating licenses; general information.

An application for an OL must include the information required by § 53.1109 and, except for an electric utility applicant, information sufficient to demonstrate to the Commission the financial qualification of the applicant to carry out, in accordance with the regulations in this chapter, the activities for which the license is sought. As applicable, the applicant must submit information that demonstrates the applicant appears to be financially qualified to cover estimated operation costs for the period of the license. The applicant must submit estimates for total annual operating costs for each of the first 5 years of operation of the facility and a financial capacity plan and must indicate any source(s) of funds available at the time of application to cover these costs. If available funding at the time of application is 50 percent or less, the applicant should include proposed license conditions to facilitate verification that funding is available prior to the start of operations.

§ 53.1369 Contents of applications for operating licenses; technical information.

Final Safety Analysis Report. The application must contain an FSAR that describes the facility and the limits on its operation and presents a safety analysis of the SSCs of the facility as a whole. The FSAR must include the following information, at a level of detail sufficient to enable the Commission to reach a final conclusion on all safety matters that must be resolved by the Commission before issuance of an OL:

- (a) **Site information.** An application for an OL for a commercial nuclear reactor must include the site information equivalent to that required for an early site permit in § 53.1146(a)(1)(iv) through (x), including all current information, such as the results of environmental and meteorological monitoring programs, which has been developed since issuance of the CP, relating to site evaluation factors identified in this part.

- (b) **Design information.** Except as specified in this paragraph (b), an FSAR for an OL for a commercial nuclear plant must include the final design information equivalent to that required for a standard design certification as defined in § 53.1239(a)(2) through (7), (a)(9), (a)(11) and (12), (a)(14) through (21), (a)(23), and (a)(25).
 - (1) The completed design, including any changes during construction, must be described.
 - (2) Where any design feature had not been fully developed or demonstrated at the time of application for the CP, the applicant must provide the analysis, research and development, test programs, gathering of experience, or a combination thereof to provide the required demonstration to fulfill the functional design criteria.
- (c) [Reserved]
- (d) **Integrity assessment program.** A description of an Integrity Assessment Program that addresses the elements described in § 53.870.
- (e) **Safeguards information.** A description of the program to protect Safeguards Information against unauthorized disclosure in accordance with the requirements in §§ 73.21 and 73.22 of this chapter, as applicable.
- (f) **Emergency response facility or facilities.** Description of location and capabilities to be established for command and control, support, and coordination of onsite and offsite, as applicable, functions during reactor accident conditions.
- (g) **Role of personnel.**
 - (1) A description of the completed assessments related to the role of personnel in ensuring safe operations considering the analyses required by § 53.730. These assessments must include the following:
 - (i) Human factors engineering design requirements of § 53.730(a);
 - (ii) Human system interface design requirements of § 53.730(b);
 - (iii) Concept of operations of § 53.730(c);
 - (iv) Functional requirements analysis and function allocation of § 53.730(d);
 - (2) A description of the program to be used for evaluating and applying operating experience as required by § 53.730(e);
 - (3) A staffing plan and supporting analyses as required by § 53.730(f).
- (h) **Training, examination, and proficiency programs.**
 - (1) A description of the training, examination, and proficiency programs required by § 53.730(g);
 - (2) A description of the training programs required by § 53.830.
- (i) **Emergency plan.** Emergency plans complying with the requirements of § 53.855.
 - (1) Include all emergency plan certifications, as applicable, that have been obtained from the State, local, and participating Tribal governmental agencies with emergency planning responsibilities that are wholly or partially within the EPZ plume exposure pathway. These certifications must state that—

- (i) The proposed emergency plans are practicable;
 - (ii) These agencies are committed to participating in any further development of the plans, including any required field demonstrations; and
 - (iii) These agencies are committed to executing their responsibilities under the plans in the event of an emergency.
- (2) If certifications cannot be obtained after sustained, good faith efforts by the applicant, then the application must contain information, including a utility plan, sufficient to show that the proposed plans provide reasonable assurance that adequate protective measures can and will be taken in the event of a radiological emergency at the site.
- (3) If complete and integrated emergency plans were approved as part of an early site permit, or submitted, reviewed, and approved as part of the CP application, new certifications that demonstrate compliance with the requirements of paragraph (i)(1) of this section are not required.
- (j) **Organization.** A description of the applicant's organizational structure, allocations of responsibilities and authorities, and personnel qualifications requirements for operation.
 - (k) **Maintenance program.** A description of a maintenance program under § 53.715.
 - (l) **Quality assurance.** A description of the QAP that demonstrates compliance with the requirements under § 53.865.
 - (m) **Radiation protection program.** A radiation protection program description under § 53.850.
 - (n) **Security program.** A physical security plan that describes how the applicant will comply with § 53.860 (and 10 CFR part 11, if applicable, including the identification and description of jobs as required by § 11.11(a) of this chapter, at the proposed facility). The plan must list tests, inspections, audits, and other means to be used to demonstrate compliance with the requirements of 10 CFR parts 11 and 73, if applicable.
 - (o) **Safeguards contingency plan.** A safeguards contingency plan in accordance with the criteria set forth in appendix C to 10 CFR part 73. The safeguards contingency plan must include plans for dealing with threats, thefts, and radiological sabotage, as defined in 10 CFR part 73, relating to the SNM and nuclear facilities licensed under this chapter and in the applicant's possession and control. Each application for this type of license must include the information contained in the applicant's safeguards contingency plan. (Implementing procedures required for this plan need not be submitted for approval.)^[1]
 - (p) **Security training and qualification.** A training and qualification plan that describes how the applicant will demonstrate compliance with the criteria set forth in § 73.100 of this chapter or appendix B to 10 CFR part 73.
 - (q) **Cybersecurity plan.** A cybersecurity plan in accordance with the criteria set forth in § 73.54 or § 73.110 of this chapter.

^[1] A physical security plan that contains all the information required in both § 73.55 or § 73.100 of this chapter and appendix C to 10 CFR part 73 satisfies the requirement for a contingency plan.

- (r) **Security, safeguards, and cybersecurity plan implementation.** A description of the implementation of the physical security plan, safeguards contingency plan, training and qualification plan, and cybersecurity plan. Each applicant who prepares a physical security plan, a safeguards contingency plan, a training and qualification plan, or a cybersecurity plan must protect the plans and other related Safeguards Information against unauthorized disclosure in accordance with the requirements of §§ 73.21 and 73.22 of this chapter.
- (s) **Fire protection program.** A description of the fire protection program under § 53.875.
- (t) **Inservice inspection/inservice testing program.** A description of the inservice inspection and inservice testing programs under § 53.880.
- (u)-(v) [Reserved]
- (w) **General employee training.** A description of the training program required to demonstrate compliance with § 53.830 and its implementation.
- (x) **Fitness-for-duty program.** A description of the FFD program required by 10 CFR part 26 and its implementation.
- (y) **Other programs.** A description and evaluation of the results of the applicant's programs, including research and development, if any, to demonstrate that any safety questions identified at the CP stage have been resolved.
- (z) **Safety design feature performance.** A description of how the performance of each safety design feature has been demonstrated capable of fulfilling functional design criteria considering interdependent effects through either analysis, appropriate test programs, prototype testing, operating experience, or a combination thereof, in accordance with § 53.440(a).
- (aa) **Technical specifications.** Proposed technical specifications prepared in accordance with the requirements of § 53.710(a).

§ 53.1372 Contents of applications for operating licenses; other application content.

In addition to the FSAR, the application must also include the following:

- (a) **Environmental report.** An environmental report in accordance with § 51.53(b) of this chapter.
- (b) **Availability controls (if not included in the FSAR).** A description of the controls on plant operations, including availability controls, to provide reasonable confidence of safe operation and that the configurations and special treatments for SR and NSRSS SSCs provide the capabilities and reliabilities required to satisfy the safety criteria of § 53.220 if not addressed by Technical Specifications under § 53.1369(aa).

§ 53.1375 Review of applications.

- (a) **Standards for review of applications.** Applications filed under this part will be reviewed according to the standards set out in 10 CFR parts 20, 26, 51, 53, 73, and 140.
- (b) **Administrative review of applications; hearings.** A proceeding on an OL is subject to all applicable procedural requirements contained in 10 CFR part 2, including the requirements for docketing (§ 2.101 of this chapter) and issuance of a notice of hearing (§ 2.104 of this chapter). All hearings on OLs are governed by the procedures contained in 10 CFR part 2.

§ 53.1381 Referral to the Advisory Committee on Reactor Safeguards.

The Commission must refer a copy of the application to the ACRS. The ACRS must report on those portions of the application that concern safety and must apply the standards referenced in § 53.1375.

§ 53.1384 Exemptions, departures, and variances.

- (a) Applicants for an OL under this part, or any amendment to an OL, may include in the application a request for an exemption from one or more of the Commission's regulations. The Commission may grant an exemption request if it determines that the exemption complies with § 53.080.
- (b) An applicant for an OL who has filed an application referencing an NRC approval, permit, license, or certification issued under this part may include in the application a request for departures, variances, or exemptions related to the subject referenced NRC approval, permit, license, or certification. In determining whether to grant the departure, variance, or exemption, the Commission must apply the same technically relevant criteria as were applicable to the application for the original or renewed approval, license, or certification.

§ 53.1387 Issuance of operating licenses.

Upon completion of the construction or alteration of a facility, in compliance with the terms and conditions of the construction permit and subject to any necessary testing of the facility for health or safety purposes, the Commission will, in the absence of good cause shown to the contrary, issue an OL or an appropriate amendment of the license, as the case may be.

- (a)
 - (1) After receiving the report submitted by the ACRS, the Commission may issue an OL if the Commission finds that—
 - (i) Construction of the facility has been substantially completed in conformity with the CP and the application as amended, the provisions of the Act, and the rules and regulations of the Commission;
 - (ii) Any required notifications to other agencies or bodies have been duly made;
 - (iii) The facility will operate in conformity with the application as amended, the provisions of the Act, and the rules and regulations of the Commission;
 - (iv) There is reasonable assurance that—
 - (A) The activities authorized by the OL can be conducted without endangering the health and safety of the public; and
 - (B) Such activities will be conducted in compliance with the regulations in this chapter.
 - (v) The applicant is technically and financially qualified to engage in the activities authorized, however, no finding of financial qualification is necessary for an electric utility applicant for an OL;
 - (vi) Issuance of the license will not be inimical to the common defense and security or to the health and safety of the public;
 - (vii) The applicable provisions of 10 CFR part 140 have been satisfied; and

(viii) The findings required by 10 CFR part 51 have been made.

(2) [Reserved]

(b) [Reserved]

(c) The OL will include appropriate provisions with respect to any uncompleted items of construction and such limitations or conditions as are required to assure that operation during the period of the completion of such items will not endanger public health and safety.

(d) The Commission will issue an OL in such form and containing such conditions and limitations, including technical specifications, as it deems necessary and appropriate.

§ 53.1390 Backfitting of operating licenses.

After issuance of an OL, the Commission may not modify, add, or delete any term or condition of the OL, except in accordance with the provisions of § 53.1590.

§ 53.1396 Duration of operating licenses.

The Commission will issue an OL under this part for the term requested by the applicant, not to exceed 40 years from the date of issuance, or for the estimated useful life of the facility if the Commission determines that the estimated useful life is less than the term requested.

§ 53.1399 Transfer of an operating license.

An OL may be transferred under § 53.1570.

§ 53.1402 Application for renewal.

The filing of an application for a renewed license must be in accordance with § 53.1595.

§ 53.1405 Continuation of an operating license.

Each OL for a facility that has permanently ceased operations continues in effect beyond the expiration date to authorize ownership and possession of the facility until the Commission notifies the licensee in writing that the license is terminated. During this period of continued effectiveness, the licensee must—

(a) Take actions necessary to decommission and decontaminate the facility and continue to maintain the facility, including, where applicable, the storage, control, and maintenance of the spent fuel in a safe condition; and

(b) Conduct activities in accordance with all other restrictions applicable to the facility in accordance with the NRC's regulations and the provisions of the OL for the facility.

§ 53.1410 Combined licenses.

Sections 53.1410 through 53.1461 set out the requirements and procedures applicable to Commission issuance of COLs for commercial nuclear plants under this part.

§ 53.1413 Contents of applications for combined licenses; general information.

An application for a COL must include the information required by § 53.1109 and, except for an electric utility applicant in regard to financial assurance required after a Commission finding under § 53.1452, the application must include information sufficient to demonstrate to the Commission the financial qualification of the applicant to carry out, in accordance with the regulations in this chapter, the activities for which the permit or license is sought. As applicable, the following should be provided:

- (a) The information that demonstrates that the applicant appears to be financially qualified to cover estimated construction costs and related fuel cycle costs, including estimates of the total construction costs and related fuel cycle costs of the facility, a financial capacity plan, and any source(s) of funds available at the time of application to cover these costs. If available funding at the time of application is 50 percent or less, the applicant should include proposed license conditions to facilitate verification that funding is available prior to the start of construction.
- (b) The applicant must submit information that demonstrates the applicant appears to be financially qualified to cover estimated operation costs for the period of the license. The applicant must submit estimates for total annual operating costs for each of the first 5 years of operation of the facility, a financial capacity plan and indicate any source(s) of funds available at the time of application to cover these costs. If available funding at the time of application is 50 percent or less, the applicant should include proposed license conditions to facilitate verification that funding is available prior to the start of operations.

§ 53.1416 Contents of applications for combined licenses; technical information.

- (a) The application must contain an FSAR that describes the facility and the limits on its operation and presents a safety analysis of the SSCs of the facility as a whole. The Commission will require, before issuance of a COL, that information normally contained in engineering documents, such as analyses, drawings, procurement specifications, or construction and installation specifications, be completed and available for audit if the more detailed information is necessary for the Commission to verify the information in the application and make its safety determination. The FSAR must include the following information, at a level of detail sufficient to enable the Commission to reach a final conclusion on all safety matters that must be resolved by the Commission before issuance of a COL:
 - (1) **Site information.** An application for a COL for a commercial nuclear reactor must include the site information required for an early site permit in § 53.1146(a)(1)(iv) through (x).
 - (2) **Design information.** An application for a COL for a commercial nuclear plant must include the design information equivalent to that required for a standard design certification as defined in § 53.1239(a)(2) through (7), (a)(9), (a)(11), (a)(12), (a)(14) through (21), and (a)(23).
 - (3) **Technical qualifications.** A description of the technical qualifications of the applicant to engage in the proposed activities in accordance with the regulations in this chapter.
 - (4) **Integrity assessment program.** A description of an Integrity Assessment Program that addresses the elements described in § 53.870.
 - (5) **Safeguards information.** A description of the program to protect Safeguards Information against unauthorized disclosure in accordance with the requirements in §§ 73.21 and 73.22 of this chapter, as applicable.

- (6) **Emergency response facility or facilities.** Description of the locations and capabilities to be established for command and control, support, and coordination of onsite and offsite, as applicable, functions during reactor accident conditions.
- (7) **Role of personnel.**
 - (i) A description of the completed assessments related to the role of personnel in ensuring safe operations considering the analyses required by § 53.730. These assessments must include the following:
 - (A) Human factors engineering design requirements of § 53.730(a);
 - (B) Human system interface design requirements of § 53.730(b);
 - (C) Concept of operations of § 53.730(c); and
 - (D) Functional requirements analysis and function allocation of § 53.730(d);
 - (ii) A description of the program to be used for evaluating and applying operating experience as required by § 53.730(e);
 - (iii) A staffing plan and supporting analyses as required by § 53.730(f).
- (8) **Training, examination, and proficiency programs.**
 - (i) A description of the training, examination, and proficiency programs required by § 53.730(g); and
 - (ii) A description of the training programs required by § 53.830.
- (9) **Emergency plan.** Emergency plans complying with the requirements of § 53.855.
 - (i) The emergency plan must include, as applicable, all emergency plan certifications that have been obtained from the State, local, and participating Tribal governmental agencies with emergency planning responsibilities. The certifications must state that—
 - (A) The proposed emergency plans are practicable;
 - (B) These agencies are committed to participating in any further development of the plans, including any required field demonstrations; and
 - (C) These agencies are committed to executing their responsibilities under the plans in the event of an emergency.
 - (ii) If certifications cannot be obtained after sustained, good faith efforts by the applicant, then the application must contain information, including a utility plan, sufficient to show that the proposed plans provide reasonable assurance that adequate protective measures can and will be taken in the event of a radiological emergency at the site.
- (10) **Organization.** A description of the applicant's organizational structure, allocations of responsibilities and authorities, and personnel qualifications requirements for operation.
- (11) **Maintenance program.** A description of a maintenance program under § 53.715.
- (12) **Quality assurance.** A description of the QAP under § 53.865.
- (13) **Radiation protection program.** A radiation protection program description under § 53.850.

- (14) **Security program.** A physical security plan that describes how the applicant will comply with § 53.860 (and 10 CFR part 11, if applicable, including the identification and description of jobs as required by § 11.11(a) of this chapter, at the proposed facility). The plan must list tests, inspections, audits, and other means to be used to demonstrate compliance with the requirements of 10 CFR parts 11 and 73, if applicable.
- (15) **Safeguards contingency plan.** A safeguards contingency plan in accordance with the criteria set forth in appendix C to 10 CFR part 73. The safeguards contingency plan must include plans for dealing with threats, thefts, and radiological sabotage, as defined in 10 CFR part 73, relating to the SNM and nuclear facilities licensed under this chapter and in the applicant's possession and control. Each application for this type of license must include the information contained in the applicant's safeguards contingency plan.^[1] (Implementing procedures required for this plan need not be submitted for approval.)
- (16) **Security training and qualification.** A training and qualification plan that describes how the applicant will demonstrate compliance with the criteria set forth in § 73.100 of this chapter or appendix B to 10 CFR part 73.
- (17) **Cybersecurity plan.** A cybersecurity plan in accordance with the criteria set forth in § 73.54 or § 73.110 of this chapter.
- (18) **Security, safeguards, and cybersecurity plan implementation.** A description of the implementation of the physical security plan, safeguards contingency plan, training and qualification plan, and cybersecurity plan. Each applicant who prepares a physical security plan, a safeguards contingency plan, a training and qualification plan, or a cybersecurity plan must protect the plans and other related Safeguards Information against unauthorized disclosure in accordance with the requirements of §§ 73.21 and 73.22 of this chapter.
- (19) **Fire protection program.** A description of the fire protection program under § 53.875.
- (20) **Inservice inspection/inservice testing program.** Descriptions of inservice inspection and inservice testing programs under § 53.880.
- (21)-(22) [Reserved]
- (23) **General employee training.** A description of the training program required to demonstrate compliance with § 53.830 and its implementation.
- (24) **Fitness-for-duty program.** A description of the FFD program under part 26 of this chapter and its implementation.
- (25) **Technical specifications.** Proposed technical specifications prepared in accordance with the requirements of § 53.710(a).
- (b) If there are SSCs of the plant for which research and development is necessary to confirm the adequacy of their design, a report which documents the resolution of any safety questions associated with such SSCs.

^[1] A physical security plan that contains all the information required in both § 73.55 or § 73.100 of this chapter and appendix C to 10 CFR part 73 demonstrates compliance with the requirement for a contingency plan.

- (c) A description of how the performance of each safety design feature has been demonstrated capable of fulfilling functional design criteria considering interdependent effects through either analysis, appropriate test programs, prototype testing, operating experience, or a combination thereof, in accordance with § 53.440(a).
- (d) If the COL application references an early site permit, then the following requirements apply:
 - (1) The FSAR need not contain information or analyses submitted to the Commission in connection with the early site permit provided that the FSAR must either include or incorporate by reference the early site permit Site Safety Analysis Report and contain, in addition to the information and analyses otherwise required, information sufficient to demonstrate that the design of the facility falls within the site characteristics and design parameters specified in the early site permit.
 - (2) If the FSAR does not demonstrate that design of the facility falls within the site characteristics and design parameters, the application must include a request for a variance that complies with the requirements of §§ 53.1188(d) and 53.1437.
 - (3) The FSAR must demonstrate that all terms and conditions that have been included in the early site permit will be satisfied by the date of issuance of the COL. Any terms or conditions of the early site permit that could not be met by the time of issuance of the COL must be set forth as terms or conditions of the COL.
 - (4) If the early site permit approves complete and integrated emergency plans, or major features of emergency plans, then the FSAR must include any new or additional information that updates and corrects the information that was provided under § 53.1146(b)(2) and discuss whether the new or additional information materially changes the bases for compliance with the applicable requirements. The application must identify changes to the emergency plans or major features of emergency plans that have been incorporated into the proposed facility emergency plans and that constitute or would constitute a change in an emergency plan that results in reducing the licensee's capability to perform an emergency planning function in the event of a radiological emergency.
 - (5) If complete and integrated emergency plans are approved as part of the early site permit, new certifications meeting the requirements of paragraph (a)(9)(i) of this section are not required.
- (e) If the COL application references a standard design approval, then the following requirements apply:
 - (1) The FSAR need not contain information or analyses submitted to the Commission in connection with the design approval, provided, however, that the FSAR must either include or incorporate by reference the standard design approval FSAR and must contain, in addition to the information and analyses otherwise required, information sufficient to demonstrate that the characteristics of the site fall within the site parameters specified in the design approval. In addition, the plant-specific information of the PRA, other SREs, or a combination thereof must use the information of the PRA, other SREs, or a combination thereof for the design approval and must be updated to account for site-specific design information and any design changes or departures.
 - (2) The FSAR must demonstrate that all terms and conditions that have been included in the design approval will be satisfied by the date of issuance of the COL.
- (f) If the COL application references a standard design certification, then the following requirements apply:
 - (1) The FSAR need not contain information or analyses submitted to the Commission in connection with the standard design certification, provided, however, that the FSAR must either include or incorporate by reference the standard design certification FSAR and must contain, in addition to the information

and analyses otherwise required, information sufficient to demonstrate that the site characteristics fall within the site parameters specified in the standard design certification. In addition, the plant-specific information of the PRA, other SREs, or a combination thereof must use the information of the PRA, other SREs, or a combination thereof for the standard design certification and must be updated to account for site-specific design information and any design changes or departures.

- (2) The FSAR must demonstrate that the interface requirements established for the design under § 53.1239(a)(24) have been met.
 - (3) The FSAR must demonstrate that all requirements and restrictions set forth in the referenced standard design certification rule must be satisfied by the date of issuance of the COL. Any requirements and restrictions set forth in the referenced standard design certification rule that could not be satisfied by the time of issuance of the COL, must be set forth as terms or conditions of the COL.
- (g) If the COL application references the use of one or more manufactured reactors licensed under § 53.1270, then the following requirements apply:
- (1) The FSAR need not contain information or analyses submitted to the Commission in connection with the ML, provided, however, that the FSAR must either include or incorporate by reference the ML FSAR and must contain, in addition to the information and analyses otherwise required, information sufficient to demonstrate that the site characteristics fall within the site parameters specified in the ML. In addition, the plant-specific information of the PRA, other SREs, or a combination thereof must use the information of the PRA, other SREs, or a combination thereof for the manufactured reactor and must be updated to account for site-specific design information and any design changes or departures.
 - (2) The FSAR must demonstrate that the interface requirements established for the design have been met.
 - (3) The FSAR must demonstrate that all terms and conditions that have been included in the ML will be satisfied by the date of issuance of the COL. Any terms or conditions of the ML that could not be met by the time of issuance of the COL, must be set forth as terms or conditions of the COL.
- (h) Each applicant for a COL under this part must protect Safeguards Information against unauthorized disclosure in accordance with the requirements in §§ 73.21 and 73.22 of this chapter, as applicable.

§ 53.1419 Contents of applications for combined licenses; other application content.

- (a) In addition to the FSAR, the application must also include the following:
- (1) *Environmental report.*
 - (i) An environmental report either in accordance with § 51.50(c) of this chapter if an LWA under § 53.1130 is not requested in conjunction with the COL application, or in accordance with §§ 51.49 and 51.50(c) of this chapter if an LWA is requested in conjunction with the COL application; or
 - (ii) If the applicant wishes to request that an LWA under § 53.1130 be issued before issuance of the COL, the information otherwise required by § 53.1130, in accordance with either § 2.101(a)(1) through (a)(4), or § 2.101(a)(9) of this chapter;

- (2) **Availability controls (if not included in the FSAR).** A description of the controls on plant operations, including availability controls, to provide reasonable confidence of safe operation and that the configurations and special treatments for SR SSCs and NSRSS SSCs provide the capabilities and reliabilities required to satisfy the safety criteria of § 53.220 if not addressed by Technical Specifications under § 53.1416(a)(25); and
- (3) **Inspections, tests, analyses, and acceptance criteria.** The proposed inspections, tests, and analyses, including those applicable to emergency planning, that the licensee must perform, and the acceptance criteria that are necessary and sufficient to provide reasonable assurance that, if the inspections, tests, and analyses are performed and the acceptance criteria met, the facility has been constructed and will be operated in conformity with the COL, the provisions of the Act, and the Commission's rules and regulations.
 - (i) If the application references an early site permit with ITAAC, the early site permit ITAAC must apply to those aspects of the COL which are approved in the early site permit.
 - (ii) If the application references a standard design certification, the ITAAC contained in the certified design must apply to those portions of the facility design which are approved in the standard design certification.
 - (iii) If the application references an ML, the ITAAC contained in the ML must apply to those portions of the facility design which are approved in the ML.
 - (iv) If the application references an early site permit with ITAAC, a standard design certification, an ML, or a combination thereof, the application may include a notification that a required inspection, test, or analysis in the ITAAC has been successfully completed and that the corresponding acceptance criterion has been met. The FEDERAL REGISTER notice required by § 53.1422 of this chapter must indicate that the application includes this notification.

(b) [Reserved]

§ 53.1422 Review of applications.

- (a) **Standards for review of applications.** Applications filed under this part will be reviewed according to the standards set out in 10 CFR parts 20, 51, 53, 73, and 140.
- (b) **Administrative review of applications; hearings.** A proceeding on a COL is subject to all applicable procedural requirements contained in 10 CFR part 2, including the requirements for docketing (§ 2.101 of this chapter) and issuance of a notice of hearing (§ 2.104 of this chapter). If an applicant requests a Commission finding on certain ITAAC with the issuance of the COL, then those ITAAC will be identified in the notice of hearing. All contested hearings on COLs are governed by the procedures contained in 10 CFR part 2.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20068, Apr. 15, 2026]

§ 53.1425 Finality of referenced NRC approvals.

If the application for a COL under this part references an early site permit, standard design certification rule, standard design approval, or ML, issued under this part, the scope and nature of matters resolved for the application and any COL issued are governed by the relevant provisions addressing finality, including §§ 53.1188, 53.1221, 53.1263, and 53.1288.

§ 53.1431 Referral to the Advisory Committee on Reactor Safeguards.

The Commission must refer a copy of the application to the ACRS. The ACRS must report on those portions of the application that concern safety and must apply the standards referenced in § 53.1422, in accordance with the finality provisions in § 53.1425.

§ 53.1434 Authorization to conduct limited work authorization activities.

- (a) If the application for a COL under this part does not reference an early site permit which authorizes the holder to perform the activities under § 53.1130(b), the applicant may not perform those activities without obtaining the separate authorization required by § 53.1130(a). Authorization may be granted only after the presiding officer in a contested hearing on the application has made the findings and determination required by § 53.1130(b)(1)(iii), and the Director, Office of Nuclear Reactor Regulation makes the determination required by § 53.1130(b)(1)(ii).
- (b) If, after an applicant has performed the activities permitted by a LWA issued under § 53.1130, the application for the COL is withdrawn or denied, then the applicant must implement the approved site redress plan.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20068, Apr. 15, 2026]

§ 53.1437 Exemptions, departures, and variances.

- (a) An applicant for a COL, or any amendment to a COL, may include in the application a request for an exemption from one or more of the Commission's regulations.
 - (1) If the request is for an exemption from any part of a referenced standard design certification rule, the Commission may grant the request if it determines that the exemption complies with any exemption provisions of the referenced standard design certification rule, or with § 53.1263 if there are no applicable exemption provisions in the referenced standard design certification rule.
 - (2) For all other requests for exemptions, the Commission may grant a request if it determines that the exemption complies with § 53.080.
- (b) An applicant for a COL who has filed an application referencing an early site permit issued under § 53.1158 may include in the application a request for a variance from one or more site characteristics, design parameters, or terms and conditions of the permit, or from the Site Safety Analysis Report. In determining whether to grant the variance, the Commission must apply the same technically relevant criteria as were applicable to the application for the original or renewed site permit. Once a COL referencing an early site permit is issued, variances from the early site permit will not be granted for that CP or COL.
- (c) An applicant for a COL who has filed an application referencing use of a manufactured reactor may include in the application a request for a departure from one or more design characteristics, site parameters, terms and conditions, or approved design of the manufactured reactor under the ML issued under § 53.1287. The Commission may grant such a request only if it determines that the departure will comply with the requirements of § 53.080, and that the special circumstances outweigh any decrease in safety that may result from the reduction in standardization caused by the departure.

- (d) Issuance of a variance under paragraph (b) of this section or a departure under paragraph (c) of this section is subject to litigation during the COL proceeding in the same manner as other issues material to that proceeding.

§ 53.1440 Issuance of combined licenses.

- (a)
 - (1) The Commission may issue a COL if the Commission finds that—
 - (i) The applicable standards and requirements of the Act and the Commission's regulations have been met;
 - (ii) Any required notifications to other agencies or bodies have been duly made;
 - (iii) There is reasonable assurance that the facility will be constructed and will operate in conformity with the license, the provisions of the Act, and the Commission's regulations;
 - (iv) The applicant is technically and financially qualified to engage in the activities authorized; however, no finding of financial qualification is necessary for an electric utility applicant for a COL;
 - (v) Issuance of the license will not be inimical to the common defense and security or to the health and safety of the public; and
 - (vi) The findings required by 10 CFR part 51 have been made.
 - (2) The Commission may also find, at the time it issues the COL, that certain acceptance criteria in one or more of the ITAAC in a referenced early site permit, standard design certification, or ML have been met. This finding will finally resolve that those acceptance criteria have been met, those acceptance criteria will be deemed to be excluded from the COL, and findings under § 53.1452(g) with respect to those acceptance criteria are unnecessary.
- (b) The Commission must identify within the COL the inspections, tests, and analyses, including those applicable to emergency planning, that the licensee must perform, and the acceptance criteria that, if met, are necessary and sufficient to provide reasonable assurance that the facility has been constructed and will be operated in conformity with the license, the provisions of the Act, and the Commission's rules and regulations.
- (c) A COL must contain the terms and conditions, including technical specifications, as the Commission deems necessary and appropriate.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20068, Apr. 15, 2026]

§ 53.1443 Finality of combined licenses.

- (a) After issuance of a COL, the Commission may not modify, add, or delete any term or condition of the COL, the design of the facility, the ITAAC contained in the license that are not derived from a referenced standard design certification or ML, except under the provisions of § 53.1452 or § 53.1590.
- (b) If the COL does not reference a standard design certification, then a licensee may make changes in the facility as described in the FSAR (as updated) and make changes in the procedures as described in the FSAR (as updated) under the applicable change processes in § 53.1550.
- (c) If the COL references a certified design, then—

- (1) Changes to or departures from information within the scope of the referenced standard design certification rule are subject to the applicable change processes in that rule; and
- (2) Changes that are not within the scope of the referenced standard design certification rule are subject to the applicable change processes in subpart I of this part, unless they also involve changes to or noncompliance with information within the scope of the referenced standard design certification rule. In these cases, the applicable provisions of this section and the standard design certification rule apply.
- (d) [Reserved]
- (e) The Commission may issue and make immediately effective any amendment to a COL upon a determination by the Commission that the amendment involves no significant hazards consideration, notwithstanding the pendency before the Commission of a request for a hearing from any person. The amendment may be issued and made immediately effective in advance of the holding and completion of any required hearing. The amendment will be processed under the procedures specified in § 53.1515.
- (f) Any modification to, addition to, or deletion from the terms and conditions of a COL, including any modification to, addition to, or deletion from the inspections, tests, and analyses, or related acceptance criteria contained in the license is a proposed amendment to the license. There must be an opportunity for a hearing on the amendment.

§ 53.1449 Inspection during construction.

- (a) **Licensee schedule for inspections, tests, or analyses.** The licensee must submit to the NRC, no later than 1 year after issuance of the COL or at the start of construction as defined at § 53.020, whichever is later, its schedule for completing the inspections, tests, or analyses in the ITAAC. The licensee must submit updates to the ITAAC schedules every 6 months thereafter and, within 1 year of its scheduled date for initial loading of fuel (or, for a fueled manufactured reactor, within 1 year of its scheduled date for initiating the removal of the features to prevent criticality required under § 53.620(d)(1)), the licensee must submit updates to the ITAAC schedule every 30 days until the final notification is provided to the NRC under paragraph (c)(1) of this section.
- (b) **Licensee and applicant conduct of activities subject to ITAAC.** With respect to activities subject to an ITAAC, an applicant for a COL may proceed at its own risk with design and procurement activities, and a licensee may proceed at its own risk with design, procurement, construction, and preoperational activities, even though the NRC may not have found that any one of the prescribed acceptance criteria are met.
- (c) **Licensee notifications —**
 - (1) **ITAAC closure notification.** The licensee must notify the NRC that prescribed inspections, tests, and analyses have been performed and that the prescribed acceptance criteria are met. The notification must contain sufficient information to demonstrate that the prescribed inspections, test, and analyses have been performed and that the prescribed acceptance criteria are met.
 - (2) **ITAAC post-closure notifications.** Following the licensee's ITAAC closure notifications under paragraph (c)(1) of this section until the Commission makes the finding under § 53.1452(g), the licensee must notify the NRC, in a timely manner, of new information that materially alters the basis for determining that either inspections, tests, or analyses were performed as required, or that acceptance criteria are met. The notification must contain sufficient information to demonstrate that, notwithstanding the new information, the prescribed inspections, tests, and analyses have been performed as required, and the prescribed acceptance criteria are met.

(3) **Uncompleted ITAAC notification.** If the licensee has not provided, by the date 225 days before the scheduled date for initial loading of fuel (or, for a fueled manufactured reactor, by the date 225 days before the scheduled date for initiating the removal of the features to prevent criticality required under § 53.620(d)(1)), the notification required by paragraph (c)(1) of this section for all ITAAC, then the licensee must notify the NRC that the prescribed inspections, tests, or analyses for all uncompleted ITAAC will be performed and that the prescribed acceptance criteria will be met prior to operation. The notification must be provided no later than the date 225 days before the scheduled date for initial loading of fuel (or, for a fueled manufactured reactor, no later than the date 225 days before the scheduled date for initiating the removal of the features to prevent criticality required under § 53.620(d)(1)), and must provide sufficient information to demonstrate that the prescribed inspections, tests, or analyses will be performed and the prescribed acceptance criteria for the uncompleted ITAAC will be met, including, but not limited to, a description of the specific procedures and analytical methods to be used for performing the prescribed inspections, tests, and analyses and determining that the prescribed acceptance criteria are met.

(4) **All ITAAC complete notification.** The licensee must notify the NRC that all ITAAC are complete.

(d) **Licensee determination of noncompliance with ITAAC.**

(1) In the event that an activity is subject to an ITAAC derived from a referenced standard design certification and the licensee has not demonstrated that the prescribed acceptance criteria are met, the licensee may take corrective actions to successfully complete that ITAAC or request an exemption from the standard design certification ITAAC, as applicable. A request for an exemption must also be accompanied by a request for a license amendment under subpart I.

(2) In the event that an activity is subject to an ITAAC not derived from a referenced standard design certification and the licensee has not demonstrated that the prescribed acceptance criteria are met, the licensee may take corrective actions to successfully complete that ITAAC or request a license amendment under subpart I.

(e) **NRC inspection, publication of notices, and availability of licensee notifications.** The NRC must ensure that the prescribed inspections, tests, and analyses in the ITAAC are performed.

(1) At appropriate intervals until the last date for submission of requests for hearing under § 53.1452, the NRC must publish notices in the FEDERAL REGISTER of the NRC staff's determination of the successful completion of inspections, tests, and analyses.

(2) The NRC must make publicly available the licensee notifications under paragraph (c) of this section. The NRC must, no later than the date of publication of the notice of intended operation required by § 53.1452(a), make publicly available those licensee notifications under paragraph (c) of this section that have been submitted to the NRC at least 7 days before that notice.

§ 53.1452 Operation under a combined license.

(a) The licensee must notify the NRC of its scheduled date for initial loading of fuel no later than 270 days before the scheduled date and must notify the NRC of updates to its schedule every 30 days thereafter.^[1] Not less than 180 days before the date scheduled for initial loading of fuel into a plant by a licensee that has been issued a COL under this part, the Commission must publish notice of intended operation in the FEDERAL REGISTER.^[2] The notice must provide that any person whose interest may be affected by operation of the plant may, within 60 days, request that the Commission hold a hearing on whether the facility as constructed complies, or on completion will comply, with the acceptance criteria in the COL, except that a hearing must not be granted for those ITAAC that the Commission found were met under § 53.1440(a)(2).

- (b) A request for hearing under paragraph (a) of this section must show, *prima facie*—
 - (1) That one or more of the acceptance criteria of the ITAAC in the COL have not been, or will not be, met; and
 - (2) The specific operational consequences of nonconformance that would be contrary to providing reasonable assurance of adequate protection of the public health and safety.
- (c) The Commission, acting as the presiding officer, must determine whether to grant or deny the request for hearing under the applicable requirements of § 2.309 of this chapter. If the Commission grants the request, the Commission, acting as the presiding officer, must determine whether during a period of interim operation there will be reasonable assurance of adequate protection to the public health and safety. The Commission's determination must consider the petitioner's *prima facie* showing and any answers thereto. If the Commission determines there is such reasonable assurance, it must allow operation during an interim period under the COL.
- (d) The Commission, in its discretion, must determine appropriate hearing procedures, whether informal or formal adjudicatory, for any hearing under paragraph (a) of this section, and must state its reasons therefor.
- (e) The Commission must, to the maximum possible extent, render a decision on issues raised by the hearing request within 180 days of the publication of the notice provided by paragraph (a) of this section or by the anticipated date for initial loading of fuel into the reactor, whichever is later.
- (f) A petition to modify the terms and conditions of the COL will be processed as a request for action under § 2.206 of this chapter. The petitioner must file the petition with the Secretary of the Commission. Before the licensed activity allegedly affected by the petition (fuel loading, low power testing, etc.) commences, the Commission must determine whether any immediate action is required. If the petition is granted, then an appropriate order will be issued. Fuel loading and operation under the COL will not be affected by the granting of the petition unless the order is made immediately effective.
- (g) The licensee must not operate the facility until the Commission makes a finding that the acceptance criteria in the COL are met, except for those acceptance criteria that the Commission found were met under § 53.1440(a)(2). If the COL is for a modular design, each reactor unit may require a separate finding as construction proceeds.
- (h) After the Commission has made the finding in paragraph (g) of this section, the ITAAC do not, by virtue of their inclusion in the COL, constitute regulatory requirements either for licensees or for renewal of the license; except for the specific ITAAC for which the Commission has granted a hearing under paragraph (a) of this section, all ITAAC expire upon final Commission action in the proceeding. However, subsequent changes to the facility or procedures described in the FSAR (as updated) must comply with the requirements in § 53.1443(e) or (f), as applicable.

^[2] For licensees installing fueled manufactured reactors under a COL, the Commission must instead publish notice of intended operation in the FEDERAL REGISTER not less than 180 days before the date scheduled for initiating the removal of the features to prevent criticality required under § 53.620(d)(1).

^[1] For licensees installing fueled manufactured reactors under a COL, the COL holder must instead notify the NRC of its scheduled date for initiating the removal of the features to prevent criticality required under § 53.620(d)(1) no later than 270 days before the scheduled date and must notify the NRC of updates to its schedule every 30 days thereafter.

§ 53.1455 Duration of combined license.

A COL is issued for a specified period not to exceed 40 years from the date on which the Commission makes a finding that acceptance criteria are met under § 53.1452(g) or allowing operation during an interim period under the COL under § 53.1452(c).

§ 53.1456 Transfer of a combined license.

A COL may be transferred under § 53.1570.

§ 53.1458 Application for renewal.

The filing of an application for a renewed license must be in accordance with § 53.1595.

§ 53.1461 Continuation of combined license.

Each COL for a facility that has permanently ceased operations continues in effect beyond the expiration date to authorize ownership and possession of the facility until the Commission notifies the licensee in writing that the license is terminated. During this period of continued effectiveness, the licensee must—

- (a) Take actions necessary to decommission and decontaminate the facility and continue to maintain the facility, including, where applicable, the storage, control and maintenance of the spent fuel, in a safe condition; and
- (b) Conduct activities in accordance with all other restrictions applicable to the facility in accordance with the NRC's regulations and the provisions of the COL for the facility.

§ 53.1470 Standardization of commercial nuclear plant designs: licenses to construct and operate nuclear power reactors of identical design at multiple sites.

- (a) Except as otherwise specified in this section, the provisions of this section apply to CP, OL, and COL applications for commercial nuclear plants of identical design (the “common design”) under this part.
- (b) Each application for a CP, OL, or COL submitted pursuant to this section must be submitted as specified in § 53.1300, § 53.1360, or § 53.1410, respectively, and § 2.101 of this chapter. Each application must state that the applicant wishes to construct a facility identical to a facility proposed for one or more sites other than the applicant's (the “common design”), and the applicant wishes to have the application considered under this section. Each application must list each of the other applications to be treated together under this section or specify that such other applications will be submitted to the NRC within 12 months of submittal of the first application.
- (c) Each application must include the information required by the applicable sections of this subpart, provided, however, that the application must identify the common design, and, if applicable, reference a standard design certification or standard design approval under this part, or the use of a reactor manufactured under this part. The FSAR for each application must either incorporate by reference or include the final safety analysis of the common design, including, if applicable, the FSAR for the referenced standard design certification, standard design approval, or the manufactured reactor.

- (d) Each application submitted pursuant to this section must contain an environmental report under § 53.1312(a)(1), § 53.1372(a), or § 53.1419(a)(1), as applicable, that complies with the applicable provisions of 10 CFR part 51, *provided, however*, that the application may incorporate by reference a single environmental report on the environmental impacts of the common design that are applicable to each site.
- (e) Upon a determination that each application is acceptable for docketing under § 2.101 of this chapter, each application will be docketed and a notice of docketing for each application will be published in the FEDERAL REGISTER, under § 2.104 of this chapter, *provided, however*, that the notice must state that the application will be processed under the provisions of this section and subpart D of 10 CFR part 2. At the discretion of the Commission, a single notice of docketing for multiple applications may be published in the FEDERAL REGISTER.
- (f) The NRC must prepare an environmental assessment or draft and final environmental impact statements for each of the applications under 10 CFR part 51. Scoping under §§ 51.28 and 51.29 of this chapter for each of the license applications may be conducted simultaneously and joint scoping may be conducted with respect to the environmental issues relevant to the common design. If the applications reference a standard design certification, then the environmental assessment or environmental impact statement for each of the applications must incorporate by reference the standard design certification environmental assessment. If the applications do not reference a standard design certification, then the NRC must prepare environmental assessments or draft and final supplemental environmental impact statements which address severe accident mitigation design alternatives for the common design, which must be incorporated by reference into the environmental assessment or environmental impact statement prepared for each application. Scoping under §§ 51.28 and 51.29 of this chapter for the supplemental environmental impact statement may be conducted simultaneously and may be part of the scoping for each of the applications.
- (g) The ACRS must report on each of the applications as required by the applicable sections of this subpart. Each report must be limited to those safety matters for each application that are not relevant to the common design. In addition, the ACRS must separately report on the safety of the common design, *provided, however*, that the report need not address the safety of a referenced standard design certification or reactor manufactured under this part.
- (h) If the design for the power reactor(s) proposed in a particular application is not identical to the others, that application may not be processed under this section and subpart D of 10 CFR part 2.
- (i) As used in this section, the design of a nuclear power reactor included in a single referenced Safety Analysis Report means the design of those SSCs important to radiological health and safety and the common defense and security.

[91 FR 15794, Mar. 30, 2026, as amended at 91 FR 20068, Apr. 15, 2026]

Subpart I—Maintaining and Revising Licensing-Basis Information

§ 53.1500 Licensing-basis information.

This subpart provides the requirements for each holder of a license for a commercial nuclear plant licensed under this part to maintain licensing-basis information as defined in § 53.020; evaluate changes to site characteristics, plant design features, and programmatic controls to determine needed approvals and revisions; and submit appropriate updates to the U.S. Nuclear Regulatory Commission (NRC).

§ 53.1502 Specific terms and conditions of licenses.

- (a) Each license issued under this part is subject to the provisions of the Atomic Energy Act of 1954, as amended, (the Act) and to all rules, regulations, and orders of the Commission. The terms and conditions of the license will be subject to amendment, revision, or modification, by reason of amendments of the Act or by reason of rules, regulations, and orders issued in accordance with the terms of the Act.
- (b) Each license issued under this part must be subject to all conditions imposed as a matter of law by sections 401(a)(2) and 401(d) of the Federal Water Pollution Control Act, as amended (33 U.S.C.A. 1341(a)(2) and (d)).
- (c) A holder of an operating license (OL) or combined license (COL) under this part may take reasonable action that departs from a license condition or a technical specification included in a license issued under this part in a national security emergency established by a law enacted by the Congress or by an order or directive issued by the President pursuant to statutes or the Constitution of the United States. The authority under this paragraph (c) must be exercised in accordance with law, including section 57e of the Act, and is in addition to the authority granted under § 53.740(h), which remains in effect unless otherwise directed by the Commission during a national security emergency. The authority under this paragraph (c) may be exercised—
 - (1) When this action is immediately needed to implement national security objectives as designated by the national command authority through the Commission; and
 - (2) No action consistent with license conditions and technical specifications that can satisfy national security objectives is immediately apparent.
- (d)
 - (1) If the NRC finds that the state of emergency preparedness does not provide reasonable assurance that adequate protective measures can and will be taken in the event of a radiological emergency (including findings based on requirements of 10 CFR part 50, appendix E, section IV.D.3) and if the deficiencies (including deficiencies based on requirements of 10 CFR part 50, appendix E, section IV.D.3) are not corrected within 4 months of that finding, the Commission will determine whether the facility must be shut down or cease operations until such deficiencies are remedied or whether other enforcement action is appropriate. In determining whether a shutdown or other enforcement action is appropriate, the Commission will take into account, among other factors, whether the licensee can demonstrate to the Commission's satisfaction that the deficiencies in the plan are not significant for the plant in question, or that adequate interim compensating actions have been or will be taken promptly, or that there are other compelling reasons for continued operation.
 - (2) If the planning standards for radiological emergency preparedness apply to offsite emergency response plans, or if the planning activities in § 50.160(b)(1)(iv)(B) apply, then the NRC will base its finding on a review of the Federal Emergency Management Agency findings and determinations as to whether State, participating Tribal, and local emergency plans are adequate and capable of being implemented, and on the NRC assessment as to whether the licensee's emergency plans are adequate and capable of being implemented. Nothing in this paragraph (d)(2) must be construed as limiting the authority of the Commission to take action under any other regulation or authority of the Commission or at any time other than that specified in this paragraph (d)(2).

§ 53.1505 Changes to licensing-basis information requiring prior NRC approval.

- (a) Sections 53.1510 through 53.1520 provide the process for a licensee to request and the NRC to issue amendments to licenses, including any conditions contained therein, technical specifications or other attachments to a license, and any orders issued by the NRC modifying a license. Sections 53.1525 and 53.1530 govern proposed changes to a commercial nuclear plant referencing a certified design or manufacturing license (ML).
- (b) A licensee may propose changing licensing-basis information established by NRC regulations by requesting an exemption in accordance with § 53.080.

§ 53.1510 Application for amendment of license.

Whenever a holder of a license under this part desires to amend the license, an application for an amendment must be filed with the Commission, as specified in § 53.040, that fully describes the changes desired and, following as far as applicable, the form prescribed for original applications. Applications for amendments involving changes to plant structures, systems, and components (SSCs), programmatic controls, or the role of plant personnel must include an assessment of the changes in relation to the safety requirements in subpart B of this part and the analyses requirements of § 53.450 as applicable, an analysis of whether the amendment involves no significant hazards consideration using the standards in § 53.1520, and a consideration of environmental factors.

§ 53.1515 Public notices; State consultation.

The Commission will use the following procedures for an application requesting an amendment to an OL or COL issued under this part.

(a) *Public notices.*

(1)

- (i) The Commission may publish in the FEDERAL REGISTER under § 2.105 of this chapter an individual notice of proposed action for an amendment for which it makes a proposed determination that no significant hazards consideration is involved, or, at least once every 30 days, publish a periodic FEDERAL REGISTER notice of proposed actions, which identifies each amendment issued and each amendment proposed to be issued since the last such periodic notice, or it may publish both such notices.
 - (ii) For each amendment proposed to be issued, the notice will
 - (A) Contain the staff's proposed determination under the standards in § 53.1520;
 - (B) Provide a brief description of the amendment and of the facility involved;
 - (C) Solicit public comments on the proposed determination; and
 - (D) Provide for a 30-day comment period.
 - (iii) The comment period will begin on the day after the date of the publication of the first notice, and, normally, the amendment will not be granted until after this comment period expires.
- (2) The Commission may inform the public about the final disposition of an amendment request for which it has made a proposed determination of no significant hazards consideration either by issuing an individual notice of issuance under § 2.106 of this chapter or by publishing such a notice

in its periodic system of FEDERAL REGISTER notices. In either event, it will not make and will not publish a final determination of no significant hazards consideration unless it receives a request for a hearing on that amendment request.

- (3) Where the Commission makes a final determination that no significant hazards consideration is involved and that the amendment should be issued, the amendment will be effective on issuance, even if adverse public comments have been received and even if an interested person meeting the provisions for intervention called for in § 2.309 of this chapter has filed a request for a hearing. The Commission need hold any required hearing only after it issues an amendment, unless it determines that a significant hazards consideration is involved, in which case the Commission will provide an opportunity for a prior hearing.
- (4) Where the Commission finds that an emergency situation exists, in that failure to act in a timely way would result in derating or shutdown of a commercial nuclear reactor, or in prevention of either resumption of operation or of increase in power output up to the plant's licensed power level, it may issue a license amendment involving no significant hazards consideration without prior notice and opportunity for a hearing or for public comment. In such a situation, the Commission will not publish a notice of proposed determination on no significant hazards consideration but will publish a notice of issuance under § 2.106 of this chapter providing for opportunity for a hearing and for public comment after issuance. The Commission expects its licensees to apply for license amendments in a timely fashion. It will decline to dispense with notice and comment on the determination of no significant hazards consideration if it determines that the licensee has abused the emergency provision by failing to make timely application for the amendment and thus itself creating the emergency. Whenever an emergency situation exists, a licensee requesting an amendment must explain why this emergency situation occurred and why it could not avoid this situation, and the Commission will assess the licensee's reasons for failing to file an application sufficiently in advance of that event.
- (5) Where the Commission finds that exigent circumstances exist, in that a licensee and the Commission must act quickly and that time does not permit the Commission to publish a FEDERAL REGISTER notice allowing 30 days for prior public comment, and it also determines that the amendment involves no significant hazards considerations, it—
 - (i)
 - (A) Will either issue a FEDERAL REGISTER notice providing notice of an opportunity for hearing and allowing at least 2 weeks from the date of the notice for prior public comment; or
 - (B) Will use local media to provide reasonable notice to the public in the area surrounding a licensee's facility of the licensee's amendment and of its proposed determination as described in paragraph (a)(1) of this section, consulting with the licensee on the proposed media release and on the geographical area of its coverage;
 - (ii) Will provide for a reasonable opportunity for the public to comment, using its best efforts to make available to the public whatever means of communication it can for the public to respond quickly, and, in the case of telephone comments, have these comments recorded or transcribed, as necessary and appropriate;
 - (iii) When it has issued a local media release, may inform the licensee of the public's comments, as necessary and appropriate;
 - (iv) Will publish a notice of issuance under § 2.106 of this chapter;

- (v) Will provide a hearing after issuance, if one has been requested by a person who satisfies the provisions for intervention specified in § 2.309 of this chapter; and
 - (vi) Will require the licensee to explain the exigency and why the licensee cannot avoid it and use its normal public notice and comment procedures in paragraph (a)(1) of this section if it determines that the licensee has failed to use its best efforts to make a timely application for the amendment in order to create the exigency and to take advantage of this procedure.
- (6) Where the Commission finds that significant hazards considerations are involved, it will issue a FEDERAL REGISTER notice providing an opportunity for a prior hearing even in an emergency situation, unless it finds an imminent danger to the health or safety of the public, in which case it will issue an appropriate order or rule under 10 CFR part 2.

(b) **State consultation.**

- (1) At the time a licensee requests an amendment, it must notify the State in which its facility is located of its request by providing that State with a copy of its application and its reasoned analysis about no significant hazards considerations and indicate on the application that it has done so.
- (2) The Commission will advise the State of its proposed determination about no significant hazards consideration normally by sending it a copy of the FEDERAL REGISTER notice.
- (3) The Commission will make the names of the Project Manager or other NRC personnel it designated to consult with the State available to the State official designated to consult about its proposed determination. The Commission will consider any comments of that State official. If it does not hear from the State in a timely manner, it will consider that the State has no interest in its determination; nonetheless, to ensure that the State is aware of the application, before it issues the amendment, it will make a good faith effort to communicate directly with that official. (Inability to consult with a responsible State official following good faith attempts will not prevent the Commission from making effective a license amendment involving no significant hazards consideration.)
- (4) The Commission will make a good faith attempt to consult with the State before it issues a license amendment involving no significant hazards consideration. If, however, it does not have time to use its normal consultation procedures because of an emergency situation, it will attempt to communicate directly with the appropriate State official. (Inability to consult with a responsible State official following good faith attempts will not prevent the Commission from making effective a license amendment involving no significant hazards consideration, if the Commission deems it necessary in an emergency situation.)
- (5) After the Commission issues the requested amendment, it will send a copy of its determination to the State.

(c) **Caveats about State consultation.**

- (1) The State consultation procedures in paragraph (b) of this section do not give the State a right—
 - (i) To veto the Commission's proposed or final determination;
 - (ii) To a hearing on the determination before the amendment becomes effective; or
 - (iii) To insist upon a postponement of the determination or upon issuance of the amendment.

- (2) These procedures do not alter present provisions of law that reserve to the Commission exclusive responsibility for setting and enforcing radiological health and safety requirements for commercial nuclear plants.

§ 53.1520 Issuance of amendment.

- (a) In determining whether an amendment to a license will be issued to the applicant, the Commission will be guided by the considerations which govern the issuance of initial licenses to the extent applicable and appropriate. If the application is for amendment of an OL or COL and involves the material alteration of a commercial nuclear plant, a construction permit (CP) will be issued before the issuance of the amendment to the license, provided however, that if the application involves a material alteration to a manufactured reactor under this part before its installation at a site, or a COL before the date that the Commission makes the finding under § 53.1452(g), no application for or issuance of a CP is required. If the amendment involves a significant hazards consideration, the Commission will give notice of its proposed action—
 - (1) Under § 2.105 of this chapter before acting thereon; and
 - (2) As soon as practicable after the application has been docketed.
- (b) The Commission will be particularly sensitive to a license amendment request that involves irreversible consequences (such as one that permits a significant increase in the amount of effluents or radiation emitted by a commercial nuclear plant).
- (c) The Commission may make a final determination, under the procedures in § 53.1515, that a proposed amendment to an OL or a COL for a commercial nuclear plant under this part involves no significant hazards consideration, if operation of the plant in accordance with the proposed amendment would not—
 - (1) Involve a significant increase in the probability or consequences of an accident previously evaluated; or
 - (2) Create the possibility of a new or different kind of an accident from any accident previously evaluated; or
 - (3) Involve a significant reduction in a margin of safety.

§ 53.1525 Revising certification information within a design certification rule.

- (a) A holder of an OL or COL who references a design certification rule issued under this part must request an exemption if proposing to change one or more elements of the certification information. The Commission may grant such a request only if it determines that the exemption will comply with the requirements of § 53.080 and that the special circumstances outweigh any decrease in safety that may result from the reduction in standardization caused by the departure.
- (b) The request for an exemption must be included with any associated license amendment request, which must be requested and processed in accordance with §§ 53.1510, 53.1515, and 53.1520.
- (c) Licensees must evaluate changes to the design as described in the Final Safety Analysis Report (FSAR) not involving changes to the certification information using the criteria in § 53.1550.

§ 53.1530 Revising information within a Final Safety Analysis Report associated with a manufacturing license.

- (a) The holder of an ML may make changes to the facility or procedures as described in the Final Safety Analysis Report (FSAR) associated with the ML without obtaining a license amendment pursuant to § 53.1510 if the change meets the criteria in § 53.1550(a)(1) and (2) using the specifications in § 53.1550(b). If needed, applications for amending an ML must be submitted and processed in accordance with §§ 53.1510, 53.1515, and 53.1520. In those cases where an ML references a design certification rule, the amendment application from the holder of the ML must also request an exemption from the design certification rule under § 53.1525 if proposing to change one or more elements of the certification information.
- (b)
 - (1) The holder of an ML must maintain records of changes to the facility or procedures made without a license amendment under paragraph (a) of this section. These records must include a written evaluation which provides the bases for the determination that the change does not require a license amendment under the criteria in paragraph § 53.1550(a)(2).
 - (2) The records of changes in the facility must be maintained until the expiration of an ML issued under this part, or the expiration of a renewed license issued under § 53.1295—whichever is later. Records of changes in procedures must be maintained for a period of 5 years.

§ 53.1535 Amendments during construction.

- (a) The holder of a CP or limited work authorization (LWA) under this part may request an amendment to the CP or LWA in order to gain Commission approval of the safety of selected design features or specifications, including proposed departures from a design certification rule or ML. Amendments to CPs or LWAs under this part must be requested and processed under §§ 53.1510 and 53.1520.
- (b) The holder of a COL under this part for which the NRC has not yet made a finding in accordance with § 53.1452(g) must request amendments required by § 53.1525 or § 53.1550 no later than 45 days from the date the licensee begins the construction of the SSCs to implement the change or departure requiring NRC approval. The licensee proceeds with such changes at its own risk recognizing that there is a possibility that the amendment will not be granted.

§ 53.1540 Updating licensing-basis information and determining the need for NRC approval.

- (a) Sections 53.1545 through 53.1565 provide the process for a holder of an OL or COL to modify licensing-basis information and to evaluate potential changes to its facilities, procedures, programs, and organizations to determine if NRC approval is required.
- (b) Definitions for the purposes of §§ 53.1545 through 53.1565—

Change means a modification or addition to, or removal from, the commercial nuclear plant or procedures that affects a design feature or related functional design criteria, method of performing or controlling the functions of design features, or an evaluation that demonstrates that intended functions will be accomplished.

Departure from a method of evaluation described in the Final Safety Analysis Report (FSAR) (as updated) used in establishing the functional design criteria for safety-related structures, systems, or components or in the safety analyses

means—

- (i) Changing any of the elements of the method described in the FSAR (as updated) unless the results of the analysis are conservative or essentially the same; or
- (ii) Changing from a method described in the FSAR to another method unless that method has been approved by NRC for the intended application.

Facility as described in the FSAR (as updated) means—

- (i) The SSCs that are described in the FSAR (as updated),
- (ii) The design and performance requirements for such SSCs described in the FSAR (as updated), and
- (iii) The evaluations or methods of evaluation included in the FSAR (as updated) for such SSCs which demonstrate that their intended function(s) will be accomplished.

Final Safety Analysis Report (as updated) means the FSAR submitted under § 53.1279, § 53.1369 or § 53.1416, as amended and supplemented, and as updated under § 53.1545, as applicable.

Procedures as described in the Final Safety Analysis Report (as updated) means those procedures that contain information described in the FSAR (as updated) such as how SSCs are operated and controlled (including assumed operator actions and response times).

§ 53.1545 Updating Final Safety Analysis Reports.

- (a) Each holder of an OL or COL under this part for which the Commission has made the finding under § 53.1452(g) must update the FSAR originally submitted as part of the application for the license every 24 months or more frequently to assure that the information included in the report contains the latest information developed. The submittal must include the effects on the content of the FSAR of—
 - (1) Changes made to the facility or procedures as described in the FSAR;
 - (2) Safety analyses and evaluations performed by the licensee either in support of approved license amendments or in support of conclusions that changes did not require a license amendment under § 53.1550;
 - (3) Updates to the probabilistic risk assessment (PRA), other systematic risk evaluations, or a combination thereof required under § 53.450(a);
 - (4) The cumulative effects of the changes to the facility or procedures on the margins to the safety criteria in §§ 53.210, 53.220, and 53.450(e) since the last FSAR update; and
 - (5) Analyses of new safety issues performed by or on behalf of the licensee at Commission request.
- (b)
 - (1) The licensee must submit revisions containing updated information to the Commission, under § 53.040, identifying the location of revised or new information.
 - (2) The submittal must include—
 - (i) A certification by a duly authorized officer of the licensee that either the information accurately presents changes made since the previous submittal, necessary to reflect information and analyses submitted to the Commission or prepared pursuant to Commission requirement, or that no such changes were made; and

- (ii) An identification of changes made under the provisions of § 53.1550 but not previously submitted to the Commission.
- (c) Each applicant for or holder of a COL under this part for which the Commission has not made the finding under § 53.1452(g) must submit an update to the FSAR annually by providing the information required in paragraphs (a)(1) through (a)(5) of this section and meeting the requirements of paragraph (b) of this section. Combined license applicants who have requested the NRC to suspend its review of the COL application and COL holders who have informed the NRC that they do not plan to pursue construction need not submit an annual update of the FSAR. If a COL applicant requests that the NRC resume its review, or a COL holder notifies the NRC that the COL holder plans to commence or resume construction, then the COL applicant or holder must submit to NRC an update to its FSAR within 90 days of the request or notification, as applicable, and annually thereafter.
- (d) The FSAR (as updated) must be retained by the licensee until the Commission terminates its license.
- (e) Each holder of an ML under this part must submit an update of the FSAR every 24 months or more frequently as necessary to facilitate dependent COL or CP applications. The submittal must include the effects of changes on the content of the FSAR as described in paragraphs (a)(1), (a)(3) through (a)(5), and (b) of this section and safety analyses and evaluations performed by the licensee either in support of approved license amendments or in support of conclusions that changes did not require a license amendment under § 53.1530.

§ 53.1550 Evaluating changes to facility as described in Final Safety Analysis Reports.

- (a) The holder of an OL or COL may make changes in the facility as described in the FSAR (as updated) and make changes in the procedures as described in the FSAR (as updated) without obtaining a license amendment pursuant to § 53.1510 only if—
 - (1) A change to the technical specifications incorporated in the license is not required; and
 - (2) The change meets all of the following criteria:
 - (i) Does not result in an increase to the frequency or consequences of an event sequence such that an event sequence not previously identified as risk significant becomes risk significant by the analyses performed in accordance with § 53.450(e).
 - (ii) Does not result in an increase to the frequency or consequences of an event sequence such that an event sequence exceeds the licensing-basis event evaluation criteria required to be established in accordance with § 53.450(e).
 - (iii) Does not involve either of the following:
 - (A) A change to the NRC-approved comprehensive risk metric(s) or associated risk performance objective under § 53.220(b); or
 - (B) An increase to the frequency or consequences of one or more event sequences such that any calculated comprehensive risk metric exceeds the associated risk performance objective established in accordance with § 53.220.
 - (iv) Does not involve a departure from a method of evaluation described in the FSAR (as updated) used in assessing design-basis accidents in accordance with § 53.450(f) unless the results of the analysis under § 53.450(f) are conservative or essentially the same, the revised method of evaluation has been previously approved by the NRC for the intended application, or the revised method of evaluation can be used under an NRC-endorsed consensus code or standard.

- (v) Does not result in a change to the safety classification of an SSC from safety-related to either non-safety-related but safety-significant or non-safety-related.
- (vi) Does not result in more than a minimal decrease in defense in depth.
- (vii) [Reserved]
- (viii) Does not result in the identification of a new design-basis accident in accordance with § 53.450(f).
- (ix) Does not result in more than a minimal increase in the consequences of any design-basis accident.

- (3) In implementing this paragraph (a), the FSAR (as updated) is considered to include FSAR changes since submittal of the last update of the FSAR under § 53.1545.
- (4) The provisions in this section do not apply to changes to the facility or procedures when the applicable regulations establish more specific criteria for accomplishing such changes.

(b)

- (1) A licensee who references a design certification rule may make departures from the standard design, without prior Commission approval, unless the proposed departure involves a change to the design as described in the rule certifying the design, in which case the requirements of § 53.1525 are applicable.
- (2) The licensee must maintain records of all departures from the certified design of the facility and these records must be maintained and available for audit until the termination of the license. The licensee must identify the location and nature of departures from licensing-basis information within supporting documents for a certified design within the updates to the Safety Analysis Report required by § 53.1545.
- (3) Licensees for which the NRC has docketed the certifications required under § 53.1070 need not retain records of departures from the design of the facility associated with SSCs that have been permanently removed from service using an NRC-approved change process.

(c)

- (1) The licensee must maintain records of changes in the facility and procedures made under paragraph (a) of this section. These records must include a written evaluation which provides the bases for the determination that the change does not require a license amendment under paragraph (a)(2) of this section.
- (2) The licensee must submit, as specified in § 53.040, a report containing a brief description of any departures and changes, including a summary of the evaluation of each. A report must be submitted at intervals not to exceed 24 months. For COLs, the report must be submitted at intervals not to exceed 6 months during the period from the date of application for a COL to the date the Commission makes its findings under § 53.1452(g).
- (3) The records of changes in the facility must be maintained until the termination of an OL or COL issued under this part, or the termination of a renewed license issued under § 53.1595—whichever is later. Records of changes in procedures must be maintained for a period of 5 years.

§ 53.1560 Updating program documents included in licensing-basis information.

- (a) Each holder under this part of an OL or COL for which the Commission has made the finding under § 53.1452(g) must biennially or more frequently update the program documents submitted as part of an application to obtain or maintain the license to assure that the information included in the documents contains the latest information developed. The submittals must include the effects on the content of the program documents of—
 - (1) Changes made in the facility, procedures, licensee's organization, or site environs;
 - (2) Safety analyses and evaluations performed by the applicant or licensee either in support of approved license amendments or in support of conclusions that changes did not require a license amendment in accordance with § 53.1550;
 - (3) Analyses of new safety issues performed by or on behalf of the licensee at Commission request; and
 - (4) Changes to the programs as a result of operating experience, corrective actions, or other reasons deemed appropriate to ensure the programs serve their underlying purpose to support the requirements in subpart B of this part or other NRC regulations.
- (b)
 - (1) The licensee must submit revisions containing updated information to the Commission, as specified in § 53.040, identifying the location of revised or new information.
 - (2) The submittal must include—
 - (i) A certification by a duly authorized officer of the licensee that either the information accurately presents changes made since the previous submittals, necessary to reflect information and analyses submitted to the Commission or prepared pursuant to Commission requirement, or that no such changes were made; and
 - (ii) An identification of changes made under the provisions of § 53.1550 but not previously submitted to the Commission.
- (c) The updated program documents must be retained by the licensee until the Commission terminates their license.

§ 53.1565 Evaluating changes to programs included in licensing-basis information.

- (a) A licensee may make changes to the facility, procedures, or organizations or address changes to site environs as described in the program documents included in licensing-basis information without obtaining prior NRC approval only if—
 - (1) A change to the technical specifications incorporated in the license is not required;
 - (2) An exemption from an NRC regulation is not required; and
 - (3) The change conforms to program-specific requirements included in regulations in this part, technical specifications, or the NRC-approved program document included and reviewed as part of a license application under subpart H or an amendment under this subpart.
- (b) In implementing this section, the program documents (as updated) include changes since submittal of the last updates of the program documents pursuant to § 53.1560.

- (c) The provisions in this section do not apply to changes to the program documents when the applicable regulations establish more specific criteria for accomplishing such changes.
- (d) To make changes to the facility, procedures, or organizations or to address changes to site environs as described in the program documents included in licensing-basis information for individual programs, the following requirements must be satisfied:

(1) ***Quality assurance program—operation.***

- (i) Each holder under this part of an OL or COL, after the Commission makes the finding under § 53.1452(g), may make a change to a previously accepted quality assurance program (QAP) description included or referenced in the Safety Analysis Report without prior NRC approval, provided the change does not reduce the commitments in the program description as accepted by the NRC. Changes to the QAP description that do not reduce the commitments must be submitted to the NRC in accordance with the requirements of § 53.1545. In addition to QAP changes involving administrative improvements and clarifications, spelling corrections, punctuation, or editorial items, the following changes are not considered to be reductions in commitment:
 - (A) The use of a quality assurance (QA) standard approved by the NRC which is more recent than the QA standard in the licensee's QAP at the time of the change;
 - (B) The use of a QA alternative or exception approved by an NRC safety evaluation, provided that the bases of the NRC approval are applicable to the licensee's facility;
 - (C) The use of generic organizational position titles that clearly denote the position function, supplemented as necessary by descriptive text, rather than specific titles;
 - (D) The use of generic organizational charts to indicate functional relationships, authorities, and responsibilities, or, alternately, the use of descriptive text;
 - (E) The elimination of QAP information that duplicates language in QA regulatory guides and QA standards to which the licensee is committed; and
 - (F) Organizational revisions that ensure that persons and organizations performing QA functions continue to have the requisite authority and organizational freedom, including sufficient independence from cost and schedule when opposed to safety considerations.
- (ii) Changes to the QAP description that do reduce the commitments must be submitted to the NRC and receive NRC approval prior to implementation, as follows:
 - (A) Changes made to the QAP description as presented in the Safety Analysis Report or in a topical report must be submitted as specified in § 53.040.
 - (B) The submittal of a change to the Safety Analysis Report QAP description must include all pages affected by that change and must be accompanied by a forwarding letter identifying the change, the reason for the change, and the basis for concluding that the revised program incorporating the change continues to satisfy the criteria of appendix B to part 50 of this chapter and the Safety Analysis Report QAP description commitments previously accepted by the NRC (the letter need not provide the basis for changes that correct spelling, punctuation, or editorial items).
 - (C) A copy of the forwarding letter identifying the change must be maintained as a facility record for 3 years.

- (D) Changes to the QAP description included or referenced in the Safety Analysis Report shall be regarded as accepted by the Commission upon receipt of a letter to this effect from the appropriate reviewing office of the Commission or 60 days after submittal to the Commission, whichever occurs first.
- (2) **Quality assurance program—siting, construction, and manufacturing.** Each holder of an LWA, early site permit, CP, ML, or COL, before the Commission makes the finding under § 53.1452(g) of this chapter, under this part may make a change to a previously accepted QAP description included or referenced in the Safety Analysis Report without prior NRC approval, provided the change does not reduce the commitments in the program description previously accepted by the NRC. Changes to the QAP description that do not reduce the commitments must be submitted to NRC within 90 days. Changes to the QAP description that reduce the commitments must be submitted to NRC and receive NRC approval before implementation, as follows:
- (i) Changes to the Safety Analysis Report must be submitted for review as specified in § 53.040. Changes made to NRC-accepted QA topical report descriptions must be submitted as specified in § 53.040.
 - (ii) The submittal of a change to the Safety Analysis Report QAP description must include all pages affected by that change and must be accompanied by a forwarding letter identifying the change, the reason for the change, and the basis for concluding that the revised program incorporating the change continues to satisfy the criteria of appendix B of part 50 of this chapter and the Safety Analysis Report QAP description commitments previously accepted by the NRC (the letter need not provide the basis for changes that correct spelling, punctuation, or editorial items).
 - (iii) A copy of the forwarding letter identifying the changes must be maintained as a facility record for 3 years.
 - (iv) Changes to the QAP description included or referenced in the Safety Analysis Report shall be regarded as accepted by the Commission upon receipt of a letter to this effect from the appropriate reviewing office of the Commission or 60 days after submittal to the Commission, whichever occurs first.
- (3) **Emergency preparedness program.**
- (i) Definitions for the purpose of paragraph (d)(3) of this section:
 - (A) **Change** means an action that results in modification or addition to, or removal from, the licensee's emergency plan. All such changes are subject to the provisions of this section except where the applicable regulations establish specific criteria for accomplishing a particular change.
 - (B) **Emergency plan** means the document(s), prepared and maintained by the licensee, that identify and describe the licensee's methods for maintaining emergency preparedness and responding to emergencies. An emergency plan includes the plan as originally approved by the NRC and all subsequent changes made by the licensee with, and without, prior NRC review and approval under paragraph (d)(3) of this section.
 - (C) **Emergency planning function** means a capability or resource necessary to prepare for and respond to a radiological emergency.

- (D) *Reduction in effectiveness* means a change in an emergency plan that results in reducing the licensee's capability to perform an emergency planning function in the event of a radiological emergency.

(ii)

- (A) Except as provided in paragraph (d)(3)(ii)(B) of this section, a holder of an OL under this part, or a COL under this part after the Commission makes the finding under § 53.1452(g), must follow and maintain the effectiveness of an emergency plan that meets the requirements in appendix E to part 50 of this chapter and the planning standards of § 50.47(b).
- (B) A holder of an OL under this part for a commercial nuclear plant consisting of small modular reactors (SMRs) or non-light-water reactors, or a holder of a COL under this part after the Commission makes the finding under § 53.1452(g) for a commercial nuclear plant consisting of either SMRs or non-light-water reactors, must follow and maintain the effectiveness of either an emergency plan that meets the requirements in § 50.160 or an emergency plan that meets the requirements in appendix E to part 50 of this chapter and the planning standards of § 50.47(b).

(iii)

- (A) Except as provided in paragraph (d)(3)(iii)(B) of this section, the licensee may make changes to its emergency plan without NRC approval only if the licensee performs and retains an analysis demonstrating that the changes do not reduce the effectiveness of the plan and the plan, as changed, continues to meet the requirements in appendix E to part 50 of this chapter and the planning standards of § 50.47(b).
- (B) A license under this part for a commercial nuclear plant consisting of either SMRs or non-light-water reactors may make changes to its emergency plan without NRC approval only if the licensee performs and retains an analysis demonstrating that the changes do not reduce the effectiveness of the plan and the plan, as changed, continues to meet either the requirements in § 50.160 or the requirements in appendix E to part 50 and the planning standards of § 50.47(b).

- (iv) The changes to a licensee's emergency plan that reduce the effectiveness of the plan as defined in paragraph (d)(3)(i)(D) of this section may not be implemented without prior approval by the NRC. A licensee desiring to make such a change must submit an application for an amendment to its license. In addition to the filing requirements of §§ 53.1510 and 53.1515, the request must include all emergency plan pages affected by that change and must be accompanied by a forwarding letter identifying the change, the reason for the change, and the basis for concluding that the licensee's emergency plan, as revised, will continue to meet either the requirements in § 50.160 to this chapter or the requirements in appendix E to part 50 of this chapter and the planning standards of § 50.47(b) of this chapter.

- (v) The licensee must retain a record of each change to the emergency plan made without prior NRC approval for a period of three years from the date of the change and shall submit, as specified in § 53.040, a report of each such change, including a summary of its analysis, within 30 days after the change is put in effect.

(vi) The licensee must retain the emergency plan and each change for which prior NRC approval was obtained pursuant to paragraph (d)(3)(iv) of this section as a record until the Commission terminates the license for the nuclear power reactor.

(vii)

(A) The licensee must provide for the development, revision, implementation, and maintenance of its emergency preparedness program. The licensee must ensure that all program elements are reviewed by persons who have no direct responsibility for the implementation of the emergency preparedness program either—

(1) At intervals not to exceed 12 months; or

(2) As necessary, based on an assessment by the licensee against performance indicators, and as soon as reasonably practicable after a change occurs in personnel, procedures, equipment, or facilities that potentially could adversely affect emergency preparedness, but no longer than 12 months after the change. In any case, all elements of the emergency preparedness program must be reviewed at least once every 24 months.

(B) The review must include an evaluation for adequacy of interfaces with State, participating Tribal, and local governments and of licensee drills, exercises, capabilities, and procedures. The results of the review, along with recommendations for improvements, must be documented, reported to the licensee's corporate and plant management, and retained for a period of 5 years. The part of the review involving the evaluation for adequacy of interface with State, participating Tribal, and local governments must be available to the appropriate State, participating Tribal, and local governments.

(4) **Security programs.**

(i) The licensee must prepare and maintain safeguards contingency plan procedures in accordance with appendix C of part 73 of this chapter for affecting the actions and decisions contained in the Responsibility Matrix of the safeguards contingency plan. The licensee may not make a change that would decrease the safeguard effectiveness of a physical security plan, or guard training and qualification plan, or cybersecurity plan submitted under subpart H or part 73 of this chapter, or of the first four categories of information (Background, Generic Planning Base, Licensee Planning Base, Responsibility Matrix) contained in a licensee safeguards contingency plan submitted under subpart H or part 73 of this chapter, as applicable, without prior approval of the Commission. A licensee desiring to make such a change must submit an application for amendment to the licensee's license under §§ 53.1510, 53.1515, and 53.1520.

(ii) The licensee may make changes to the plans referenced in paragraph (d)(4)(i) of this section without prior Commission approval if the changes do not decrease the safeguards effectiveness of the plan. The licensee must maintain records of changes to the plans made without prior Commission approval for a period of 3 years from the date of the change, and must submit, as specified in § 53.040, a report containing a description of each change within 2 months after the change is made. Prior to the safeguards contingency plan being put into effect, the licensee must have—

(A) All safeguards capabilities specified in the safeguards contingency plan available and functional;

- (B) Detailed procedures developed according to appendix C to part 73 of this chapter available at the licensee's site; and
- (C) All appropriate personnel trained to respond to safeguards incidents as outlined in the plan and specified in the detailed procedures.
- (iii) The licensee must provide for the development, revision, implementation, and maintenance of its safeguards contingency plan. The licensee must ensure that all program elements are reviewed by individuals independent of both security program management and personnel who have direct responsibility for implementation of the security program either—
 - (A) At intervals not to exceed 12 months; or
 - (B) As necessary, based on an assessment by the licensee against performance indicators, and as soon as reasonably practicable after a change occurs in personnel, procedures, equipment, or facilities that potentially could adversely affect security, but no longer than 12 months after the change. In any case, all elements of the safeguards contingency plan must be reviewed at least once every 24 months.
- (iv) The review must include a review and audit of safeguards contingency procedures and practices, an audit of the security system testing and maintenance program, and a test of the safeguards systems along with commitments established for response by local law enforcement authorities. The results of the review and audit, along with recommendations for improvements, must be documented, reported to the licensee's corporate and plant management, and kept available at the plant for inspection for a period of 3 years.

§ 53.1570 Transfer of licenses.

- (a) No commercial nuclear plant license issued under this part, or any right thereunder, shall be transferred, assigned, or in any manner disposed of, either voluntarily or involuntarily, directly or indirectly, through transfer of control of the license to any person, unless the Commission gives its consent in writing.
- (b)
 - (1) An application for transfer of a license must include—
 - (i) As much of the information described in §§ 53.1109, 53.1306, 53.1366, and 53.1413 with respect to the identity and technical and financial qualifications of the proposed transferee as would be required by those sections if the application were for an initial license. The Commission may require additional information such as data respecting proposed safeguards against hazards from radioactive materials and the applicant's qualifications to protect against such hazards.
 - (ii) A statement of the purposes for which the transfer of the license is requested, the nature of the transaction necessitating or making desirable the transfer of the license, and an agreement to limit access to Restricted Data or Classified National Security Information pursuant to § 53.1115. The Commission may require any person who submits an application for license pursuant to the provisions of this section to file a written consent from the existing licensee or a certified copy of an order or judgment of a court of competent jurisdiction attesting to the person's right (subject to the licensing requirements of the Act and these regulations) to possession of the facility or site involved.
 - (2) [Reserved]

- (c) After appropriate notice to interested persons, including the existing licensee, and observance of such procedures as may be required by the Act or regulations or orders of the Commission, the Commission will approve an application for the transfer of a license, if the Commission determines—
 - (1) That the proposed transferee is qualified to be the holder of the license; and
 - (2) That transfer of the license is otherwise consistent with applicable provisions of law, regulations, and orders issued by the Commission pursuant thereto.

§ 53.1575 Termination of licenses.

- (a) When the holder of an OL or COL under this part has determined to permanently cease operations the licensee must, within 30 days, submit a written certification to the NRC, consistent with the requirements of § 53.1070.
- (b) Once fuel has been permanently removed from the reactor system, the licensee must submit a written certification to the NRC that meets the requirements of § 53.1070.
- (c)
 - (1) Upon docketing of the certifications for permanent cessation of operations and permanent removal of fuel from the reactor system, or when a final legally effective order to permanently cease operations has come into effect, the license no longer authorizes operation of the reactor or emplacement or retention of fuel into the reactor system.
 - (2) Activities associated with decommissioning will be carried out in accordance with the requirements and procedures in subpart G of this part.
 - (3) The Commission shall terminate the license if it determines that—
 - (i) The remaining dismantlement has been performed in accordance with the approved license termination plan required in subpart G of this part; and
 - (ii) The final radiation survey and associated documentation, including an assessment of dose contributions associated with parts released for use before approval of the license termination plan, demonstrate that the facility and site have met the criteria for decommissioning in subpart E of 10 CFR part 20.
- (d) A holder of a CP or COL under this part may request the termination of the license as well as licenses issued by the NRC under parts 30, 40, or 70 of this chapter prior to plant operations. Such requests may support an immediate NRC approval of the site for unrestricted use.

§ 53.1580 Information requests.

Each licensee under this part must at any time before termination of the license, upon request of the Commission, submit, as specified in § 53.040 written statements, signed under oath or affirmation, to enable the Commission to determine whether or not the license should be modified, suspended, or revoked. Except for information sought to verify licensee compliance with the current licensing basis for that facility, the NRC must prepare the reason or reasons for each information request prior to issuance to ensure that the burden to be imposed on respondents is justified in view of the potential safety significance of the issue to be addressed in the requested information. Each such justification provided for an evaluation performed by the NRC staff must be approved by the Executive Director for Operations or his or her designee prior to issuance of the request.

§ 53.1585 Revocation, suspension, modification of licenses and approvals for cause.

A license or standard design approval issued under this part may be revoked, suspended, or modified, in whole or in part, for any material false statement in the application or in the supplemental or other statement of fact required of the applicant; or because of conditions revealed by the application or statement of fact of any report, record, inspection, or other means which would warrant the Commission to refuse to grant a license or approval on an original application; or for failure to manufacture a reactor, or construct or operate a facility in accordance with the terms of the license, provided, however, that failure to make timely completion of the proposed construction or alteration of a facility under a CP under this part shall be governed by the provisions of § 53.1342(b); or for violation of, or failure to observe, any of the terms and provisions of the Act, regulations, license, approval, or order of the Commission.

§ 53.1590 Backfitting.

(a)

- (1) Backfitting means the modification of or addition to systems, structures, components, or design of a facility; or the design approval for a facility; or the procedures or organization required to design, construct or operate a facility; any of which may result from a new or amended provision in the Commission's regulations or the imposition of a regulatory staff position interpreting the Commission's regulations that is either new or different from a previously applicable staff position after the date of the commercial nuclear plant license issued under this part.
- (2) Except as provided in paragraph (a)(4) of this section, the Commission shall require a systematic and documented analysis pursuant to paragraph (b) of this section for backfits which it seeks to impose.
- (3) Except as provided in paragraph (a)(4) of this section, the Commission shall require the backfitting of a facility only when it determines, based on the analysis described in paragraph (b) of this section, that there is a substantial increase in the overall protection of the public health and safety or the common defense and security to be derived from the backfit and that the direct and indirect costs of implementation for that facility are justified in view of this increased protection.
- (4) The provisions of paragraphs (a)(2) and (a)(3) of this section are inapplicable and, therefore, backfit analysis is not required and the standards in paragraph (a)(3) of this section do not apply where the Commission or staff, as appropriate, finds and declares, with appropriate documented evaluation for its finding, either—
 - (i) That a modification is necessary to bring a facility into compliance with a license or the rules or orders of the Commission, or into conformance with written commitments by the licensee; or
 - (ii) That regulatory action is necessary to ensure that the facility provides adequate protection to the health and safety of the public and is in accord with the common defense and security; or
 - (iii) That the regulatory action involves defining or redefining what level of protection to the public health and safety or common defense and security should be regarded as adequate.
- (5) The Commission must always require the backfitting of a facility if it determines that such regulatory action is necessary to ensure that the facility provides adequate protection to the health and safety of the public and is in accord with the common defense and security.

- (6) The documented evaluation required by paragraph (a)(4) of this section must include a statement of the objectives of and reasons for the modification and the basis for invoking the exception. If immediately effective regulatory action is required, then the documented evaluation may follow rather than precede the regulatory action.
- (7) If there are two or more ways to achieve compliance with a license or the rules or orders of the Commission, or with written licensee commitments, or there are two or more ways to reach a level of protection which is adequate, then ordinarily the applicant or licensee is free to choose the way which best suits its purposes. However, should it be necessary or appropriate for the Commission to prescribe a specific way to comply with its requirements or to achieve adequate protection, then cost may be a factor in selecting the way, provided that the objective of compliance or adequate protection is met.
- (b) In reaching the determination required by paragraph (a)(3) of this section, the Commission will consider how the backfit should be scheduled in light of other ongoing regulatory activities at the facility and, in addition, will consider information available concerning any of the following factors as may be appropriate and any other information relevant and material to the proposed backfit:
 - (1) The statement of the specific objectives that the proposed backfit is designed to achieve;
 - (2) The general description of the activity that would be required by the licensee or applicant in order to complete the backfit;
 - (3) The potential change in the risk to the public from the accidental off-site release of radioactive material;
 - (4) The potential impact on radiological exposure of facility employees;
 - (5) The installation and continuing costs associated with the backfit, including the cost of facility downtime or the cost of construction delay;
 - (6) The potential safety impact of changes in plant or operational complexity, including the relationship to proposed and existing regulatory requirements;
 - (7) The estimated resource burden on the NRC associated with the proposed backfit and the availability of such resources;
 - (8) The potential impact of differences in facility type, design or age on the relevancy and practicality of the proposed backfit;
 - (9) Whether the proposed backfit is interim or final and, if interim, the justification for imposing the proposed backfit on an interim basis.
- (c) No licensing action will be withheld during the pendency of backfit analyses required by the Commission's rules.
- (d) The Executive Director for Operations shall be responsible for implementation of this section, and all analyses required by this section shall be approved by the Executive Director for Operations or his or her designee.

§ 53.1595 Renewal.

Licenses may be renewed by the Commission upon expiration of the period of the license.

Subpart J—Reporting and Other Administrative Requirements

§ 53.1600 General information.

Each applicant and licensee under this part must ensure that U.S. Nuclear Regulatory Commission (NRC) inspectors have unfettered access to sites and facilities licensed or proposed to be licensed in § 53.1610, must maintain records and make reports to the NRC in accordance with requirements in §§ 53.1620 through 53.1650, must satisfy financial qualification and reporting requirements in §§ 53.1660 through 53.1700, and must obtain and maintain required financial protections in case of an accident in §§ 53.1720 and 53.1730.

§ 53.1610 Unfettered access for inspections.

- (a) Each applicant for or holder of a manufacturing license (ML), operating license (OL), combined license (COL), construction permit (CP), or early site permit must permit inspection, by duly authorized representatives of the Commission, of its records, premises, activities, and of licensed materials in possession or use, related to the license or CP or early site permit as may be necessary to effectuate the purposes of the Atomic Energy Act of 1956, as amended, (the Act) and the Energy Reorganization Act of 1974, as amended.
- (b)
 - (1) Each holder of an ML, OL, COL, or CP must, upon request by the Director, Office of Nuclear Reactor Regulation, provide rent-free office space for the exclusive use of the Commission inspection personnel. Heat, air conditioning, light, electrical outlets, and janitorial services must be furnished by each licensee and each holder of a CP. The office must be convenient to and have full access to the facility and must provide the inspectors both visual and acoustic privacy.
 - (2) For a site or facility with an assigned resident inspector, the space provided must be adequate to accommodate a full-time inspector, a part-time secretary, and transient NRC personnel and must be generally commensurate with other office facilities at the site. For sites or facilities assigned multiple resident inspectors, additional space may be requested. The office space that is provided must be subject to the approval of the Director, Office of Nuclear Reactor Regulation. All furniture, supplies, and communication equipment will be furnished by the Commission.
 - (3) For a site or facility without an assigned resident inspector, temporary space to accommodate periodic or special inspections must be provided. The office space must be generally commensurate with other office accommodations at the site.
 - (4) The licensee or permit holder must afford any NRC resident inspector assigned to that site, or other NRC inspectors identified by the Regional Administrator as likely to inspect the facility, immediate unfettered access, equivalent to access provided regular plant employees, following proper identification and compliance with applicable access control measures for security, radiological protection, and personal safety.
 - (5) The licensee or permit holder must ensure that the arrival and presence of an NRC inspector, who has been properly authorized facility access as described in paragraph (b)(4) of this section, is not announced or otherwise communicated by its employees or contractors to other persons at the facility unless specifically requested by the NRC inspector.

§ 53.1620 Maintenance of records, making of reports.

- (a) Each holder of an ML, OL, COL, CP, or early site permit must maintain all records and make all reports, in connection with the activity, as may be required by the conditions of the license or permit or by the regulations and orders of the Commission in effectuating the purposes of the Act and the Energy Reorganization Act of 1974, as amended. Reports must be submitted in accordance with § 53.040.
- (b) [Reserved]
- (c) Records that are required by the regulations in this part, by license condition, or by technical specifications must be retained for the period specified by the appropriate regulation, license condition, or technical specification. If a retention period is not otherwise specified, these records must be retained until the Commission terminates the facility license or, in the case of an early site permit, until the permit expires.
- (d)
 - (1) Records which must be retained under this part may be the original or a reproduced copy or a microform if the reproduced copy or microform is duly authenticated by authorized personnel and the microform is capable of producing a clear and legible copy after storage for the period specified by Commission regulations. The record may also be stored in electronic media with the capability of producing legible, accurate, and complete records during the required retention period. Records such as letters, drawings, and specifications, must include all pertinent information such as stamps, initials, and signatures. The licensee must maintain adequate safeguards against tampering with, and loss of records.
 - (2) If there is a conflict between the Commission's regulations in this part, license condition, or technical specification, or other written Commission approval or authorization pertaining to the retention period for the same type of record, the retention period specified in the regulations in this part for such records shall apply unless the Commission, under § 53.080 of this part, has granted a specific exemption from the record retention requirements in the regulations in this part.
- (e) Each licensee must notify the Commission as specified in § 53.040 of this part, of successfully completing power ascension testing or startup testing as applicable within 30 calendar days of completing the testing.

§ 53.1630 Immediate notification requirements for operating commercial nuclear plants.

- (a) *General requirements.*^[1]
 - (1) Each holder of an OL under this part or a COL under this part after the Commission makes the finding under § 53.1452(g), must notify the NRC Headquarters Operations Center via the Emergency Notification System (ENS) of—
 - (i) The declaration of any of the Emergency Classes specified in the licensee's approved Emergency Plan; or
 - (ii) Those non-emergency events specified in paragraph (b) of this section that occurred within 3 years of the date of discovery.

^[1] Other requirements for immediate notification of the NRC by licensed operating commercial nuclear plants are contained elsewhere in this chapter, in particular §§ 20.1906, 20.2202, 72.216, 73.77, and 73.1200 of this chapter.

- (2) If the ENS is inoperative, the licensee must make the required notifications via commercial telephone service, other dedicated telephone system, or any other method which will ensure that a report is made as soon as practical to the NRC Headquarters Operations Center at the numbers specified in appendix A to part 73 of this chapter.
- (3) The licensee must notify the NRC immediately after notification of the appropriate State or local agencies and not later than 1 hour after the time the licensee declares one of the Emergency Classes.
- (4) The licensee must activate the data links with the NRC as specified in their emergency plans after declaring an Emergency Class for events of actual or potential substantial degradation of plant safety or security, probable risk to site personnel life, or site equipment damage caused by hostile action. The data links may also be activated by the licensee during emergency drills or exercises if the licensee's computer system has the capability to transmit the exercise data.
- (5) When making a report under paragraph (a)(1) of this section, the licensee must identify—
 - (i) The Emergency Class declared; or
 - (ii) Paragraph (b)(1), "One-hour reports," paragraph (b)(2), "Four-hour reports," or paragraph (b)(3), "Eight-hour reports," as the paragraph of this section requiring notification of the non-emergency event.
- (6) In lieu of submitting a report required under paragraph (b)(2) or (b)(3) of this section through the Emergency Notification System, the licensee may submit the report using other methods, provided the licensee submits the report to the NRC Headquarters Operations Center within the required timeframe and confirms receipt of the report by the NRC.

(b) **Non-emergency events** —

- (1) **One-hour reports.** If not reported as a declaration of an Emergency Class under paragraph (a) of this section, the licensee must notify the NRC as soon as practical and in all cases within one hour of the occurrence of any deviation from the plant's Technical Specifications authorized under § 53.740(h) of this part.
- (2) **Four-hour reports.** If not reported under paragraphs (a) or (b)(1) of this section, the licensee must notify the NRC as soon as practical, and in all cases within 4 hours of the occurrence of any of the following:
 - (i) The initiation of any commercial nuclear plant shutdown required by the plant's Technical Specifications.
 - (ii) Any event or condition that results in actuation of the reactor protection system when the reactor is critical except when the actuation results from and is part of a pre-planned sequence during testing or reactor operation.
 - (iii) [Reserved]
 - (iv) Any event or condition that results in an unplanned movement of, change of state in, or chemical interaction involving a significant amount of radioactive material within the commercial nuclear plant.
 - (v) [Reserved]

- (3) ***Eight-hour reports.*** If not reported under paragraphs (a), (b)(1), or (b)(2) of this section, the licensee must notify the NRC as soon as practical and in all cases within 8 hours of the occurrence of any of the following:
 - (i) Any event or condition that results in—
 - (A) The condition of the commercial nuclear plant, including its principal safety barriers, being seriously degraded; or
 - (B) The commercial nuclear plant being in a condition not analyzed under § 53.450 that significantly degrades plant safety.
 - (ii)-(iv) [Reserved]
 - (v) Any event that results in a major loss of emergency assessment capability, offsite response capability, or offsite communications capability (e.g., significant portion of control room indication, ENS, or offsite notification system).
- (c) ***Follow-up notification:*** With respect to the notifications made under paragraphs (a) and (b) of this section, in addition to making the required initial notification, each licensee, must during the course of the event—
 - (1) Immediately report:
 - (i) Any further degradation in the level of safety of the plant or other worsening plant conditions, including those that require the declaration of any of the Emergency Classes, if such a declaration has not been previously made; or
 - (ii) Any change from one Emergency Class to another; or
 - (iii) A termination of the Emergency Class.
 - (2) Immediately report:
 - (i) The results of ensuing evaluations or assessments of plant conditions,
 - (ii) The effectiveness of response or protective measures taken, and
 - (iii) Important information related to plant behavior that is not understood.
 - (3) Maintain an open, continuous communication channel with the NRC Headquarters Operation Center upon request by the NRC.

§ 53.1640 Licensee event report system.

- (a) ***Reportable events.***
 - (1) Each commercial nuclear plant licensee holding an OL under this part or a COL under this part after the Commission makes the finding under § 53.1452(g), must submit a Licensee Event Report (LER) for any event of the type described in this paragraph (a) within 60 days after discovery of the event. In the case of an invalid actuation reported under § 53.1640(a)(2), other than automatic reactor shutdown when the reactor is critical, the licensee may, at its option, provide a telephone notification to the NRC Operations Center within 60 days after discovery of the event instead of submitting a written LER. Unless otherwise specified in this section, the licensee must report an event if it occurred within 3 years of the date of discovery regardless of the plant mode or power level, and regardless of the significance of the structure, system, or component that initiated the event.

(2) The licensee must report—

(i)

- (A) The completion of any commercial nuclear plant shutdown required by the plant's Technical Specifications.
- (B) Any operation or condition which was prohibited by the plant's Technical Specifications except when—
 - (1) The Technical Specification is administrative in nature;
 - (2) The event consisted solely of a case of a late surveillance test where the oversight was corrected, the test was performed, and the equipment was found to be capable of performing its specified safety functions; or
 - (3) The Technical Specification was revised prior to discovery of the event such that the operation or condition was no longer prohibited at the time of the event.
- (C) Any deviation from the plant's Technical Specifications authorized under § 53.740(h).

(ii) Any event or condition that resulted in—

- (A) The condition of the commercial nuclear plant, including its principal safety barriers, being seriously degraded; or
- (B) The commercial nuclear plant being in a condition not analyzed under § 53.450 that significantly degrades plant safety.

(iii) Any natural phenomena or other external condition that posed an actual threat to the safety of the commercial nuclear plant or significantly hampered site personnel in the performance of duties necessary for the safe operation of the commercial nuclear plant.

(iv) Any event or condition that resulted in inadvertent operation of any structures, systems, and component classified as safety-related (SR) for an identified safety function under § 53.460 or the unplanned sole reliance on an SR system for those systems that are in constant operation, except when—

- (A) The actuation resulted from and was part of a pre-planned sequence during testing; or
- (B) The actuation was invalid and—
 - (1) Occurred while the system was properly removed from service; or
 - (2) Occurred after the safety function had been already completed.

(v) Any event or condition that could have prevented the fulfillment of the safety functions identified under § 53.230.

(vi) Events covered in paragraph (a)(2)(v) of this section may include one or more procedural errors, equipment failures, and/or discovery of design, fabrication, construction, and/or procedural inadequacies. However, individual component failures need not be reported pursuant to paragraph (a)(2)(v) of this section if any other equipment was operable and available to perform the required safety function.

(vii)

- (A) Any event or condition that as a result of a single cause could have prevented the fulfillment of any of the safety functions identified under § 53.230.
- (B) Events covered in paragraph (a)(2)(vii)(A) of this section may include cases of procedural error, equipment failure, and/or discovery of a design, analysis, fabrication, construction, and/or procedural inadequacy. However, licensees are not required to report an event pursuant to paragraph (a)(2)(vii)(A) of this section if the event results from—
 - (1) A shared dependency among trains or channels that is a natural or expected consequence of the approved plant design; or
 - (2) Normal and expected wear or degradation.

(viii)

- (A) Any airborne radioactive release that, when averaged over a time period of 1-hour, resulted in airborne radionuclide concentrations in an unrestricted area that exceeds 20 times the applicable concentration limits specified in appendix B to 10 CFR part 20, table 2, column 1.
- (B) Any liquid effluent release that, when averaged over a time period of 1-hour, exceeds 20 times the applicable concentrations specified in appendix B to 10 CFR part 20, table 2, column 2, at the point of entry into the receiving waters (*i.e.*, unrestricted area) for all radionuclides except tritium and dissolved noble gases.
- (ix) Any event that posed an actual threat to the safety of the commercial nuclear plant or significantly hampered site personnel in the performance of duties necessary for the safe operation of the plant, including fires, toxic gas releases, or radioactive releases.

(b) **Contents.** The LER must contain—

- (1) A brief abstract describing the major occurrences during the event, including all component or system failures that contributed to the event and significant corrective action taken or planned to prevent recurrence.
- (2)
 - (i) A clear, specific narrative description of what occurred so that knowledgeable readers conversant with the design of commercial nuclear plants, but not familiar with the details of a particular plant, can understand the complete event.
 - (ii) The narrative description must include the following specific information as appropriate for the particular event:
 - (A) Plant operating conditions before the event.
 - (B) Status of systems, structures, or components that were inoperable at the start of the event and that contributed to the event.
 - (C) Dates and approximate time of the occurrences.
 - (D) The cause of each component or system failure or personnel error, if known.
 - (E) The failure mode, mechanism, and effect of each failed component, if known.
 - (F) [Reserved]

- (G) For failures of components with multiple functions, include a list of systems or secondary functions that were also affected.
 - (H) For failure that rendered a component or system classified as SR or non-safety-related but safety-significant inoperable, an estimate of the elapsed time from the discovery of the failure until the component or system was returned to service.
 - (I) The method of discovery of each component or system failure or procedural error.
 - (J) For each human performance related root cause, the licensee must discuss the cause(s) and circumstances.
 - (K) Automatically and manually initiated safety system responses.
 - (L) The manufacturer and model number (or other identification) of each component that failed during the event.
- (3) An assessment of the safety consequences and implications of the event. This assessment must include—
- (i) The availability of systems or components that could have performed the same function as the components and systems that failed during the event, and
 - (ii) For events that occurred when the reactor was shut down, the availability of systems or components that are needed to shut down the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident.
- (4) A description of any corrective actions planned as a result of the event, including those to reduce the probability of similar events occurring in the future.
- (5) Reference to any previous similar events at the same plant that are known to the licensee.
- (6) The name and contact information of a person within the licensee's organization who is knowledgeable about the event and can provide additional information concerning the event and the plant's characteristics.
- (c) **Supplemental information.** The Commission may require the licensee to submit specific additional information beyond that required by paragraph (b) of this section if the Commission finds that supplemental material is necessary for complete understanding of an unusually complex or significant event. These requests for supplemental information will be made in writing and the licensee must submit, as specified in § 53.040, the requested information as a supplement to the initial LER.
- (d) **Submission of reports.** Licensee Event Reports must be prepared on NRC Form 366 and submitted to the NRC, as specified in § 53.040.
- (e) **Report legibility.** The reports and copies that licensees are required to submit to the Commission under the provisions of this section must be of sufficient quality to permit legible reproduction and micrographic processing.

§ 53.1645 Reports of radiation exposure to members of the public.

- (a) Each holder of an OL, and each holder of a COL after the Commission has made the finding under § 53.1452(g), must submit radiological reports as required by 10 CFR part 20, as well as an Annual Radioactive Effluent Release Report and an Annual Radiological Environmental Operating Report. The

Annual Radioactive Effluent Release Report must specify the quantity of each of the principal radionuclides released to unrestricted areas in liquid and in gaseous effluents and an estimate of the dose received by the maximally exposed member of the public in an unrestricted area from effluents and direct radiation from contained sources during the previous calendar year. The Annual Radiological Environmental Operating Report must provide data on measurable levels of radiation and radioactive materials in the environment, must include an evaluation of the relationship between quantities of radioactive material released in effluents and resultant radiation doses to individuals from principal pathways of exposure, and must include the results of environmental monitoring during the previous calendar year. These reports must also include any other information as may be required by the Commission to estimate maximum potential annual radiation doses to the public. The reports must be submitted as specified in § 53.040 by May 15 of each successive year. If the total effective dose equivalent to members of the public in unrestricted areas during the reporting period is greater than the design objectives established under § 53.425, the report must specify the causes for exceeding the design objective and describe any corrective actions. On the basis of these reports and any additional information the Commission may obtain from the licensee or others, the Commission may require the licensee to take action as the Commission deems appropriate.

- (b) If during any calendar quarter the radiation exposure to a member of the public in the unrestricted areas, calculated on the same basis as the respective design objective exposure, exceeds one-half of the annual design objective exposure, the licensee must submit a report as specified in § 53.040. The report shall specify the causes for exceeding one-half the annual design objective exposure in a quarter and describe corrective actions that the licensee will take to maintain radiation exposure to levels within the design objectives for the remainder of the year. The report shall be submitted within 30 days from the end of the quarter when one-half of the annual design objective exposure was exceeded.

§ 53.1650 Facility information and verification.

- (a) In response to a written request by the Commission, each applicant for a CP or license and each recipient of a CP or a license must submit facility information, as described in § 75.10 of this chapter, on International Atomic Energy Agency (IAEA) Design Information Questionnaire forms and site information on DOC/NRC Form AP-A and associated forms;
- (b) As required by the Additional Protocol, must submit location information described in § 75.11 of this chapter on DOC/NRC Form AP-1 and associated forms; and
- (c) Must permit verification thereof by the IAEA and take other action as necessary to implement the US/IAEA Safeguards Agreement, as described in part 75 of this chapter.

§ 53.1660 Financial requirements.

Sections 53.1670 through 53.1700 set out the requirements and procedures related to financial qualifications and related reporting requirements.

§ 53.1670 Financial qualifications.

Except for an electric utility applicant for a license to operate a commercial nuclear plant, an applicant for a CP, OL, or COL under this part must appear to be financially qualified for the activities for which the permit or license is sought.

§ 53.1680 [Reserved]

§ 53.1690 Licensee's change of status; financial qualifications.

- (a) An electric utility licensee holding an OL or COL (including a renewed license) for a commercial nuclear plant, no later than seventy-five (75) days prior to ceasing to be an electric utility in any manner not involving a license transfer under § 53.1399 or § 53.1456 must provide the NRC with the financial qualifications information that would be required for obtaining an initial OL under this part. The financial qualifications information must address the first full 5 years of operation after the date the licensee ceases to be an electric utility.
- (b)
 - (1) Any holder of a license issued under this part must notify the appropriate NRC Regional Administrator, in writing, immediately following the filing of a voluntary or involuntary petition for bankruptcy under any chapter of title 11 (Bankruptcy) of the United States Code by or against—
 - (i) The licensee;
 - (ii) An entity (as 11 U.S.C. 101(14) defines that term) controlling the licensee or listing the license or licensee as property of the estate; or
 - (iii) An affiliate (as 11 U.S.C. 101(2) defines that term) of the licensee.
 - (2) This notification must indicate—
 - (i) The bankruptcy court in which the petition for bankruptcy was filed; and
 - (ii) The date of the filing of the petition.

§ 53.1700 Creditor regulations.

- (a) Pursuant to section 184 of the Act, the Commission consents, without individual application, to the creation of any mortgage, pledge, or other lien upon any facility not owned by the United States which is the subject of a license or upon any leasehold or other interest in such facility; provided—
 - (1) That the rights of any creditor so secured may be exercised only in compliance with and subject to the same requirements and restrictions as would apply to the licensee pursuant to the provisions of the license, the Act, and regulations issued by the Commission under the Act; and
 - (2) That no creditor so secured may take possession of the facility pursuant to the provisions of this section prior to either the issuance of a license from the Commission authorizing such possession or the transfer of the license.
- (b) Any creditor so secured may apply for transfer of the license covering such facility by filing an application for transfer of the license under § 53.1570. The Commission will act upon such application under subpart I of this part.
- (c) Nothing contained in this regulation shall be deemed to affect the means of acquiring, or the priority of, any tax lien or other lien provided by law.
- (d) As used in this section—
 - (1) **License** includes any license under this part, which may be issued by the Commission with regard to a facility.

- (2) **Creditor** includes, without implied limitation, the trustee under any mortgage, pledge or lien on a facility made to secure any creditor, any trustee or receiver of the facility appointed by a court of competent jurisdiction in any action brought for the benefit of any creditor secured by such mortgage, pledge or lien, any purchaser of such facility at the sale thereof upon foreclosure of such mortgage, pledge, or lien or upon exercise of any power of sale contained therein, or any assignee of any such purchaser.
- (3) **Facility** includes, but is not limited to, a site which is the subject of an early site permit under this part, and a reactor manufactured under an ML under this part.

§ 53.1710 Financial protection.

Sections 53.1720 and 53.1730 set out the requirements and procedures related to licensees obtaining and maintaining insurance to cover stabilization and decontamination activities in the event of an accident and financial protection in accordance with part 140, "Financial Protection Requirements and Indemnity Agreements," of this chapter.

§ 53.1720 Insurance required to stabilize and decontaminate plant following an accident.

Each commercial nuclear plant licensee under this part must take reasonable steps to obtain insurance available at reasonable costs and on reasonable terms from private sources or to demonstrate that it possesses an equivalent amount of protection covering the licensee's obligation, in the event of an accident at the licensee's commercial nuclear reactor, to stabilize and decontaminate the plant and the plant site at which such an accident may occur, provided that—

- (a) The insurance required by this section must have a minimum coverage limit for each commercial nuclear plant site of \$1.06 billion, an amount based on plant-specific estimates of costs to stabilize and decontaminate a plant, or whatever amount of insurance is generally available from private sources, whichever is less. The required insurance must clearly state that, as and to the extent provided in paragraph (d)(1) of this section, any proceeds must be payable first for stabilization of the plant and next for decontamination of the plant and the plant site. If a licensee's coverage falls below the required minimum, the licensee must within 60 days take all reasonable steps to restore its coverage to the required minimum. The required insurance may, at the option of the licensee, be included within policies that also provide coverage for other risks, including, but not limited to, the risk of direct physical damage.
- (b)
 - (1) With respect to policies issued or annually renewed, the proceeds of such required insurance must be dedicated, as and to the extent provided in this paragraph (b), to reimbursement or payment on behalf of the insured of reasonable expenses incurred or estimated to be incurred by the licensee in taking action to fulfill the licensee's obligation, in the event of an accident at the licensee's plant, to ensure that the plant is in, or is returned to, and maintained in, a safe and stable condition and that radioactive contamination is removed or controlled such that personnel exposures are consistent with the occupational exposure limits in 10 CFR part 20. These actions must be consistent with any other obligation the licensee may have under this chapter and must be subject to paragraph (d) of this section. As used in this section, an "accident" means an event that involves the release of radioactive material from its intended place of confinement within the commercial nuclear plant such that there is a present danger of release off site in amounts that would pose a threat to the public health and safety.

(2) The stabilization and decontamination requirements set forth in paragraph (d) of this section must apply uniformly to all insurance policies required under this section.

(c) The licensee shall report to the NRC on April 1 of each year the current levels of this insurance or financial security it maintains and the sources of this insurance or financial security.

(d)

(1) In the event of an accident at the licensee's plant, whenever the estimated costs of stabilizing the licensed plant and of decontaminating the plant and the plant site exceed one tenth of the minimum insurance under paragraph (a) of this section, the proceeds of the insurance required by this section must be dedicated to and used, first, to ensure that the licensed plant is in, or is returned to, and can be maintained in, a safe and stable condition so as to prevent any significant risk to the public health and safety and, second, to decontaminate the plant and the plant site in accordance with the licensee's cleanup plan as approved by order of the Director, Office of Nuclear Reactor Regulation. This priority on insurance proceeds must remain in effect for 60 days or, upon order of the Director, for such longer periods, in increments not to exceed 60 days except as provided for activities under the cleanup plan required in paragraphs (d)(3) and (d)(4) of this section, as the Director may find necessary to protect the public health and safety. Actions needed to bring the plant to and maintain the plant in a safe and stable condition may include one or more of the following, as appropriate:

(i) Shutdown of the reactor(s) and other processes at the plant;

(ii) Establishment and maintenance of long-term cooling with stable decay heat removal;

(iii) Maintenance of sub-criticality;

(iv) Control of radioactive releases; and

(v) Securing of structures, systems, or components to minimize radiation exposure to onsite personnel or to the offsite public or to facilitate later decontamination or both.

(2) The licensee must inform the Director, Office of Nuclear Reactor Regulation in writing when the plant is and can be maintained in a safe and stable condition so as to prevent any significant risk to the public health and safety. Within 30 days after the licensee informs the Director that the plant is in this condition, or at such earlier time as the licensee may elect or the Director may for good cause direct, the licensee must prepare and submit a cleanup plan for the Director's approval. The cleanup plan must identify and contain an estimate of the cost of each cleanup operation that will be required to decontaminate the reactor sufficiently to permit the licensee either to resume operation of the reactor or to apply to the Commission under subpart G of this part for authority to decommission the reactor and to surrender the license voluntarily. Cleanup operations may include one or more of the following, as appropriate:

(i) Processing any contaminated materials generated by the accident and by decontamination operations to remove radioactive materials;

(ii) Decontamination of surfaces inside the plant buildings to levels consistent with the Commission's occupational exposure limits in 10 CFR part 20, and decontamination or disposal of equipment;

(iii) Decontamination or removal and disposal of internal parts, damaged fuel from the reactor coolant or fuel systems, or related process or waste systems; and

(iv) Cleanup of the reactor coolant or fuel systems or related process or waste systems.

- (3) Following review of the licensee's cleanup plan, the Director will order the licensee to complete all operations that the Director finds are necessary to decontaminate the reactor sufficiently to permit the licensee either to resume operation of the reactor or to apply to the Commission under subpart G of this part for authority to decommission the reactor and to surrender the license voluntarily. The Director must approve or disapprove, in whole or in part for stated reasons, the licensee's estimate of cleanup costs for such operations. Such order may not be effective for more than one year, at which time it may be renewed. Each subsequent renewal order, if imposed, may be effective for not more than 6 months.
- (4) Of the balance of the proceeds of the required insurance not already expended to place the plant in a safe and stable condition under paragraph (b)(1) of this section, an amount sufficient to cover the expenses of completion of those decontamination operations that are the subject of the Director's order must be dedicated to such use, provided that, upon certification to the Director of the amounts expended previously and from time to time for stabilization and decontamination and upon further certification to the Director as to the sufficiency of the dedicated amount remaining, policies of insurance may provide for payment to the licensee or other loss payees of amounts not so dedicated, and the licensee may proceed to use in parallel (and not in preference thereto) any insurance proceeds not so dedicated for other purposes.

§ 53.1730 Financial protection requirements.

Commercial nuclear plant licensees must satisfy the applicable provisions of part 140, "Financial Protection Requirements and Indemnity Agreements," of this chapter.

Subparts K-L [Reserved]

Subpart M—Enforcement

§ 53.9000 Violations.

- (a) The Commission may obtain an injunction or other court order to prevent a violation of the provisions of—
 - (1) The Atomic Energy Act of 1954, as amended (the Act);
 - (2) Title II of the Energy Reorganization Act of 1974, as amended; or
 - (3) A regulation or order issued under those Acts.
- (b) The Commission may obtain a court order for the payment of a civil penalty imposed under Section 234 of the Act:
 - (1) For violations of—
 - (i) Sections 53, 57, 62, 63, 81, 82, 101, 103, 104, 107, or 109 of the Act;
 - (ii) Section 206 of the Energy Reorganization Act of 1974, as amended;
 - (iii) Any rule, regulation, or order issued under the sections specified in paragraph (b)(1)(i) of this section;
 - (iv) Any term, condition, or limitation of any license issued under the sections specified in paragraph (b)(1)(i) of this section.
 - (2) For any violation for which a license may be revoked under section 186 of the Act.

§ 53.9010 Criminal penalties.

- (a) Section 223 of the Act provides for criminal sanctions for willful violation of, attempted violation of, or conspiracy to violate, any regulation issued under sections 161b, 161i, or 161o of the Act. For purposes of section 223, all the regulations in part 53 are issued under one or more of sections 161b, 161i, or 161o, except for the sections listed in paragraph (b) of this section.
- (b) The regulations in 10 CFR part 53 that are not issued under sections 161b, 161i, or 161o for the purposes of section 223 are as follows: §§ 53.000, 53.015, 53.020, 53.040, 53.080, 53.090, 53.100, 53.110, 53.120, 53.600, 53.725, 53.726, 53.735, 53.760, 53.775, 53.790, 53.795, 53.820, 53.910, 53.1000, 53.1050, 53.1100, 53.1103, 53.1106, 53.1109, 53.1112, 53.1115, 53.1118, 53.1120, 53.1121, 53.1124, 53.1140, 53.1144, 53.1146, 53.1149, 53.1155, 53.1158, 53.1164, 53.1170, 53.1173, 53.1176, 53.1179, 53.1188, 53.1200, 53.1206, 53.1209, 53.1210, 53.1212, 53.1215, 53.1218, 53.1221, 53.1230, 53.1236, 53.1239, 53.1241, 53.1242, 53.1245, 53.1248, 53.1251, 53.1254, 52.1257, 52.1260, 53.1263, 53.1270, 53.1276, 53.1279, 53.1282, 53.1285, 53.1286, 53.1287, 53.1288, 53.1291, 53.1293, 53.1295, 53.1300, 53.1306, 53.1309, 53.1312, 53.1315, 53.1318, 53.1324, 53.1330, 53.1333, 53.1336, 53.1348, 53.1360, 53.1366, 53.1369, 53.1372, 53.1375, 53.1381, 53.1384, 53.1387, 53.1390, 53.1396, 53.1401, 53.1405, 53.1410, 53.1416, 53.1419, 53.1422, 53.1425, 53.1431, 53.1437, 53.1440, 53.1443, 53.1452, 53.1455, 53.1456, 53.1458, 53.1461, 53.1470, 53.1500, 53.1510, 53.1515, 53.1520, 53.1525, 53.1530, 53.1535, 53.1540, 53.1560, 53.1585, 53.1590, 53.1595, 53.1600, 53.1660, 53.1670, 53.1700, 53.1710, 53.1730, 53.9000, 53.9010.